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CHE 102 — Past Questions & Materials

by Renance

Course: CHE 102

Institution: Federal University of Technology, Akure

Sources in this pack: 7

Includes past questions and lecture notes

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Paper 1 of 7 — 2nd SemesterCHE102Past QuestionsCHE 102 - Practice Exam Questions BY PHOTOCHEM.pdf

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CH102 Practice Exam #1 ____ 17. The addition of bromine (Br₂) to an alkene

results in an alkane because one bond of the multiple

WAY longer than actual exam, bond is broken.

answers are at the end

____ 18. One of the halogenation reactions occurs

True/False when a halogen, a member of group VIIA, reacts with

Indicate whether the sentence or statement is true or alkene.

false.

____ 19. Cyclic compounds do not undergo

____ 1. Organic chemistry can be defined as the halogenation reactions.

chemistry of carbon-containing compounds.

____ 20. Markovnikov's rule states that when a

____ 2. Organic compounds can contain other hydrogen and a different molecule (OH-, Br-) add to a elements such as oxygen or nitrogen. double bond the hydrogen goes to the end of the double bond that already had more hydrogens

____ 3. Organic compounds are always derived from a

living source. ____ 21. An alkene with one multiple bond can be converted to an alkane by hydration.

____ 4. An unlimited number of carbon atoms can

bond together to form chains. ____ 22. Polymers are compounds that are composed of repeating units chemically bound to each other.

____ 5. Most organic compounds contain ionic bonds.

____ 23. The chemical properties of alkynes are very

____ 6. There are more inorganic compounds than different from those of alkenes. organic compounds.

____ 24. 2-butyne can exist as cis- and trans- isomers.

____ 7. Most inorganic compounds have a higher

melting point than organic compounds. ____ 25. Two moles of hydrogen gas would be required to convert one mole of 2-butyne into butane.

____ 8. Inorganic compounds are never found in living

systems. ____ 26. Benzene is an alkene with more than one multiple bond.

____ 9. Isomers differ from each other in the number

of neutrons present in each. ____ 27. Phenyl is the name given to benzene when it is a substituent of a larger molecule.

____ 10. Isomers always have the same empirical

(molecular) formula. ____ 28. Aromatic compounds dissolve well in a

nonpolar solvent.

___ 11. Isomers have identical physical and chemical properties. Multiple Choice

Identify the letter of the choice that best completes the

___ 12. Carbon-carbon bonds are usually strong and statement or answers the question. covalent.

___ 29. What is the reason why there are so many

___ 13. A characteristic of the alkanes is that they burn organic compounds?

in air. a. Organic compounds are produced by living things.

b. There are cyclic organic compounds, and none of the

___ 14. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2$ is the formula for a saturated inorganic compounds are cyclic.

hydrocarbon. c. Carbon atoms can form stable covalent bonds with other carbon atoms.

___ 15. Alkenes must have at least two carbon atoms. d. Organic compounds can be larger than inorganic compounds.

___ 16. Alkenes only have one double bond.

___ 30. Which applies to the manner in which by IUPAC rules.

carbon bonds in alkanes? c. The naming system for hydrocarbons requires the

a. The bonds are formed with carbon using an sp^3 configuration. identification of the longest carbon chain.

b. The bonds between carbon and other atoms are ionic. d. The naming system for organic compounds requires

c. carbon to carbon bonds are expected to be double bonds. and IUPAC name and a common name.

d. The bonds formed by carbon all use p orbitals.

___ 37. Identify the correct statement. Cycloalkanes

___ 31. Which statement referring to functional groups a. can have as few as two carbons.

is correct? Functional groups are b. are unsaturated hydrocarbons.

a. where nothing happens in a chemical reaction c. have more hydrogens than straight chain alkanes with

b. the parts of molecules used to classify organic molecules the same number of carbons

c. where most chemical reaction occur d. these are all wrong

d. both b and c

___ 38. Which of the following terms applies to the

___ 32. Identify the correct statement: The alkanes alkanes?

represent one family of organic compounds composed of a. hydrophobic carbon compounds b. hydrophilic

a. in which the halogens are a major component c. soluble

b. in which there is at least one double bond between d. bases carbons

c. that are composed only of carbon and hydrogen

d. in which hydrogens can be bonded to other hydrogens ___ 39. What is a characteristic shared by the alkanes?

a. solids

b. polar

___ 33. Which of the following formulas represents an organic compound? c. flammable
d. colored

a. CH₄

b. CH₄CH₂CH₃

c. CH₃CH₃CH₃ ___ 40. Which of the following is used for alkanes?

d. they all do a. fuel

b. lubricant

c. solvent

___ 34. Which of the following is an alcohol? d. all of the above

a. CH₃CH₂COOH

b. CH₃CH₂OH

c. CH₃COCH₃ ___ 41. Which of the following formulas represents a

d. CH₃CH₂Br saturated hydrocarbon?

a. C₂H₆

b. C₂H₄

___ 35. What is meant by the term "saturated" when describing hydrocarbons? c. C₃H₆

d. C₅H₁₀

a. The hydrocarbons are saturated with water and are wet.

b. The hydrocarbons are saturated with carbons in its

parent chain. ___ 42. All of the members of the alkane series of

c. The hydrocarbons are saturated with hydrogens and hydrocarbons have the general formula
have only single bonds between carbons. a. C_nH_{2n}

d. The compound is pure b. C_nH_n

c. C_nH_{2n+2}

d. C_nH_{2n-2}

___ 36. The naming of organic compounds is specific.

Which is an incorrect statement?

a. The smaller of the compounds have been known long enough to have common names.

b. The naming system of organic compounds is set out

___ 43. Which of the following structural formulas represents an unbranched alkane? b. butane
c. hexane

a. d. octane

b.

___ 49. Which is the longest carbon chain?

c.

d.

___ 44. Which of the following compounds exhibits structural isomerism? a. 9
b. 6

- a. propane c. 7
- b. methane d. 10
- c. butane
- d. ethane

____ 50. The IUPAC name for is

____ 45. How many structural isomers are possible for a. 2-ethylbutane butane (molecular formula C_4H_{10})? b. 2-methylbutane

- a. 2 c. 3-methylpentane
- b. 3 d. 3-ethylbutane
- c. 4
- d. 5

____ 51. The compound has

____ 46. Which of the following compounds is a the IUPAC name structural isomer of 2-methylbutane? a. 4-isopropyloctane

- a. propane b. 5-isopropyloctane
- b. w-methylpropone c. 5-sec-butyloctane
- c. butane d. 4-propyloctane
- d. pentane

____ 52. The structural formula of 1-bromo-2,2-dimethylpropane is

____ 47. A structural isomer of is a.

- a.
- b.
- b.
- c. c.
- d.
- d.

____ 48. The name for the following structure will end with

- a. ethane

____ 53. A general formula for cycloalkanes is c. carbon dioxide and carbon monoxide

- a. $C_{2H_{2n+2}}$ d. carbon monoxide and carbon
- b. C_nH_{2n}
- c. C_nH_{2n-2}
- d. C_nH_{2n-4}

____ 60. Alkanes react readily with

- a. sodium hydroxide
- b. hydrochloric acid
- c. sulfuric acid
- d. none of the above

____ 54. The IUPAC name of

contains an ethyl group located at position

____ 61. Name a difference between a saturated and an

a. 1 unsaturated hydrocarbon.

b. 2

a. Saturated hydrocarbons are composed of only carbon

c. 3

and hydrogen, and unsaturated hydrocarbons include

d. 4

other atoms than just carbon and hydrogen.

b. Saturated hydrocarbons do not contain multiple bonds

between carbons, but unsaturated hydrocarbons do

contain multiple bonds.

____ 55. A correct IUPAC name for is c. Unsaturated hydrocarbons are flammable, but saturated

a. 1-bromo-3-ethyl-1-methylcyclopropane hydrocarbons aren't flammable.

b. 1-bromo-2-ethyl-1-methylcyclopropane d. Saturated hydrocarbons are essentially insoluble.

c. 1-bromo-1-ethyl-2-methylcyclopropane Unsaturated hydrocarbons are soluble.

d. 2-bromo-1-ethyl-2-methylcyclopropane

____ 62. What number would be used to indicate the

double bond position in the IUPAC name for $\text{CH}_3\text{CH}=\text{CHCH}_3$?

= CHCH_3 ?

a. 1

____ 56. The component is ____-1,2-

dimethylcyclopropane. b. 2

c. 3

a. cis

b. trans d. 4

c. trans

d. trans ____ 63. In the IUPAC name for the following

compound, the -Br group is located at position

____ 57. An alkane with 10 carbon atoms is a ____ at

room temperature.

a. solid a. 1

b. liquid b. 2

c. gas c. 3

d. not possible to tell d. 4

____ 58. Alkanes are ____ in water and ____ than

water. ____ 64. The IUPAC name of is

a. insoluble, less dense a. 3-ethyl-1-butene

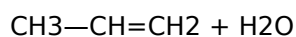
b. soluble, less dense b. 2-ethyl-2-butene

c. insoluble, more dense c. 3-ethyl-3-butene

d. soluble, more dense d. 2-ethyl-1-butene

___ 59. Complete combustion of pentane produces

- a. carbon monoxide and water
- b. carbon dioxide and water



___ 65. The IUPAC name of is

- a. 2-methyl-1,4-pentadiene
- b. 2-methyl-2,4-dipentene
- c. 4-methyl-1,3-pentadiene
- b. $\text{CH}_3\text{CH}_2\text{CH}_2\text{—OH}$
- d. 4-methyl-2,4-pentadiene
- c. $\text{CH}_3\text{CH}_2\text{CH}_3$
- d. $\text{CH}_3\text{CH}_2\text{CH}_2\text{—SO}_4$

___ 71. Select the major product that would result

___ 66. A correct IUPAC name for is from the reaction:

- a. 5-bromo-1,3-cyclohexadiene
- b. 2-bromo-1,3-cyclohexadiene
- c. 2-bromo-1,4-cyclohexadiene
- $\text{CH}_3\text{CH}_2\text{CH=CH}_2 + \text{HBr} \rightarrow$
- d. 3-bromo-1,5-cyclohexadiene
- a.

- b. $\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

___ 67. Which of the following compounds could exist c.
in the form of two geometric isomers?

- a. d. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$

b.

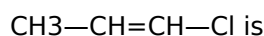
___ 72. The conversion of cyclohexene to cyclohexane

c. requires the reagent(s)

- a. HCl
- b. H_2O and H_2SO_4
- d. c. H_2 and H_2SO_4
- d. H_2 and Pt

___ 68. Which of the following compounds is trans-3-

___ 73. The addition polymer produced from
hexene?



a.

a.

b.

b.

c.

c.

d.

d.

___ 69. Which of the following represents an addition

___ 74. The monomer of teflon (polytetrafluoroethene) reaction?

is

a. $\text{HX} + \text{C}_4\text{H}_8 \rightarrow \text{C}_4\text{H}_9\text{X}$

b. $\text{X}_2 + \text{C}_3\text{H}_6 \rightarrow \text{C}_3\text{H}_5\text{X} + \text{HX}$

a.

c. $\text{C}_4\text{H}_8 \rightarrow \text{C}_4\text{H}_6 + \text{H}_2$

d. more than one response is correct b.

c.

___ 70. Select the major product that would result d. $\text{CF}_2=\text{CF}_2$ from the reaction:

___ 75. Which is the formula for an alkene? b.

a. CH_3CHCH_2

b. $\text{CH}_3\text{CH}_2\text{CH}_2$

c. $\text{CH}_3\text{CH}_3\text{CH}_2$ c.

d. all of these.

d.

___ 76. Which is the formula for an alkyne?

a. $\text{CH}_3\text{CH}_2\text{CCH}_2$

b. $\text{CH}_3\text{CH}_2\text{C}_2\text{CH}_3$

c. $\text{CH}_3\text{CH}_2\text{CHCH}_2$ ___ 82. Acetylene (ethyne gas) is commercially useful

d. $\text{CH}_3\text{CH}_2\text{CCH}_2$ as:

a. a starting material for plastics

b. a fuel for torches

___ 77. Which is a difference between butane and c. an ingredient in pesticides

butene? d. a component in paint formulations

a. butane burns and butene does not

b. the presence of a double bond

c. they are isomers ___ 83. What is the characteristic of aromatic

d. the presence of a triple bond compounds that is responsible for them being named aromatic compounds?

a. The compounds have a pleasant smell.

___ 78. Which is a difference between butene and b. These compounds contain a benzene ring. cyclobutene? c. It must contain a hydrocarbon chain that is either

a. They are isomers. saturated or unsaturated and at least 3 carbons long.

b. Cyclobutene has 2 double bonds, butene does not. d. There is more than one correct response.

c. The phase variance of the infrared transconductance

is asymmetric.

d. Cyclobutene has fewer hydrogens than butene. ___ 84. Which of the following compounds is not considered to be aromatic?

a.

____ 79. Which is a difference between butyne and cyclobutyne?

b.

a. Cyclobutyne can't exist!

b. Butyne's multiple bond is not as strong as that of cyclobutyne. c.

c. Cyclobutyne burns much hotter than butyne because of the greater unsaturation.

d.

d. Both b and c are differences between the molecules.

____ 80. The correct IUPAC name for the compound

$\text{CH}_2=\text{C}-\text{C}-\text{CH}_2-\text{CH}_2-\text{Br}$ is ____ 85. Which of the following structures is wrong?

a. 4-bromopentyne a.

b. 1-bromo-2-pentyne

c. 1-bromo-3-pentyne

d. 5-bromo-2-pentyne b.

____ 81. Select the product of the following reaction: c.

d.

a.

Answer Section 21. T 42. C 64. D

TRUE/FALSE 22. T 43. B 65. C

1. T 23. T 44. C 66. A

2. T 24. F 45. A 67. D

3. F 25. T 46. D 68. C

4. T 26. T 47. D 69. A

5. F 27. T 48. B 70. A

6. F 28. T 49. A 71. C

7. T MULTIPLE CHOICE 50. C 72. D

8. F 29. C 51. A 73. B

9. F 30. A 52. A 74. D

10. T 31. D 53. B 75. A

11. F 32. C 54. C 76. C

12. T 33. D 55. B 77. B

13. T 34. B 56. A 78. D

14. F 35. C 57. B 79. A

15. T 36. D 58. A 80. D

16. F 37. D 59. B 81. B

17. T 38. A 60. D 82. B

18. T 39. C 61. B 83. B

19. F 40. D 62. B 84. C

20. T 41. A 63. C 85. B

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d)Nuceophi
li
candr
eac
tswi
thba
ses
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25.I her

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tonet
oas
econda
rya
lcoholbye
it
herNa
BH4orLAH,
I
whywilyoupr
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que c
ousaidi
nthef
ir
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tep?
)I
a twi
llr
educ
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tonet i
oaprma
rya
lcoholi
nst
ead
b)The
rewi
llbenor
eac
ti
on
)Ac
c idwi

Ilha
ver
eac
tedwi
tht
her
eage
ntbe
for
ethec
arbonylha
sany
c
hanc
etounde
rgor
educ
ti
on+++
tc
d)l a
use
sas
eri
ouse
xpl
osi
ont
hatc
ans
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yonf
ir
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a
26.Rect
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vingt
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niumhydr
ide(
LAH)mus
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nine
rts
olve
ntsbe
caus
e
a
)LAHi
sina
cti
vei
nanyot
hers
olve
nts
e
b)Theprse
nceofr
eac
ti
ves
olve
ntswi
llpr
oduc
ela
rgea
mountofs

ide

r

eac

ti

onpr

oduc

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)LAHwi

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27.Whywilyoupr

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arbonylgr
oup?
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)LAHi
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b)Na
BH4i
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lowt
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eca
rbonylgr
oup+++
c
)Na
BH4i
snota
sse
lec
ti
vea
sLAH
d)LAHi
smor
eenvi
ronme
nt-
fr
iendl
ytha
nLAH
ta
28.Wha r
ethepr

oduc
tsoft e
herac
ti
onofCH₃CH₂C(
O)OCH₃wi
thLAH
f
oll
owe
dbyt
rea
tme
ntwi
tha
cid?
a
)1-
propa
nola
ndme
tha
nol
b)Pr
opa
nala
ndme
tha
nol
c
)Et
hanola
nde
tha
noi
cac
id
d)2-
propa
nola
ndme

tha

nol

c

29.Whihoft

hes

ere

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ntswi

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veas

at a

urte

dal

coholwi

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nuns

atur

ate

d

a

lde

hyde

a

)Na H₃O⁺

BH₄/

b)LAH/H₃O⁺

c

)H₂/

Rane

yNi

cke

l+++

d)CH₃MgBr

30.Wha

tisXi

nthi

sre

act

ion?

a

)CH₃CHCHCH₂COH+++

b)CH₃CH₂CH₂CH₂COH

c

)CH₃CHCH₂CH₂COH

d)CH₃CH₂CH₂CHCOH

31.Wha

tist

hef

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tonofH₃O⁺i

i nt e

herac

ti

onofRMgXwi

thc

arbonyl

c

ompounds

a

)Topr

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b)Toc

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)Tone

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ti
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dium
d)Tor
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ethec
arbonylgr
oupt
oal
cohol
a
32.Rect
ionsi
nvol
vingRMgXmus
tbec
arr
iedouti
nine
rts
olve
ntsl
ikee
the
r
be
caus
e
)Sol
a vent
sha
vinga
cidi
cpr
otonswi
l e

lrac
twi
tha
ndde
str
oyt
her
eage
nt
f
beor
eitc
oul
dre
actwi
tht
hec
arbonylc
ompound+++
a
b)Rect
ives
olve
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twi
tht
her
eage
ntt
ofor
munde
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oduc
tswi
tht
hec

arbonylc
ompounds
)RMgX,be
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ryr
eac
ti
vet
owa
rds
olve
ntsha
ving
a
cidi
cpr
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ns
sav
d)RMgXi erys
ens
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eage
nta
ndbe
come
sde
act
iva
tedupon
ont
c actwi
thr
eac
ti
ves
olve

nts

c

33.Whihoft

hes

egr

oupsdoe

snotpos

ses

sana

cidi

cpr

oton?

a

)-C $\equiv$ CH

b)-OH

c

)-O-

CH₃+++d)-NH₂

ti

34.Wha s

(ar

e)t

hepr

oduc

t(s

)oft

histr

eac

ti

on?

a

)3-

met

hyl

-3-

pent

anone

b)3-

met

hyl

-3-

pent

anonea

ndme

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c

)3-

met

hyl

pent

an-

3-ola

ndme

tha

nol

d)3-

met

hyl

pent

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3-ol

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35.Wha st

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ordeorc
arboxyl
at
esa
lt
a
)Al
dehyde
b)Al
kanone
c
)Al
kanoi
cac
id
d)Al
cohol
c
36.Whihoft
hes
ere
age
ntswi
lloxi
diz
epr
i r
maya
lcohol
sdi
ret
cl
yto
lde
hyde
s?
a
)Cr

2072-
b)MnO4-
c
)PCC+++
d)HI
O4
da
37.Oxi t
ionofs
econda
rya
lcohol
sine
it
hera
cidi
corba
sicc
ondi
ti
onswi
llgi
ve
a
)Al
dehyde
s
b)Ke
tone
s+++
c
)Oz
oni
des
d)Ca
rboxyl
ica
cids
ti
38.Wha st
hepr

oduc
tyouwoul
dobt
ai
nifyouoxi
diz
ete
rti
arya
lcoholi
nei
the
r
di
chr
oma
teorpe
rma
nga
nat
e?
a
)Al
kanoneonl
y
b)Nopr
oduc
t+++
c
)Al
kanonea
ndAl
kana
l
d)Al
kana
lonl
y
de
39.I i
ntf

yXa
ndYa
sdi
ff
ere
ntc
las
sesofa
lcohol
spr
ese
nti
nthes
truc
tur
edr
awn
a s2oa
)Xi lcohol s3oa
,Yi lcohol
s2oa
b)Xi lcohol s1oa
,Yi lcohol
c s3oa
)Xi lcohol s2oa
,Yi lcohol
s3oa
d)Xi lcohol s1oa
,Yi lcohol
40.At
ermi
nalke
tonei
sot
her
wis
eknowna
s
a
)Al
dehyde

b)Al
kanone
c
)Al
kanoi
cac
id
d)Fr
eon
a
41.Thecrbonylgr
oupi
sca
pabl
eofunde
rgoi hef
ngt oll
owi
ngr
eac
ti
onse
xce
pt
a
)Oxi
dat
ion
b)Re
duc
ti
on
c
)Nuc
leophi
li
caddi
ti
on
d)El
imi

nat
ion+++
e
42.Theprse
rva
ti
veus
edf
orbi
ologi
cals
pec
ime
nsc
ont
ai
ns
a
)Et
hanol
b)Me
tha
nal
c
)For
mica
cid
d)Tr
ioxa
ne
43.Topr
epa
ref
orma
li
nlwoul
ddi
ssol
ve
a
)73gf

orma
lde
hydei
n17gwa
ter
b)37gf
orma
lde
hydei
n63gwa
ter
c
)37gf
orma
lde
hydei
n100gwa
ter
d)63gf
orma
lde
hydei
n37gwa
ter
44.Apur
esour
ceofme
tha
nali
s
a
)Dr
yfor
mal
dehyde
b)Tr
ioxa
ne+++
c
)Di
oxa

ne
d)Ac
etone
e
45.Actonei
sus
edi
nor
ga cl
ni a
bor
ator
iesf l
orcea
ninggl
aswa
s r
ebe
caus
eiti
s
a
)Ve
ryc
hea
p
b)Hi
ghl
yvol
at
il
eandpol
ar+++
c
)Col
our
les
sandnon-
vis
cous
d)Col

our
les
sands
oapy
c
46.Whihoft
hes
ecoul r
dwoodvani
she
scont
ai
n?
a
)Pr
opa
nol
b)2-
but
anone
c
)Be
nze
ne
d)Et
hanol
e
47.CH₃CHOrac
tsf
ir
stwi
the
thyll
it
hium(
CH₃CH₂Li
)andt
henwi
thdi
lut
eac

id
t
ogi
ve
a
)et
hane
b)but
an-
2-ol
c
)but
ane
d)nor
eac
ti
on
48.Li
thi
umr
eac
tswi
thn-
but
ylc
hlor
idet
ogi
ve
) n-
a but
ane
b)n-
but
yll
it
hium+++
c
)chl
ori
ne

d)hydr
oge
n
ma
49.For I
dehyde(
HCHO)r
eac
tsf
ir
stwi
thCH₃MgCla
ndt
henwi
thdi
lut
eac
idt
o
gi
vea
a
)pr
ima
rya
lcohol
b)s
econda
rya
lcohol
c
)te
rti
arya
lcohol
d)qua
ter
nar
yal
cohol
50.Theoxi

dat
ionofCH₃CH₂OHwi
tha
cidi
cdi
chr
oma
te(
CrO₇2-)gi
2 ves
a
)ake
tone
b)e
thyla
lcohol
c
)ana
lde
hyde
d)a
nac
id+++
e
51.CH₃CH₂OHrac
tswi
thpy
ri ni
di umc
hlor
ochr
oma
te(
PCC)t
ogi
ve
a
)ake
tone
b)a
nal

cohol
c
)ana
lde
hyde
d)a
nac
id
52.PBr
3re
act
swi
the
tha
nolt
ogi
ve
a
)tr
ibr
omoe
tha
ne
b)br
omoe
tha
nol
c
)br
omoe
tha
ne+++
d)nor
eac
ti
on
onyle
53.Thi hlor
ide
,Cl
S(O)

Cl,r
eat
cswi
thi
sopr
opyla
lcohol
,(CH3)
2CHOH,t
o
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ve
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)2-
chl
oropr
opa
ne+++
b)1-
chl
oropr
opa
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c
)2-
propa
nol
d)a
cet
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54.lopr
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2 e
CHOH,rat
cswi
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edH₂SO₄t
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a
)Pr
opa
ne
b)2-
propa
nol
c
)Pr
ope
ne+++
d)Pr
opyne
316
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55.Al h
yde
sha
ve—— b
oil
ingpoi
nt ha
st n/
asa
lcohol
sofc
ompa
rabl
e
mol
ecul
arwe
ight
a
)hi

ghe
r
b)l
owe
r+++
c
)aboutt
hes
ame
d)e
xac
tl
ythes
ame
56.Ac
etonei
sthec
ommonna
mef
or
a
)1-
but
anone
b)1-
propa
none
c
)2-
propa
none
d)2-
but
anone
c
57.Whihma
ter
iali
smos
tre
aci

tvet
owa
rdnuc
leo
phi
li
caddi
ti
on?
a
)Ac
eta
lde
hyde
b)di
met
hylke
tone
c
)di
is
opr
opylke
tone
d)di
-t
ert
-but
ylke
tone
da
58.Theoxi t
ionof1-
propa
nolwi
thc
hromi
cac
idyi
el
ds

a
)ana
lcohol
b)a
nal
dehyde
c
)ake
tone
d)a
nac
id+++
da
59.Theoxi t
ionof2-
propa
nolwi
thc
hromi
cac
idy
el
ds
a
)ana
lcohol
b)a
nal
dehyde
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)ake
tone
d)a
nac
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60.Theoxi
dat
ionof1-
propa
nolwi
thPCCgi

ves
a
)ana
lcohol
b)a
nal
dehyde
c
)ake
tone
d)a
nac
id
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61.Theoxi t
ionof—— wi
thoz
onef
oll dbyr
owe educ
ti
onwi
th(
CH3)
2S
yi
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dsonepr
oduc
t,1,
5-pe
nta
nedi
al
a
)1,
5-pe
nta
nedi
ol
b)2,

4-pe
nta
nedi
one
c
)Cyc
lope
nte
ne+++
d)c
ycl
ope
nta
ne
c
62.Theaid-
cat
al
yze a
dhydrti
onofc
ycl
ode
cyney
iel
ds
a
)Cyc
lode
canone
b)c
ycl
ode
cadi
ene
c
)cyc
lode
cane
diol
d)c

ycl
ode
cane
dia
l
63.Re
duc
ti
onofa
cet
i c
caidwi
thLi
(t
ri
-e
trt
-but
oxy)
AlHyi
el
ds
a
)ana
cid
b)a
nal
dehyde
c
)ana
lcohol
d)ake
tone
e
64.Therac
ti
onofa
cet
ylc
hlor
idewi

the
xce
ssGr
igna
rdr
eage
ntf
oll
owe
dby
a
cidi
fi
cat
ionyi
el
ds
a
)ake
tone
b)a
nes
ter
c
)as
econda
rya
lcoho
l
d)at
ert
ia
rya
lcohol
t
65.Li umdi
hi met
hyl
cupr
ate(
Gil

manr
ea nt
ge)r
eac
tswi
t c
haet
ylc
hlor
idet
ogi
ve
a
)ake
tone
b)a
nes
ter
c
)as
econda
rya
lcoho
l
d)at
ert
ia
rya
lcohol
66.Thec
yani
dei
on-C≡Nr
eac
tswi
thake
tonet
ogi
ve
a
)ana

cid
b)ac
yanohydr
in+++
c
)ana
lde
hyde
d)a
nami
ne
c
67.Theaid-
cat
al
yze
daddi
ti
onofwa
tert
ofor
mal
dehydegi
ves
a
)ake
tal
b)ahe
mia
cet
al
c
)ana
cet
al
d)ahydr
ate
c
68.Theaid-
cat
al

zye
daddi
ti
onof2molofa
lcoho
lto1molofa
lde
hydeyi
el
ds
a
)ake
tal
b)ahe
mia
cet
al
c
)ana
cet
al
d)ahydr
ate
69.Ammoni
are
act
swi
tha
nal
dehydet
ogi
vea
n
a
)al
cohol
b)a
mide
c
)imi
ne+++

d)a
mine
e
70.Therac
ti
onofaWi
tt
igr
eage
ntwi
thke
tone
sgi
ve
a
)Al
kene
s+++
b)ke
tone
s
c
)al
dehyde
s
d)a
lka
nes
71.Si
lve
rion,Ag+,r
eac
tswi
tha
lde
hyde
stof
orm
a
)ar
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eci
pit
at
e
b)as
il
ver“
mir
ror
”+++
c
)abl
ues
olut
ion
d)t
her
eisor
eac
ti
on
me
72.Di t
hyl
ami
ner
eac
tswi
thake
tonet
ogi
vea
n
a
)ami
de
b)i
mide
c
)ami
ne

d)e
nami
ne+++
73.Ca
rbo
cat
ionsa
re
a
)ca
rbona
tomswi
thane
gat
ivec
har
ge
b)c
arbona
tomswi
thapos
it
ivec
har
ge+++
c
)ca
rbona
tomswi
thnoc
har
ge
d)a
nya
tomwi
thapos
it
ivec
har
ge
a

74.A—— crboc

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a

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b)pr

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conda

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75.Cabo

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o+++
b)wi
thdr
awe
lec
transf
rom
c
)dona
tepr
oto
nst
o
d)wi
thdr
awpr
otonsf
rom
76.Wha
tist
hena
meoft
hisc
ompound?
a
)2,
5-he
pta
nedi
one
b)2,
5-he
pta
dio
ne
c
)3,
6-he

pta
dio
ne
d)3,
6-he
pta
nedi
one
77.Thec
orr
ectna
mef
ort
hisc
ompoundi
s
a
)3-
met
hyl
-6-
oxohe
pta
nal
b)3-
met
hyl
-1,
6-he
pta
nedi
one
c
)2-
ket
o-5-
met
hyl
-he
pta
nal

d)5-
met
hyl
-2,
7-he
pta
nedi
one
ti
78.Wha sr
espons
ibl
efort
hehi
ghboi
li
ngpoi
ntsofa
lde
hyde
sandke
tone
s?
)Hydr
a oge
nbondi
nge
xis
ti t
ngbewe
ent
hemol
ecul
es
nt
b)l e
rmol
ecul
arf
orc
esofa

tt
rac
ti
onbe
twe
enmol
ecul
es
)Pol
c ari
nte
rac
tonb
i etwe
enmol
ecul
es+++
s
d)Dipe
rsi
onf
orc
esbe
twe
enmol
ecul
es
r
79.TheMakovni
kovr
ules
tat
est
hatahydr
oge
nat
ombondst
othec
arbona
tom
t

hata
lre
adyi
sbonde
dto—— hydr
oge
nat
oms
a
)ane
qua
lnumbe
rof
b) t
hegr
eat
estnumb
erof
c
)thel
eas
tnumbe
rof
d) no
r
80.Cabo
cat
ionsunde
rgor
ear
range
mentr
eac
ti
ons
a
)togi
vemor
est
abl
eca

rboc
ati
ons
b)byhydr
ides
hif
ts
c
)byme
thyls
hif
ts
d)f
ora
lloft
hea
bov
ere
asons
ke
81.Al n
esunde
rgoa
nac
id-
caa
tl
yze
daddi
ti
onofwa
tert
ogi
ve
a
)al
kyne
s
b)a
lka
nes

c
)et
her
s
d)a
lcohol
s+++
82.Bor
aner
eac
tswi
tha
lke
nes
)t
a ogi
vet
hea
nti
- r
Makovni
kova
ddi
ti
onpr
oduc
t
b)byac
onc
ert
edme
cha
nis
m
c
)togi
ves
yna
ddi
ti
on

d)a
lloft
hea
bove
ol
83.Di sa
res
ynt
hes
ize e
dbyrac
ti
nga
nal
kenewi
th
a
)wa
ter
b)bor
ane
c
)oz
one
d)os
miumt
et
roxi
de+++
oner
84.Oz eac
tswi
th2
, di
3- me
thyl
-2-
but
enet
ogi
ve

a
)oneke
tone
b)t
wodi
ff
ere
ntke
tone
s
c
)onea
lde
hyde
d)oneke
tona
ndonea
lde
hyde
a
85.Therpi
dconve
rsi
onbe
twe
ena
nenola
ndi
tske
tof
ormi
sca
ll
ed
a
)re
sona
nce
b)c
ati
oni

cre
arr
ange
ment
c
)ta
utome
riz
ati
on+++
d)hydr
oge
nat
ion
o
86.Theslubi
li
tyofs
omea
ldhyde
e sandke
tone
sinwa
teri
sduet
o
)Hydr
a oge
nbondi
ngbe
twe
ent
hemol
ecu
lesofwa
tera
ndt
hos
eof
a
lde

hyde
sorke
tone
s
a
b)Polri
nte
rac
ti
ona oge
ndhydr nbondi
ngbe
twe
ent
hemol
ecul
esof
wa
tera
ndt
hos
eofa
lde
hyde
sorke
tone
s+++
)Thei
c nte
rmol
ecul
arf
orc
e t
sbewe
ent
hemol
ecul
esofwa
tera
ndt

hos
eof
a
lde
hyde
sorke
tone
s
s
d)Dipe
rsi
ona
ndVa
nde
rWa
alf
ore
csbe
twe
ent
hemol
ecul
esofwa
ter
a
ndt
hos
eofa
lde
hyde
sorke
tone
s
de
87.Al h
yde
sar
esa
idt
obemor
era

ect
ivet
hanke
tone
sbe
caus
e
)Thec
a arbony
lca
rboni
nthef
orme
rismor
epol
art
hant
hatoft
hel
at
te
r
a
b)Thecrbony
lca
rboni
nthel
at
te
ri es
smor t
eri
cal
lyhi
nde
redf
rom
l
nuceophi
li
cat

tc
akt
hani
nthef
orme
r+++
)Thec
c arbony
lca
rboni
nthef
or ri
me sat
ermi
nalonehe
ncebe
tt
er
ont
c actwi
thnuc
leophi
le
sunl
ikei
nthel
at
te
rwhe
ret
hec
arbonyl
a
crboni
snote
asi
lya
vai
la
blef
ora

tt
ack
e
d)Thett
rahe
dra
lst
ruc
tur
eoft or
hef me
rpos
ses
seshi
ghe
rpot
ent
ia
l
e
ner
gyt
hant
hel
at
te
r
c
88.Whihoft
hes
eha
sthemos
tsa
tbl
ece
ntr
alc
arbon?
a
)RCOR'
b)RCOH

c
)RCH3
d)CH4
nt
89.Thei e
rme
dia
tef
orme
ddur
ingt e
herac
ti
ono
foz t
onewihuns
atur
ate
d
c
ompoundsi
sca
ll
ed
a
)Oxi
diz
inga
gent
b)Oz
oni
de+++
c
)Tr
ioxa
ne
d)Pe
roxi
de
a
90.Rect

ionofa
lke
newi
thoz
onei
nthep
ese
nceofhydr
oge
npe
roxi
dewi
ll
pr
oduc
e
a
)Twoa
cids
b)Ke
tonea
nda
cid
c
)Twoke
tone
s
d)Al
loft
hea
bove
onol
91.Oz ysi
sisave
ryi
mpor
tantr
eac
ti
oni
nche

mis
trybe
caus
e
a
)Iti
snott
oxi
c
tc
b)l a
nbeus
edt
ocha
rac
ter
izec
ompl
exmol
ecul
es+++
)l
c tt
ake
spl
ac tr
ea oomt
empe
rat
ure
ti
d)l same
ansofr
educ
ingc
ompl
exmol
ecul
es
c
92.Whihoft

hes
eisnotawa
yofpr
epa
ringa
Ide
hyde
sandke
tone
s?
a
)Hydr
ati
onofa
lkyn
es
b)Re
duc
ti
onofa
cidc
hlor
ide
s
c
)Ac
yla
ti
onr
eac
ti
on
d)Ami
nat
ionr
eac
ti
on+++
93.Wha
tist
hepr

oduc
toft
histr
eac
ti
on?
a
)He
pta
n-1,
6-di
oica
cid+++
b)Cyc
lohe
pta
n-1,
6-di
oica
cid
c
)Et
hanoi
cac
ida
ndpe
nta
ne
d)Cyc
lohe
pta
noi
cac
id
94.Howma
nyi
some
rswi
llt
hisc
ompoundha

ve?

a

)1

b)2

c

)3+++

d)4

95.Whi

choft

hes

eha

sthehi

ghe

stboi

li

ngpoi

nt?

a

)But

ane

b)But

ana

l

c

)But

anol

d)But

anone

96.Wha

tist

hepr

oduc

tyouwoul

dge

tfr

omt

hehy

dra

ti

onof $\text{CH}_3\text{C}\equiv\text{CCH}_2\text{CH}_3$

us

ingme
rcur
ysa
lt
a
)CH₃COCH₂CH₂CH₃
b)CH₃CH₂COCH₂CH₃
c
)CH₃COCH₂CH₂CH₃a
ndCH₃CH₂COCH₂CH₃+++
d)CH₃CH₂C(
OH)
CH₂CH₃
97.Ar
oma
ti
cke
tone
sar
epr
epa
redby
a
)Wi
tt
igr
eac
ti
ons
b)Fr
iede
l-Cr
afta
cyl
at
ion+++
c
)Cl
emme
nse
nre

act
ion
d)Me
rcur
ati
onr
eac
ti
on
nohydr
98.Cya insc
anbehydr
olys
edi
nthep
ese
nceofa
cidt
opr
oduc
e
a
)Compoundswi
thbot
hhydr
oxyla
ndc
arboxyl
icgr
oups
b)Be
ta-
Ami
noa
lcoh
ols
c
)Al
dehyde
s
d)Ke

tone

s

ti

99.Wha st

hepr

oduc

toft

her

eac

ti

onofwa

terwi

thke

tone

s?

a

)Ac

eta

l

b)Hydr

ate

c

)He

mia

cet

al

d)Ge

rani

ol

100Wha

tist

hef

ina

lpr

oduc

toft

her

eat

ci

onofa

lcoholwi

tha
lde
hydec
all
ed?
a
)He
mia
cet
al
b)Ac
eta
l+++
c
)Hydr
ate
d)Ke
tal
SECTI
ON2
1.
A.He
x-1-
en-
6-yne
B.He
x-1-
en-
1-yne
C.He
pt-
1-e
n-6-
yne
D.He
pt-
6-e
n-1-
yne
2.
A.4-

Ethyl
-3,7-di
met
hyloc
tane
B.5-
Ethyl
-2,6-
dime
thyloc
tane
C.4-
Ethyl
-3,6-
dime
thylhe
pta
ne
D.5-
Ethyl
-3,7-
dime
thylhe
pta
ne
3.
A.6-
chl
orohe
pt-
2-e
n-4-
ol
B.2-
chl
orohe
pt-
5-e
n-4-
ol+++
C.2-

chl
orohe
pt-
6-e
n-4-
ol
D.2-
chl
oro-
4-hydr
oxyhe
pt-
5-e
ne
4.
A.4-
ter
tBut
ylhe
xana
l+++
B.3-
ter
tbut
ylhe
xana
l
C.4-
Ethyl
-5,5-
dime
thylhe
xana
l
D.3-
ter
tBut
ylhe
xana
l
5.

A.5-

cyc

lot

exylc

arboxyl

ica

cid

B.3-

cyc

lohe

xylc

arboxyl

ica

cid

C.2-

Ethyl

-3-

cyc

lohe

xylbut

anoi

cac

id

D.2-

Met

hyl

-4-

cyc

lohe

xylbut

anoi

c

a

cid+++

6.

Et

A.1- hyl

-5-me

thyl

- i

2-sopr

opylc

ycl

ohe

xane

hyl

B.4-Et -3-

isopr

opyl

-1-

met

hylc

ycl

ohe

xane

hyl

C.2-Et -1-

isopr

opyl

-5-

met

hylc

ycl

ohe

xane

Et

D.1- hyl

-2-

isopr

o -

pyl4-

met

hylc

ycl

ohe

xane

7.

c

A.3-hl

oro-

2-hydr

oxybi

c l
yco[
2,2,
2]-
oct
ane
c
B.6-hl
orobi
cyc
l 2,
o[2,
2]oc
tan-
2-ol
c
C.2-hl
orobi
cyc
lo[
2,2,
2]oc
tan-
2-ol
c
D.5-hl
orobi
cyc
lo[
2,2,
2]oc
tan-
3-ol
8.
c
A.5-hl
orobi
cyc
lo[
2,2,
0]he

x-2-

ene

B.2-

chl

orobi

cyc

lo[

2,2,

0]he

x-5-

ene

C.2-

chl

orobi

cyc

lo[

2,2,

0]he

x-2-

ene

D.5-

chl

orobi

cyc

lo[

2,2,

0]he

x-1-

ene

9.

A.4-

Ethyls

pir

o[4,

5]nona

ne

B.1-

Ethyls

pir

o[4,

4]nona

ne+++

C.4-

Ethyls

pri

o[4,

4]oc

tane

D.1-

Ethyls

pir

o[5,

5]nona

ne

10.

A.2-

isopr

opyl

-8-Et

hyls

pir

o[5,

4]de

cane

Et

B.2- hyl

-6-

isopr

opyls

pir

o[4,

5]de

cane

Et

C.3- hyl

-7-

isopr

opyls

pir

o[4,

5]de

cane

D.3-
 Ethyl
 -10-
 isopr
 opyls
 pir
 o[4,
 5]de
 cane
 11.hexe
 neand2-hexene re:
 A.Functionalgroupis
 ome
 rs
 B.Cha i
 ni somers
 C.Alkenei somers
 D.Posit
 ioni somers
 12.Theset
 wocompoundsa re:
 A.Posi
 tioni somers
 B.Functiona lg
 roupis
 ome r
 s+++
 C.Ethe
 r-a l
 c oholi
 somers
 D.Chaini somers
 13.The
 sea
 re:
 A.Cycl
 e-alkeneisomer
 s
 B.Funct
 ionalgroupis
 omers

C.Chai

n-cycleisomer

s+++

D.Posi

ti

oni s

ome rs

14.Thena

meoft

hisc

ompoundi s

A.tr

ans -

2-me t

hylhe x- 3- e

ne

B.Trans-3-methylhe x- 3-ene

C.Cis-3-me t

hylhe x-

3- e ne+++

D.Trans- 2-methylhe x- 2-ene

nyc

15.Howma hir

alcentr

e saet

r he r

eint hes tr

uct

ure

A.1

B.2

C.3

D.4+++

16.Anopt

ic

ali

some

rha

salleexcept

A.hasac hi

ralcent

re

B.rot

ate

sp l

a nepol

ari

sedl

ight

C.hasami rr

orima ge

D.issuperimposableonitsmi r

ror

ima ge

17.Acompoundhas8carbonatomsonl ysixoftheca r

bonsare

me

thylr

adic

als

.Thena meofthec ompoundi s:

A.1,2-hexamethylethane

B.2,4-tet

ramethylbutane

C.2,2,3,

3-tet

ramethyl

butane+++

D.1,2,3,

4-tet

ramethylbutane

18.Namethecompound

A.2-Ethyl -

4-c yc

lope nt

ylhexane

B.3-Me thyl-4-cyclopentylhept

ane

C.3-Cyc lopentyl

-5-

me t

hyl heptane+++

D.3-Me thyl-6-cyclopentyl

heptane s

19.Na

met

hec

ompound

A.4-Methylhex-

2-en-

5-

yne

B.4-Methylhe

x-1-yne-

4

-ene

C.4-methylhex-

4-en-

1-

yne+++

D.4-methylhex-

2-en-

1-

yne

20.Thi

scompoundi

s

A.2-Et

hyl-

4-me t

hylcyc

lohexene

B.3-Et

hyl-

5-me t

hyl

c

ycl

ohexene+++

C.5-Et

hyl-

3-me t

hylcyl

ohexene
 D.1-Et
 hyl-
 5-me t
 hylcyc
 lohex-2-e
 ne
 i
 21.Chani
 some
 rsa
 re:c ha
 ompoundst thavethesa
 memol e
 c ul
 ar
 f
 ormulabutdif
 feri
 n---
 A.Optic
 alar
 r a
 nge ment
 B.Functi
 onalgroups
 C.Spati
 alar
 rangeme ntofat
 oms
 D.Bondingarra
 nge mentofradi
 cal
 s+++
 22.Givet
 henameof :
 4-
 vinylheptene
 4-
 all
 ylhexene

3-
propylhexa-1,5-die
ne+++

4-
propylhe xa-1,5-di
ene

nys
23.Howma truc

tur
a s
liome

rsc
anyouwr

it
eforC5H12

A.4
B3+++

A.2
B.5

nys
24.Howma atur

ate
dcc
yle
sca
nyouwr

it
efr
omC5H10

A.5
B.4
C.2
D.3+++

c
25.Whihe

xpe
rime
ntwi

l s
ldit
ingui

shbe

twe

en

A. Ignit

ion

B. Sodium

C. Bromi nat

ion+++

D. Solubili

tytes

t

26.The suff

ix of the compound:

$\text{CH}_3\text{-CH}_2\text{-CH}_2\text{CH=C-COOH}$

$\text{CH}_2\text{-CH}_3$

A. He xeno i

ca ca

id

B. He x-2-

enoicaci

d+++

C. He x-4-

enoicaci

d

D. He

x-3-

enoi

cac

id

27.The pref

ix of this compound:

$\text{CH}_3\text{ Br}$

$\text{CH}_3\text{-CH-C-CH}_3$

CH_3

A. 3-Bromo-1,3-dimethyl

B. 2,3-

dimethyl-

3-bromo

C. 2-Bromo-2,3-di

me thyl+++

D. 2-Br

omo- 2-

Isoprophyl

28.Na

met

hec

ompound

A. Spi

ro{4,

5}dec-

2-ene+++

B. Sp i

ro{4,

6}dec-2-ene

C. Sp i

ro{5,

6}dec-

2-ene

D. Sp i

ro{4,

5}dec-1-ene

29.Pi

ckt

hec

orr

ectna

mef

or:

A.Spir

o{4,5}nor

iane

B.2-i

sopr

opylspi

ro{4,

5}no

nane

C.3- i

sopr

opylspi

ro{4,

5}no

nane

D. 7-i

sopr

opylspi

ro{4,

5}no

nane

30.Thi

scompoundi

s:

A. 1-ethylbi

c yc

lo[2,1,1]

-hexane

B. 2-Ethylbicycl

o[211] -hexa

ne+++

C. 5- Ethylbicycl

o[2 ,

1,1]-he

xane

D. 3-Ethylbic

yc l

e[2,1,1]-hexane

31.Namet

hecompound

CH₃

CH₃-C -CH-CH=CH₂

||

CH₃ CH=CH₂

A. 2,

2-Di me t

hyl-

3-Vinylpente

ne

B. 2,

2-Dimet

hyl-3-

Vi nylpent

-5-

ene

C. 3-ter

tbut

ylpent

a-1,4-die

ne+++

D. 4,

4-Dimet

hyl-3-

Vi nylpe

ntene

32.Thena

meoft

hisc

ompoundi

s:

Cl

A.5-c

hloro-3-met

hylpente

ne

B.5-c

hloro-3-met

hylpent-

3-ene

C.1-c

hloro-2-met

hylpent-

2-ene

D.1-c

hloro-3-met

hylpent-

2-ene+++

33.Thi

scompoundi

s

Cl

A.3-

chloro-4-e

thylhept-

3-ene+++

B.3-chloro-3-ethylhept-3-ene

C.4-chloro-3-propylhex-3-ene

D.5-chloro-4-ethylhept-4-ene

34. What

t is

the

IUPAC name

of

the

compound?

A. Cyclohexane

1,2-dimethylcyclohexane

1,3-dimethylcyclohexane

B. Cyclohexane

1,2-dimethylcyclopentane

C. Cyclohexane

1,2-dimethylcyclohexane

1,3-dimethylcyclohexane

1,4-dimethylcyclohexane

D. Cyclohexane

1,2-dimethylcyclohexane

35. The IUPAC name of

the

compound is

CH₃-CH₂-CH₂-CH₂-O-CH₂-CH₃

A. Ethylbutyl ether

her

B. Butylethyl ether

her

C. Butoxyethane

her

D. Ethoxybutane

36. What

choft

hes

ecompoundsha

sac

hir

alc

arbon?

A.Chleropenta

ne

B.3-

chloropentane

C.2-chloropent

a ne+++

D.Chloro-2,2-dimethyl

propa

ne

he

37.Oneoft s

ecompound

swi

llnotbeopt

ic

all

yac

ti

ve

A. Chl

orobromolodome ta

ne

B. Chl

oro-2-met

hylpropane

C. 2-ch

loro-2-

me t

hylpropane

D. 2-

chl

oropropane+++

38.Wha

tist

hel

UPACna

meoft

hisc

ompound?

A. 4-Ethyloct-

6-en-3-yne

B.4-Ethyloct

-2-en-5-yne+++

C. 5-Ethyloct-

2-en-5-yne

D. 4-Ethyloct-

2-en-3-yne

39.

a

. b.

aandba r

e :

A.Functionalgroupis

ome

rs

B.Tautome r

s

C.Alken eis

ome rs

D.Pos i

tionisome r

s+++

40.Thi

scompoundis:

CH₃

CH₃-C-CH₂-CH=CH₂

CH₃

A.2,

2-di

met

hylpent

ene

B. 4,

4-di

met
hylpent
ene+++
C.2,2-
dimet
hylpe
nt-
4-ene
D.4,
4-di
met
hylpent
-4e
ne
41.Thena
meoft
hisc
ompoundi
s:
CH
3
CH
3
A.1,
3-di
met
hylhexan
e
B.1,
3-di
met
hylcycl
ohex-
2-e
ne
C. 1,
3-di
met
hylcycl
ohexe
ne+++

D.1,
 3-di
 met
 hylhex-
 1-ene
 42.Na
 met
 hec
 ompoundbe
 low:
 O
 A.2-Me thyl-
 2-isopr
 opylbut anone
 B.2-Me thyl-
 2-isopr
 opylbutan-2-
 one
 C.3,3,4-
 trimethylpent
 an-3-one
 D.3,3,4-t
 rimethylpenta
 n-2-one+++
 43.Na
 met
 heCompound:
 O
 2,
 A. 1, 7-
 tri
 me t
 hyl-
 3-met
 hoxybicycl
 e[2,2,
 10]hepta
 ne
 2,
 B. 1, 7-

tri
 me t
 hylbiyc
 c le[
 2,2,
 1]hepta
 ne s
 C.1,2,
 7-tr
 imethyl
 -5-
 me hoxybi
 t c
 yclo[2,
 2,1]hept
 anes+++
 D. 1,2,7-
 tri
 me t
 hyl-
 4-mehox
 t ybicycl
 e[2,2,
 1]hepta
 nes
 44.Wha
 tist
 hena
 meoft
 hec
 ompound?
 OH
 A.1,
 3-di
 met
 hylpent
 -3-
 e nol
 B.1,
 3-di
 met

hylpe

nt-

2-enol

C. 2,

3-di

met

hylpent

-3-

e nol

D.1,

3-di

met

hylpent

en-

3- ol

45.Gi

vet

hena

meoft

hec

ompound.

A. 3-s e

cbutylhept- 1-e n-4-yne

B. 5-is

opropylhe pt-1-en-3-yne

C. 3-isopropylhe pt-6-en-3-yne

D. 4-vingl

-3-methyl oet-3-yne

46. $\text{CH}_2=\text{CHCl}$ i s

A.Allychloride

B.Ethylc hloride

C.Vinylc hloride+++

D.Chloroe thyl i

dene

47. Na

met

hec

ompound

A. 2,

2-di

met

hylpe

nte

ne

B. 4,

4-di

met

hylpe

nte

ne+++

C. 2,

2-di

met

hylpe

nt-

4-e

ne

D. 4,

4-di

met

hylpe

nt-

2-e

ne

48. Thediffer

entpar

tsofas

yst

e t

mai

cna

mei

ncl

udea

lloft

hes

eexc

ept

A. suf

fix

B. Pri

ncipalchai

n

C. Str

aightchai

n+++

D. pre

fix

49. Thepr

efi

xfort

hisc

ompound

A.3-

But

yl-5-methyl

B.5-

but

yl-

7-me thyl

C.3-

met

hyl-4-butyl

D. 5-

but

yl-

3-me thyl

50. Thena

meoft

hisc

ompoundi

s:

A. 2-

met

hyl

hept

yne

B. 2-

met

hyl

hept

-3-

yne

C. 2-

met

hyl

hept

-4-

yne

D. 6-

met

hyl

hept

-3-

yne

I

51.Themet

ingpoi

ntofme

mbe

rsofc

arboxyl

ica

cidhomol

ogouss

eri

es

Ai

ncr

eas

esupt

hes

ervi

ces

Bde

cre

ase

supt

hes

eri

es

hee

Ct venc

arbona
cidsha
vehi
ghe
rme
lt
ingpoi
ntst
hant
hei
rimme
dia
te
hi
ghe
randl
owe
rme
mbe
rs+++
hownode
Ds fini
tet
rendi
nthes
eri
es
l
52.Themei
tngpoi
ntso
fst
ruc
tur
ali
some
r a
sofcrb
oxyl
ica
cids

eri
esa
re.....
the
ir
nor
malc
hai
ncount
erpa
rt
A)Lowe
rtha
n+++
B)hi
ghe
rtha
n
C)t
hes
amea
s
D)notde
fine
d
r
53. Cabo
xyl
ica
ci vehi
dsha ghe
rboi
li
ngpoi
ntsduet
o
A)t
hea
lkylc
hai
n

B)hydr

oge

nbondi

ng+++

C)c

arboxylgr

oup

D)c

arboxyl

ica

cidbondi

ng

l

54. HelVol

ardZe

li

nsk

yre

act

ioni

s

A)r

egi

ose

lec

ti

ve+++

B)r

egi

ospe

cif

ic

C)s

ter

eos

ele

cti

ve

D)s

ter

eos

pec
ifi
c
l
55. Hel
-Vol
ardZe
li
nskyr
eac
ti
ons
ubs
ti
tut
esonl
y
A) α -
car
bona
tom
B) β -
car
bona
tom
C) α -
hydr
oge
n+++
D) β -
hydr
oge
n
nHe
56. I II
-Vol
ard-
Zel
ir
skyac
ata

lyt
ica
mountof
.....i
sne
ede
d
A)s
odi
um
B)pot
ass
ium
C)r
edphos
phor
us+++
D)br
omoni
umi
on
c
57. Whihoft
hes
ewi
llr
educ
ecr
aboxyl
ica
cidt
oal
cohol
A)Li
AH4+++
B)NaBH₄
C)H₂/
Pd
D)H₂/
Ni
58. Cl

CH₂COOH
smor
eac
idi
ctha
nCH₃COOHbyr
eas
onof
A)r
esona
ncee
ffe
ct
B)c
onj
uga
ti
vee
ffe
ct
C)c
ummul
at
ivee
ffe
ct
D)i
nduc
ti
vee
ffe
ct+++
nt
59. Whe hehydr
oge
noft
he-COOH i
sre
pla
cedbya
nal

kylgr
oup(
R)i
t
be
come
s(A)a
nami
de
B)a
nes
ter
C)ak
etone
D)a
nal
kane
ndr
60. Thee esul
toft
her
educ
ti a
onofcrboxyl
ica
cidbybor
ane(
B2H4)i
n
A)a
nal
dehyde
B)ak
etone
C)a
nal
cohol
D)a
nal
kane
nz

61. Be e

nec

anbedi

ff

ere

nti

ae

tdf

roma

nal

kenebyt

her

eac

ti

on

A)i

tbur

nswi

ths

oot

ylummi

niousf

lame

B)i

tboi

I t78oC

sa

C)i

tdoe

snotde

col

our

isebr

omi

newa

ter

D)i

tme

It

sa 5oC

t5.

62. Whi

choft

hes

egr

oupswi

llbegoodme

tadi

rec

ti

nggr

oup

A)-NH₂

B)-

CH₃

C)-NO₂+++

D)-Cl

63. Be

nze

nei

sins

olubl

ein

A)c

hlor

ofor

m

B)e

tha

nol

C)wa

ter

D)a

cet

one

64. Be

nze

nec

anbepr

oduc

edf

rom

A)e
thyl
ene
B)a
cet
yle
ne+++
C)me
thyl
ene
D)pr
opyl
ene
65. Be
nze
nei
sre
sis
tantt
ooxi
dai
tona
ndr
educ
ti
onbe
caus
edoubl
ebondsa
re:
A)t
hre
e
B)c
ummul
at
ed
C)a
ddi
ti
ve

D)c

onj

uga

ted+++

66. Thena

meoft

hisc

ompound

CH₂COOH

A)a

cet

icbe

nze

nea

cid

B)be

nzyla

cid

C)be

nzoi

cac

id

D)2-

phe

nyla

cet

ica

cid+++

67. Be

nze

ner

ingunde

rgoe

s

A)s

ubs

ti

tut

ionr

eac

ti

ons+++

B)oxi

dat

ionr

eac

ti

on

C)a

ddi

ti

onr

eac

ti

on

D)r

educ

ti

onr

eac

ti

on

68. Be

nze

nei

smos

tav

ail

abl

efr

omt

his

A)pe

trol

eum+++

B)na

tur

alpr

oduc

ts

C)wood

D)s

awdus

t

69. Be

nze

nei

soxi

die

zdr

eadi

lywi

th

A)oz

one

B)oxyge

n

C)c

hlor

ine

D)pot

ass

iumpe

rma

nga

nat

e

70. Ac

etophe

nonei

ses

ail

ypr

epa

redby

A)Fr

(iede

I-Cr

aft

sbe

nzoyl

at

ion

B)Fr

(iede

I- yt

Cr sa

lkyl

at

ion

C)Fr

(iede

I-Cr

aft

sac

yla

ti

on+++

D)phe

nyl

at

ionofa

cet

one

SECTI

ON3

1. Thepe

riodi

cta

blei

sac

har

ttha

tar

range

sel

eme

nta

ccor

dingt

o

A.At

omi

cma

ssm

B.At

omi

cnumbe

rZ+++

C.Mol

ecul

arma

ss

D.Noneoft

hea

bove

2. Thepr

inc

ipa

lus

eofdi

amondi

sin

A.Li

ght

ing

B.J

ewe

lry+++

C.Le

adpe

nci

l

D.Sa

w

3. Thea

bil

it

yofme

tal

stoc te

onduc l

ect

ri

ci

tyi

sduet

o

A.Sur

fac

ete

nsi

on

B.Smoot

hsur

fac

e

e

C.Prse

ncoff

re l

emobiee

lec

trons

D.Hi

ght

ens

ion

4. Ther

eas

onwhygr

aphi

tei e

susda

sel

ect

rode

sisduet

o;

A.Abs

enc

eo

reemobi

lee

lec
trons

e
B.Prse
nceoff
re l
emobiete

lec
trons

C.Ri
gids
truc
tur
e

D.Fl
exi
bles
truc
tur
e

5. Whi
choft
hes
eel
eme
ntsi l
snotaka
li

ne-e
art
hme
tal

A.Ba

B.Sr

C.Mg

D.Pd+++

6. Whi
choft
hes
eel

eme

ntsi

snota

nal

kal

ime

tal

A.Li

B.Na

C.Ba

D.Cs

7. Wha

tisa

nal

loy?

A.A misci

blemi

xtur

eoft

woo

rt e

hreme

tal

satt

hei

rmol

te

n

s

tat

es+++

B.Twononme

tal

s

C.Nonme

tal

s

D.Ga

ses

8. TheMor

sel

ey'

sequa

ti

oni

s;

2

A.E=mc

B. v =a

(Z-b)+++

2

C.PV=mc

D.PV/

T =c

ons

tant

t

9. Meal

soni

oni

zat

ionbe

comec

har

gedwi

ths

mal

le

r

A.Si

zes+++

B.Si

te

s

C.Sour

ces

D.Mome

ntum

10.Di

amondha

swhi

chhybr

idi

zat

ion?

A.Sp

B.Sp2

C.Sp3 + + +

D.Sp_d

11.I

sgr

aphi

tec

onve

rti

blet

odi

amond?

A.Ye

S + + +

B.No

C.Unc

ert

ai

n

D.Noneoft

hea

bove

12.El

ect

rol

ysi

sofwa

terpr

oduc

es;

A.Hydr

oge

nandoxyge

n +++

B.Ni

tr

oge

nandc

oke

C.Col

anut

D.Bi

tt

erc

ola

13.Wha

tisac

rys

tal

ol

A.A s i

dwi

thl

at

ti

cepoi

ntsoc

cupi

edbyi

ons

,at

omsormol

ecul

e

B.Gl

ass

C.Col

droom

D.Li

qui

d

14.Exa

mpl
eofa
mor
phouss
oli
dis
A.Gol
d
B.Gl
ass +++
C.Si
lve
r
D.Mi
te
s
15.Wha
tisame
tal
A.A mal
lea
ble
,duc
tl
ie
,lus
trouss
oli
dtha
tconduc
tse
lec
tri
ci
tya
nd
he
at.+++
B.I
nsul

at

or

C.Pe

rfor

ator

D.PI

at

e

16.Thec

har

act

eri

st

ic l

sofaka

lime

talc

ompoundsi

stha

tthe

yar

e;

t

A.Whiea

ndf

ormc

olour

les

ssol

uti

oni

nwa

ter+++

B.l

nsol

ubl

einwa

ter

C.Bl

acki

nco
lour
D.Noneoft
hea
bove
17.Whi
choft
hes
ema
ter
ial
sisnotac
ova
lentic
rys
tal
A.Coppe
r+++
B.Gr
aphi
te
C.Di
amond
D.Qua
rtz
18.Pe
riod1c
ont
ai
ns;
A.Ha
ndHe
B.Ur
ani
uma
ndo
the
rac
ti
nide

s

C.La

ntha

nide

s

D.Ha

loge

ns

19.Gr

ouple

leme

ntsa

re;

A.Coppe

r,s

il

vera

ndgol

d

B.I

ron,ma

nga

nes

eandc

oba

It

C.Lit

hium,s

odi

um,pot

ass

ium,r di

ubi um,c

aes

iuma

ndf

ranc

ium

D.Hydr

oge

nandhe
li
um
20.Ani
sot
opeha
s;
e
A.Elme
ntwi
tht
woormor
edi
fe
fre
nta
tomi
cma
sse
s+++
e
B.Elme
ntwi
thdi
fe
fre
nta
tomi
cnumbe
r
C.El
eme
ntwi
thoutma
ss
e
D.Elme
ntwi
thhi
ghe

lec
trona
ffi
nit
y
21.Gr
oup18e
leme
ntsl
ikeAr
,He
,Nea
ndKra
reknowna
s;
A.Nobl
ega
ses
B.Nobl
eme
tal
s
C.Pr
eci
ouss
tone
s
D.Al
kal
i
22.Suc
ros
e,i
cec
ube
,sul
phur
,phos
phor
usa

ndi
odi
nea
reknowna
s
A.Me
tal
li
ccr
yst
al
s
B.Mol
ecu
larc
rys
tal
C.J
asmi
ne
D.I
oni
ccr
yst
al
s
23.Gr
oup17e
leme
ntsl
ikel
,Br
,F,Cla
reknowna
s
A.Hydr
oge
n
B.Ha
loge

ns+++

C.Pe

troge

ns

D.Andr

oge

ns

24.TheI

at

ti

cepoi

ntsofi

oni

ccr

yst

al

sar

eoc

cupi

edby;

A.Mol

ecu

les

B.Tube

rs

C.Cha

rge

dions

D.Chr

oni

cl

es

25.Whi

choft

hisi

sthepr

ope

rtyofdi

amond?

A.Ha

rdne

ss+++

B.Sof

tne

ss

C.Br

it

tl

ene

ss

D.Duc

ti

li

ty

t

26.Meal

li

ccys

r t

al

scont

ai

n;

A.Mol

ecu

les

r

B.Chage

dionsoft

heme

tali e

nasaorc

loudofe

lec

trons+++

C.But

te

r

D.Ya

m

e
27.Elct
rova
le ompoundsa
ntc rehe
ldt
oge
the
rby;
A.Mol
ecul
arf
orc
es
B.El
ect
rova
lentf
orc
es+++
C.Cova
lentf
orc
es
D.Mi
li
ta
ryf
orc
es
28.Thema
jorus
eofgr
aphi
tei
s;
A.El
ect
rode
s+++

B.Por

tal

C.Pr

ime

r

D.Eme

ral

d

29.Di

amondi

sus

edpr

ima

ril

yas

A.J

ewe

lry+++

B.Pa

veme

nt

C.Food

D.Fa

rm

30.Whi

choft

hes

eel

eme

ntswi

lll

ibe

rat

ehydr

oge

nfr

omc

oldwa

ter

A.Phos

phor

us

B.Sul

phur

C.Pot

ass

ium+++

D.Pur

ine

31.El

ect

rol

ysi

sis

A.Useofele

ctr

ica

lcur

rentt

obrea

kelect

rol

yti

ccompoundsi

nmol

te

n

oraque

oussta

test

othei

rcons

ti

tue

ntsele

ments

B.Us

eofha

mme

rtobr

eaks

ubs

tanc

es

C.Us

eofa

dhe

sivet

obi

nds

urf

ace

s

D.Us

eofwa

tert

ospr

ay.

32.Whi

choft

hes

eisnota

nal

kal

iel

eme

nt?

A.I

it

hium

B.Gol

d+++

C.Sodi

um

D.Pot

ass

ium

33.Whi

choft

hes

eel
eme
ntswi
l s
ldipl
acehydr
oge
nfr
omc
oldwa
ter
A.Rubi
dium+++
B.Coppe
r
C.Gol
d
D.Tungs
ten
34.Thegr
oupi
nape
riodi
cta
bles
har
es;
mi
A.Si l
arelec
tro
nicc
onf
igur
ati
onph
ysi
cala
ndc
hemi

cal

pr

ope

rti

es+++

B.Di

ff

ere

ntpr

ope

rti

es.

C.Di

ff

ere

ntmobi

li

ty

D.Di

ff

ere

ntt

ool

s.

35.Wha

tfor

cehol

dsc

ova

len

tcompoundst

oge

the

r?

A.I

oni

c

B.Va

nde

rWa

al+++

C.Cubi

c

D.Re

cta

ngul

ar

36.Whi

choft

hes

ear

eal

lot

rope

sofc

arbon?

A.Di

amonda

ndgr

aphi

te

B.Pr

oct

era

ndGa

mbI

e

C.Si

Ive

r

D.Sol

ar

37.Gol

disa

A.Bul

le

t

B.Ma

It

C.Me

tal

D.Gl

ass

38.Whi

choft

hes

eist

hec

har

act

eri

st

icofTr

ans

it

ione

leme

nts

r

A.Patl

yfi

le

ldd-

orbi

ta

ls+++

B.Ful

lyf

il

le

dli

fec

ycl

e

C.Nonme

tal

s

D.Pol

ari

zabl

eel

ect

rode

s

39.Whi

choft

hes

eisnott

hec

har

act

ers

it

icofl

ant

ha de

ni sa

nda

cti

nide

A.Nonme

tal

s+++

B.Ra

dioa

cti

ve

r

C.Pati

al

lyf

il

le

df-

orbi

ta

ls

D.Synt

het

ice
leme
nts
40.Whi
choft
hes
eel
eme
nt snotat
si rans
it
ionme
tal
A.I
ron
B.Ma
nga
nes
e
C.Zi
nc
D.Fl
uor
ine
41.Whi
choft
hes
eel
eme
ntsi
snotaha
loge
n?
A.Fl
uor
ine
B.Chl
ori
ne
C.I

odi
ne
D.Bor
on+++
42.Whi
choft
hes
esubs
tanc
esi ondu
snotagoodc ctorofe
lec
tri
ci
ty?
A.Coppe
r
B.Si
lve
r
C.I
ron
D.Di
amond+++
43.Whyi
sgr
aphi
teagoodc
onduc
t l
orofeec
tri
ci
ty?
A.I
tcont
ai
nsmobi
lene
lec

trons

B.I

tisr

igi

d

C.I

tfl

ows

D.I

tisaga

s.

44.Whe

nane

leme

ntwi

t ghe

hhi lec

trona

ffni

i t

yga

insa

nee

lct

ront

obe

comea

n

a

nion;i

tss

ize

A.De

cre

ase

s+++

B.Doe

snotc

hange

C.I

ncr

eas

es

D.Notc

ert

ai

n.

45.Thei

onsi

nioni

ccr

ysa

tll

at

ti

ceha

ve;

A.Sa

mes

ize

B.Di

ff

ere

nts

ize

s+++

C.Unknowns

ize

s

D.La

rges

ize

s.

46.Whi

choft

hes

es t

ubsanc

esi

samor

phous

A.Gl

ass

B.Sodi

umc

hlor

ide

C.Ca

lci

umni

tr

ate

D.Di

amond

47.Themol

ecul

arc

rys

tal

sha

vet

hispr

ope

rty;

A.Me

lta

tbe

low1000C+++

B.Ve

ryha

rds

oli

d

C.Fl

ows

D.Fl

uid

48.Ca

ndi

amondbec

onve

rte

dtogr

aphi

te

A.Ye

s

B.No+++

C.Notc

ert

ai

n

D.Pos

sibl

e

l

49.Haog

ensa

rehi

ghl

yra

ect

ivebe

caus

ethe

y;

A.Ha

vehi

ghe

lec

troa

ffi

nit

y+++

B.Lowi

oni

zat

ione

ner

gy
C.Lowboi
li
ngpoi
nt
D.I
ner
t
50.Ther
eac
ti
vit
yofa
lka
lie
leme
ntsi
sduet
o;
A.Lowi
oni
zat
ione
ner
gy+++
B.Hi
ghe
lec
trona
ffi
nit
y
C.I
ner
tne
ss
D.Nots
ure
nde
51.Me lee

v'sPe

riod

icTa

blewa

sba

sedon:

A.At

omi

cma

ss+++

B.I

oni

zat

ione

ner

gy.

C.At

omi

cra

dius

D.I

oni

cra

diu

s.

52.Na

Cli

sani

oni

ccompoundbe

caus

e

A.I

onsa

rehe

ldi

nla

tt

ic

e+++

B.I

tisc

ova

lentmol

ecul

e

I

C.Themet

ingpoi

nti

sbe

lowf

ree

zingpoi

nt.

D.I

tisf

orme

dbys

har

ingofe

lec

trons

53.Mor

sel

ey'

sPe

ri cTa

odi blewa

sra

the

rba

sedon

A.Cova

lentr

adi

us

B.At

omi

cnumbe

r+++

C.El

ect

rons

har

ing

D.El

ect

rodepot

ent

ia

ls

t

54.Meals

truc

tur

eha

sas

eaofde

loc

ali

zede

lec

tro

nsf

unc

ti

oni

ngf

or:

A.Conduc

ti

onofe

lec

tri

ci

ty+++

B.Cova

lentbondi

ng

C.Mol

ecu

le

D.El

ect

rons

har

ing.

55.Whya

reme

mbe

rsoft

hes

amegr

oupha

vings

imi

la

rphys

ica

landc

hemi

cal

pr

ope

rti

es?

yha

A.The ves

imi

la

rel

ect

roni

cconf

igur

ati

on+++

B.The

ycondu

cthe

at

ya

C.The r

eab

let

oconduc

tel

ect

ri

ci

ty

D.Nonea

tal

l

ka

56.Al l

ime

tal

sar

ehi

ghl

yel

ect

ropos

it

ivebe

caus

ethe

y

dt

A.Hol i

ght

lyt

hei

rout

ermos

tel

ect

rons

s

B.Eail
yre
lea
set
hei
rout
ermos
tel
ect
ronsf
ori
oni
zat
ion+++

C.Ha
vemoi
st
ure

D.The
yar
ecova
lent

57.I
ce,Su
lphura
ndi
odi
nea
ree
xampl
esof

A.Mol
ecu
larc
rys
tal
s+++

B.Sol
idwi
thhi

ghme

lt

ingpoi

nts

C.NoI

at

ti

ce

D.Noneoft

hea

bove

a

58.Grphi

te

,di

amonda r

ndquatza

ree

xampl

esof

A.Br

it

tl

ema

ter

ial

s

B.St

rongl

ycova

lentic

rys

tal

s+++

C.Nor

igi

dst

ruc

tur

es

D.Cr

udes

oli

ds

59.Di

amondha

s:

p3h

A.s ybr

idi

sat

ion+++

B.Sphybr

idi

zat

ion

C.Spdhybr

idi

zat

ion

D.Spdfhybr

idi

zat

ion.

60.Gr

aphi

tei

smos

tl

yus

eda

s:

A.Gl

assc

utt

er

B.El

ect

rode

C.J

ewe

lry

D.Br

it

tl

ema

ter

ial

61.Di

amondi

sani

mpor

tant

A.J

ewe

lry+++

B.Le

adpe

nci

l

C.Lubr

ica

nt

D.Br

ail

le

62.Whi

choft

hes

eel

eme

ntsi

snota

nal

kal

ime

tal

A.Gol

d+++

B.Sodi

um

C.Ca

esi

um

D.Rubi

diu

m

63.Whi

choft

hes

eel

eme

ntsi

snota

nal

kal

inee

art

hme

tal

s?

A.Sul

phur

B.Ca

lci

um

C.Ba

rium

D.Ma

gne

sium.

64.Cova

lentic

ompoundsa

rehe

ldt

oge

the

rby:

A.va

nde
rwa
alf
orc
es+++
B.Col
la
gen
C.Gum
D.I
oni
cfor
ces
65.Chl
ori
ne,f
luor
ine
,br
omi
ne odi
,i nea
nda
sta
ti
nebe
longt
othegr
oupof
A.Cha
lcoge
n
B.Ha
loge
n+++
C.Tr
ans
it
ionme
tal

s

D.I

nne

rtr

ans

it

ionme

tal

s.

66.La

ntha

nide

sandAc

ti

nide

sha

ve:

r

A.Pati

al

lyf

il

ldout

e ermos

tf-

orbi

ta

ls+++

B.Noor

bit

al

C.S-

orbi

ta

lon

ly

D.Noneoft

hea

bove

67.Nobl

ega
sesha
vet
hef
oll
owi
ngpr
ope
rti
es:
c
A.Vaanc
iesi
nthe
i nt
ri he
irs
-or
bit
al
s
c
B.Novaanc
iesi
nthe
i e
routrmos
ts-orp-
orbi
ta
ls+++
C.Va
canc
iesi
np-
orbi
ta
ls
D.Va
canc
iesi

nd-
orbi
ta
ls.
e
68.Elme
ntsi
nPe
riod7i
nthePe
riodi
cta
blea
reuns
tabl
ewi
t hor
hs tha
lf-
li
ves
be
caus
ethe
yar
e:
A.Synt
het
ica
nde
leme
nts
B.Na
tur
all
yoc
cur
ri
ng
C.St
abl

e

D.Noneoft

hea

bove

I

69. Hei

um,xe

non,ne

on,kr

ypt

ona

nda

rgoni

nGr

oup18a

rec

all

ed:

A.Nobl

ega

ses

B.Cha

lcoge

ns

C.Ha

loge

ns

D.Ac

ti

nide

s

70.I

sgr

aphi

tec

onve

rti

blet

odi

amond?

A.Nots

ure

B.No

C.Ye

s+++

D.Unc

ert

ai

n

SECTI

ON4

c

24.Whihoft

hef

oll

owi

nga

lka

nesha

sthel

owe

stboi

li

ngpoi

nt?

A.(

CH3)

2CHCH(

CH3)

2

B.(

CH3CH2)

2CHCH3

C.(

CH3)

3CCH2CH3+++

D.CH3(

CH2)

4CH3

c

25. Whihoft

hef

oll

owi

nga

lka

nesha

sthehi

ghe

stboi

li

ngpoi

nt?

A.(

CH₃)

2CHCH(

CH₃)

2

B.(

CH₃CH₂)

2CHCH₃

C.(

CH₃)

3CCH₂CH₃

D.CH₃(

CH₂)

4CH₃+++

26. Whichoft

hef

oll

owi

ngc oundsha

omp sthehi

ghe

stboi

li

ng

poi

nt?

A.(

CH₃)

3CBr

B.CH₃CH₂CH₂CH₂Br+++

C.(

CH₃)

2C(

CH₃)

2

D.

27.Whichoft

hef

oll

owi

ngc oundsha

omp sthel

owe

stboi

li

ng

poi

nt?

A.(

CH₃)

3CBr

B.CH₃CH₂CH₂CH₂Br

C.(

CH₃)

2C(

CH₃)

2++++

D.

e

28.Prdi

ctt

hepr

oduc

torpr t

oducsoft

histr

eac

ti

on.

A. $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CNa}+++$

B. $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CNH}_2$

NH_2

C. $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CNa}$

D. $\text{CH}_3\text{CH}_2\text{CH}=\text{CNa}$

e

29. Prdi

ctt

hepr

oduc

torpr t

oducsoft

histr

eac

ti

on.

A. (CH_3CH_2)

$3\text{Br}+\text{HBr}$

B. (CH_3CH_2)

$3\text{CBr}+\text{HBr}+++$

C. (CH_3CH_2)

$3\text{HBr}+\text{HBr}$

D. (CH_3CH_2)

$2\text{Br}+\text{HBr}$

e

30. Prdi

ctt

hepr

oduc

torpr t

oducsoft

histr

eac

ti

on

A. +++

B.

C.

D.

31. Wha

tar

ehydr

oca

rbons

A.Compound scont

aini

ngonl ytheelme

e ntsc

rbonc al

led

hydroc

a r

bons

B.Compound scont

aini

ngonl ytheele

mentsca

rbona ndoxygen

arecal

ledhydroca

rbons

C.Compound scont

aini

ngt heelement

scar

bon,hydrogenand

oxygenarecall

edhydrocarbons

D.Compound scont

aini

ngonl ytheele

mentsca

rbona ndhydroge

n

arecal

ledhydroca

rbons+++

32.Choos eonewhichdoesnotbel

ongtothi

sset:C₂H₂,C₃H₄,

C₃H₆

A. C_3H_6 + + +

B. C_2H_2

C. C_3H_6

D. C_3H_4

33. An organic

compound A with

the formula C_2H_6O give

the

compound B with

the same number of carbon

atoms when

di

oxidized

and, which fur

thermally

undergoes oxida

tion to a

car

bonic

acid.

Ident

ify the

compound A.

A. Acetic

B. Ethanol + + +

C. Aldehyde

D. Ketone

34. What

is the product

of the

reac

tion be

tween

eth

anol

and

ac

etic

acid

and

ac

etic

eds sulphuricac

ida

t170 C?

A.Ethene

B.Propeneandwat

er

C.Etheneandwat

er+++

D.Butene

namol

35.I ecul

eofahydr

oc abon,t

r henumberof'

C'a

toms

s'5'

Whatwi

llbeit

sfor

mul

ai fi

tisanal

kyne

A.C₅H₁₀

B.C₅H₁₂

C.C₅H₈+++

D.C₅H₁₄

twoul

36.Wha dbet hepr

oduc

tofr

eac

ti

ont

histr

eac

ti

onbe

low

A.Et

hane

B.C.Be

nze

ne

C.C.Ac

eti

cac

id+++

D.D.Et

hanol

37.Whic

hofthefoll

owi

ngwil

Inotgivea

ddi

ti

onr

eac

ti

ons

Pr

opane

,propyne

,pr

ope

ne,benz

ene.

A.Pr opane+++

B.Pr oypne

C.Pr opene

D.Be nz e

ne

38.Whi

choneoft

hef ol

lowingcompoundsisnotexpec

tedt

obe

ompl

c e

tel
ysol
ubleinwa te
ratroomtemperat
ure.

A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

B. CH_3COOH

C. $\text{CH}_3(\text{CH}_2)_1$
6COOH

D. CH_3+++

39.The funct

ionalgr

oup ofamol

ecule does not a

ffe

cta

compound's

A.sol

ubil

it

y

B.melti

ngpoint

C.rea

cti

on with other mole

cule

s

D.car

bonchain lengt

h+++

ombus

40.The c ti

on of a hydroc

arbon will pr

oducema

nda

tor

y:

A. SO_2

B. CO_2+++

C.CO

D.SO3

41.Which of the following is a substance

a. e

ncspr

esent

sonlypr

ima

ry

car

bonatoms in tss

truc

ture

A. me t

ha ne

B. a cetylene

C. e t

ha ne+++

D. pr opane

42. The for

mul a of the compound be l

ow repre

sent

s

A. pri

ma r

yalcohol

B. sec

ondary alcohol

C. te

rti

aryal

cohol+++

D. quat

ernar

ya l

cohol

43. What c

anr

esul

t by the reduct

ionofanaldehyde?

A.apri

maryalcohol+++

B.asec

ondaryalcohol

C.acar

boxyli

ca ci

d

D.acompoundwi thami xe

dfunct

iona l

cohol-al

dehyde

44.Whatc

anr

esul

tbyther educt

ionofaketone?

A.apri

maryalcohol

B.asec

ondaryalcohol+++

C.acar

boxyli

ca ci

d

D.acompoundwi thami xe

dfunct

iona l

cohol-ket

one

l

45.Theacoholi

cf e

rme

ntt

ai

oni

sar

eac

ti

onof

A.dehydrat

ion

B.hydrol

ysis

C.est

erif

ica

tion

D.catal

yzedenzymati

cal

ly+++

46.H₂doesnotre

actwithanyoftheisome r

softhefoll

owing

mole

cularf

ormulas:

A.C₂H₄

B.C₂H₂

C.C₆H₆

D.C₄H₁₀+++

nwhi

47.I c

hofthefoll

owingreact

ionsaC- Cbondi sbroke

n:

A.CH₃-CN+2H₂ CH₃-CH₂-NH₂

B.C₆H₅-CH₃+3H₂ C₆H₁₁-CH₃

C.C₆H₅-NO₂+3H₂ C₆H₅-NH₂+2H₂O

D.C₆H₅-CH₃+H₂ C₆H₆+CH₄+++

48.Organi

ccompoundsc ancont

anot

i herel

ement

ssucha

s
 oxygenorni
 troge
 n.Tr ueorFal
 se+++
 A.Tr ue
 49.Organi
 ccompoundsa r
 ea l
 waysdei
 rvedf
 romalivi
 ngsour
 ce.
 TrueorFal
 se+++
 A.Fa l
 se
 50.Anunl imite dnumbe rofc a
 rbona t
 omisc anbondt ogetherto
 formc hains.Tr ueorFa lse+++
 A.Tr ue
 51.Mo storgani cc ompoundsc ontai
 ni onicbonds .Tr ueorFa l
 se
 A.Fa lse
 52.The r
 ea remor einor ganiccompoundst hanor gani c
 compounds .Tr ueorFa lse+++
 A.Fa lse
 53.Mo stinorga nicc ompoundsh aveahi ghe rme lti
 ngpoi ntthan
 organiccompounds .Tr ueorfalset+++
 A.Tr ue
 54.Inorganicc ompoundsa reneverfoundi nl ivi
 ngs ys t
 ems .True
 orfals
 e+++
 A.Fa lse

55. Isomersa lwa ysha vet hes a
 mee mpi ri
 ca l(mole cula r
)f or
 mul a
 True or false+++

A. True

56. Isomersha vei de nti
 c alphysic
 alandc he mi ca
 lpr ope rt
 ie s
 .True
 or false
 e+++

A. False

57. Ca r
 bon-c ar
 bon bondsa re usuall
 ys tr
 onga ndc ova lent. True or
 false
 e+++

A. True

58. Ac haracteristi
 coft hea l
 kanesisthatthe ybur nina ir
 .True or
 false
 e+++

A. True

59. What is the reason why there are some nylons and nitrile compounds?

A. Organic compounds are

easy to be made by

various

ways

B. The reaction of

organic compounds, and none of

the other

organic

compounds are recycled

c
 C. Carbon atom sc a
 n form sta
 ble covalent
 bond with
 other car
 bon atoms + + +
 D. Organic compounds can be
 ionic or
 covalent
 c
 compound
 60. Which sta
 tement refers
 to the func
 tion of
 organic groups
 in a
 molecule
 rec
 t?
 Func
 tion of
 organic groups
 is
 A. when nothing happens
 in a chemical
 reaction
 on
 B. the part
 of a molecule
 used to
 attach
 to an organic
 compound
 molecule
 is
 C. when a chemical
 reaction
 occurs
 D. both B and C + + +
 de

61. I nt
 if
 ythe corrects
 tat
 e ment: Thea l
 kanes repres
 e ntone
 f
 amil
 yof organic compoundsc omposed ofc a
 rbon
 c
 ompounds
 A. in whi ch the halogens a re a major compone nt
 B. in whi ch the
 eisa tle
 a stonedouble bondbe tween
 ca
 rbons
 C. that are compos ed only ofc ar
 bona ndhydroge n+++
 D. in whi chhydrogensca nbe bonde d to otherhydrogens

62. Whi
 ch of the f
 oll
 owi
 ng isa
 nal
 cohol
 A. $\text{CH}_3\text{CH}_2\text{COOH}$
 B. $\text{BCH}_3\text{CH}_2\text{OH}+++$
 C. CH_3COCH_3
 D. $\text{CH}_3\text{CH}_2\text{Br}$

63. Wha t is meant by the ter
 m" sa
 turat
 ed" whendescrib
 ing
 hydrocar
 bons?

A.Thehyd rocar
bonsaresat
urt
aedwithwatera
nda r
ewet.

B.Thehyd rocar
bonsaresat
ura
tedwithcar
bonsinit
spar
ent
chai
n.

C.Thehyd rocar
bonsaresaur
t a
tedwithhydr
ogensandhave
onl
ys i
nglebondsbat
weencarbons+++

D.Thec ompoundi spur
e

64.Whichofthef
oll
owingf
ormu
la e
srpr
ese
ntsas
atur
ate
d
hydr
ocar
bon?

A.C₂H₆+++

B.C₂H₄

C.C₃H₆

D.C₅H₁₀

65.What is a character

er

st

cs

red by the alkanes

A. solids

B. polar

C. fl

amenable

D. colored

66. Which of the

following compounds exhibit

the following properties?

t

tr

t

al

i

somer

ism?

A. propane

B. methane

C. butane

D. ethane

67. An alkane

with 10 carbon atoms

has

how many isomers?

o

t

empirical

formula

A. C₁₀H₂₂

D

B. liquid

C. gas

D. not possible to tell

I

68. Alkanes react with

a dil

with

A. Asodium hydroxide

de

B. hydrochloric acid

C. sulfuric acid

d

D. none of these

69. What would be the product(s) of the reaction of 2-methyl-2-butanol with concentrated sulfuric acid?

s) of

his

each

ion?

A. Et

hane

B. Et

hanol

C. Et

hyne

D. None of these

hea

bove

70. What would be the product(s) of the reaction of 2-methyl-2-butanol with concentrated sulfuric acid?

twoul

dbet

hepr

oduc

t(s

) of

hisr

eac

ti

on?

A. CH₃-

CH₃

B. CH₃-

CH₂-

Br

C. CH₃-

CH₂-

NaBr

D.CH₃-

CH₃+2Na

Br+++

71.Wha

twoul

dbet

hepr

oduc

t(s

)oft

histr

eac

ti

on?

A.CH₄

B.CH₃COOH

C.CH₄+Na

OH

D.CH₄+Na₂CO₃+++

72.Wha

twoul

dbet

hepr

oduc

t(s

)oft

histr

eac

ti

on?

A.H₂O+CO₂

B.CH₃OH

C.2CH₃OH+++

D.CH₃COOH

73.Wha

twoul

dbetheproduc

t(s

)oft

histr

eac

ti

on?

A.A.HCHO

B.H₂O+CO₂

C.CH₃OH

D.HCHO+H₂O+++

74.Wha

twoul

dbet

hepr

oduc

t(s

)oft

histr

eac

ti

on?

A.CH₃COOH+H₂O

B.2CH₃COOH+2H₂O +++

C.CH₃CH₂OH

D.CH₃CH₂COOH

75.CH₃Clca

nbeus

edi

nthemanuf

act

ureof

A.si

li

cones

B.methylcel

lul

ose

C.Synthet

icr

ubber

D.All

76.Whichofthefol

lowi
ngc
anbeus
e sl
da oc
ala
nes
the
ti
c?
A.Chlor
oethane+++
B.Tri
chl
orofl
uoromet
hane
C.Chlor
oform
D.Bromotri
fl
uoromet
hane
i
77.Thefr
stt
hreeme mber
sar s
egaes
,thene
xtf
our
tee
nar
e
i
lqui
dsa
ndt
hehi
ghe

rone
saes
r ol
ids
.Tr
ueorf
als
e+++
A.True
r
78.Stai
ghtchai
nal
kene
shavehigherboi
li
ngpoi
ntt
hani
some
ric
br
anchedcha
inc
ompounds.Trueorfa
lse+++
A.True
79.Thisre
act
ionispos
sibl
eint
hepr
ese
nceoft
he.....
cat
alys
t
A.Pt
/Ni

B.Ni/

Pd

C.Pt

/Pd/Ni+++

D.Cu/Pt/Pd

80.Wha

twoul

dbet

hepr

oduc

t(s

)oft

hizr

eac

ti

on?

A. +++

B.

C.

D.

81.Wha

twoul

dbet

hepr

oduc

t(s

)oft

hizr

eac

ti

on?

A. +++

B.

C.

D.

I

82.Boii

ngpoi

nta

rylgr

oupi

sint

hefol

lowi

ngor

der

:Ar-I>Ar

-Br>Ar-Cl>Ar-F.TrueorFal

se+++

A.Tr

ue

SECTI

ON5

1. Wha

tar

ethepr

oduc

tsoft

histr

eac

ti

on;2Na+2H2O?

A.2Na

OH+H2 +++

B.N2+HNO3

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B.Na

Cl+H₂O

C.K₂O

D.MgCl

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B.MgO+H2

C.ZnCl

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$H + H_2O \rightarrow$

A.Li

3N

B.Li

F

C.Li

Cl

D.Li

$OH + H_2 \rightarrow$

40.Ba

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on,Na

$2O + H_2SO_4 \rightarrow$

A.Na

$2SO_4 + H_2O \rightarrow$

B.Na

OH

C.Na

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 $C_2 + 2H_2O \rightarrow$ A. $C_2H_2 + Ca$

(OH)

2+++

B.Ca

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C.Ca

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D.Ca

3(PO₄)

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 $2+2\text{HNO}_3\rightarrow$

A. Sr

(NO₃) $2+2\text{H}_2\text{O}+++$

B. Ba

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C.Ba

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 $\text{C}_2+\text{N}_2\rightarrow$

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Compiled by O'yemi

CHE 102 COMPILED TEST QUESTIONS BY O'YEMI

Q means question and A is the answers

1. Organic Chemistry

Hydrocarbons

Q: General formula for:

- Alkanes: $C_n H_{2n+2}$
- Alkenes: $C_n H_{2n}$
- Alkynes: $C_n H_{2n-2}$

Q: Alkanes undergo what type of reaction?

A: Substitution (e.g., with halogens under UV light).

Q: Convert alkyne to alkene:

Options:

- a) Halogenation
- b) Hydration
- c) Hydrogenation

A: c) Hydrogenation (using H_2 /Lindlar's catalyst).

EM

Q: Ozonolysis of alkenes yields:

A: Cleavage of the double bond → aldehydes/ketones.

Q: Hybridization of:

'Y

- Ethene (C_2H_4): sp^2 (trigonal planar)
- Methane (CH_4): sp^3 (tetrahedral)
- Ethyne (C_2H_2): sp (linear, 180° bond angle).

Q: Keto-enol forms exhibit:

A: Tautomerism (dynamic equilibrium).

Q: General formula for aromatic compounds?

A: $C_n H_n$ (e.g., benzene: C_6H_6).

Functional Groups

Q: Name $CH_3-O-CH_2CH_3$.

A: Methoxyethane (ether).

Q: "Ester" is also called:

A: Carboxylic acid derivative ($R-COO-R'$).

Q: Reduce aldehyde ($R-CHO$) using $NaBH_4$:

A: Forms primary alcohol ($R-CH_2OH$).

2. Physical Chemistry

Q: Percentage of O_2 liberated in combustion (given 66.7%)?

A: Example: For $H_2O_2 \rightarrow 66.7\% O_2$ (varies by compound).

Q: Effective distillation requires:

A: $>25^\circ C$ difference in boiling points.

3. Inorganic Chemistry

Q: Valence electrons of nitrogen?

A: 5 (Group 15: $2s^2 2p^3$).

Q: Hybridization of:

● Carbene ($:CH_2$): sp^2 (lone pair in p-orbital).

● BCl_3 : sp^2 (trigonal planar).

Q: Which salt can't be used in Lassaigne's test?

A: Fluorine (forms insoluble CaF_2).

Q: Elements undetected by Lassaigne's test:

A: Oxygen, hydrogen (detects N, S, halogens). I

Q: Lewis structure of:

EM

● CO : $:C \equiv O:$ (triple bond + lone pairs).

● MgF_2 : $[Mg^{2+}][:F:]^-[:F:]^-$ (ionic).

'Y

4. Analytical Chemistry

O

Q: Separate petrol:

A: Fractional distillation (based on boiling points).

Q: Isomers for C_5H_{12} ?

A: 3 (n-pentane, isopentane, neopentane).

5. Practical Applications

Q: Use of acetic acid?

A: Food preservation (vinegar), plastics, textiles.

Q: CO_2 turns limewater:

A: Milky (forms $CaCO_3$ precipitate).

Geometry

Q: Bond angle in alkynes:

A: 180° (sp hybridization, linear geometry).

Q: Number of isomers for

C_5H_{12}

A: 3 (n-pentane, isopentane, neopentane).

For more questions regarding this document,

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07050376975

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CHM 102 – Organic Chemistry II

Genius Point Academy

STUDY TIP: “Organic chemistry isn’t about memorizing reactions; it’s about following the electrons. Understand the mechanism, and every reaction makes sense!”

Mindset for Success:

Patience with curved arrows, clarity in stereochemistry, and consistency in practice.

1. Benzene undergoes substitution re- 3. Which agent is used to make benzene action more easily than addition reaction react with concentrated nitric acid to give because nitrobenzene?

(a) it has a cyclic structure (a) Concentrated H₂ SO₄

(b) it has three double bonds (b) FeCl₃ catalyst

(c) Lindlar’s catalyst

(c) it has six hydrogen atoms

(d) Ultraviolet light

(d) there is delocalization of electrons

Answer: (a)

Answer: (d)

Explanation: Concentrated H₂ SO₄

Explanation: Benzene possesses a protonates HNO₃ , generating the nitronium highly stable ring of delocalized pi electrons ion needed for nitration.

(aromaticity). Addition breaks this stable

electron cloud; substitution preserves it. 4. Which agent is used to make benzene react with acetyl chloride to give acetophe-

2. Benzene reacts with concentrated none?

HNO₃ in the presence of concentrated

H₂ SO₄ to give nitrobenzene. This is an ex- (a) Ultraviolet light ample of

(b) AlCl₃ catalyst

(a) Electrophilic addition

(c) Platinum catalyst

(b) Nucleophilic addition (d) Al₂ O₃ catalyst

(c) Electrophilic substitution Answer: (b)

(d) Nucleophilic substitution

Explanation: Friedel-Crafts acylation;

Answer: (c) AlCl_3 generates the acylium ion (CH_3CO^+) from acetyl chloride.

Explanation: The nitronium ion NO_2^+ , an electrophile, substitutes for a hydrogen on the benzene ring — classic electrophilic aromatic substitution. (a) Ultraviolet light

Motto: Logic and Mechanisms Guide Our Pathways. Success is earned! 1

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(b) Fe catalyst (c) Cl^+

(c) Nickel catalyst (d) HCl

(d) Al_2O_3 catalyst Answer: (c)

Answer: (b)

Explanation: FeCl_3 polarizes $\text{Cl}-\text{Cl}$, effectively generating a chloronium ion

Explanation: Iron forms FeBr_3 in situ, (Cl^+) for electrophilic attack.

which polarizes Br_2 creating a Br^+ electrophile for substitution. 9. Benzene + $\text{CH}_3\text{CH}_2\text{Cl} \rightarrow$

– Ethylben-

zene + HCl . The catalyst is

6. In Friedel-Crafts acetylation, the role of AlCl_3 is to (a) LiAlH_4

(a) Form a CH_3CO^+ ion (b) AlCl_3

(b) Function as a Lewis base (c) Na

(c) Chlorinate the aromatic ring (d) KOH

(d) Withdraw electrons from the ring Answer: (b)

Answer: (a)

Explanation: Friedel-Crafts alkylation uses AlCl_3 to generate an ethyl carbocation

Explanation: AlCl_3 accepts a chloride electrophile.

ion from acetyl chloride, producing the re-

active acylium ion electrophile. 10. Benzene reacts with H_2 at 150°C , 30 atm, Ni catalyst to give

7. The active electrophile in the nitra-

(a) Cyclohexane

tion of benzene is

(b) Cyclohexene

(a) NO_2^-

(c) n-Hexane

(b) NO^+

(d) No reaction

(c) $\text{NO} + 2$

Answer: (a)

(d) HNO_2

Answer: (c) Explanation: Under vigorous catalytic

hydrogenation conditions, benzene is fully

Explanation: The nitronium ion reduced to cyclohexane.

(NO_2) is generated from the mixed acids

FeCl_3

and attacks the benzene ring. 11. Benzene + $\text{Cl}_2 \xrightarrow{\text{FeCl}_3}$?

(a) Hexachlorobenzene

8. In chlorination of benzene, FeCl_3 gen-

erates (b) Chlorobenzene

(a) Cl^- (c) Hexachlorocyclohexane

(b) $\text{Cl}^\cdot$ (d) Benzyl chloride

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Answer: (b) Answer: (c)

Explanation: Electrophilic aromatic Explanation: The acid generates a

substitution replaces one H with Cl, yield- stable isopropyl carbocation that alkylates
ing chlorobenzene. benzene to isopropylbenzene (cumene).

AlCl_3

12. Benzene + acetic anhydride $\xrightarrow{\text{AlCl}_3}$

3 H SO

15. Benzene + propene $\xrightarrow{\text{H}_2\text{SO}_4}$

4

(a) n-Propylbenzene

(a) Acetophenone

(b) Benzophenone

(b) Benzophenone

(c) Cumene

(c) Phenylacetic acid

(d) Phenyl acetate (d) No reaction

Answer: (a) Answer: (c)

Explanation: Friedel-Crafts acylation Explanation: Propene protonation fol-

with acetic anhydride gives acetophenone lows Markovnikov's rule; the resulting iso-
(methyl phenyl ketone). propyl carbocation yields cumene.

AlCl_3

13. Benzene + benzoyl chloride $\xrightarrow{\text{AlCl}_3}$

3

? 16. Ozonolysis of benzene gives

- (a) Benzyl chloride (a) Formic acid
- (b) Benzaldehyde (b) Glyoxal
- (c) Benzal chloride (c) Formaldehyde
- (d) Benzophenone (d) Glycine

Answer: (d)

Answer: (b)

Explanation: Benzoyl chloride acyla-

Explanation: Ozonolysis cleaves all
tion produces benzophenone, a ketone with
three double bonds, producing three
two phenyl groups.
molecules of glyoxal (ethanedial).

H SO

14. Benzene + isopropyl alcohol $\xrightarrow{2}$

4

17. Gammexane is

- (a) Hexachloroethane
- (a) n-Propylbenzene
- (b) Benzophenone (b) DDT
- (c) Isopropylbenzene (c) Hexachlorocyclohexane
- (d) No reaction (d) TNT

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Answer: (c) Answer: (b)

Explanation: Gammexane (Lindane) Explanation: Halogens are inductively
is the gamma-isomer of hexachlorocyclohex- deactivating but resonance-donating, mak-
ane, produced by UV-catalyzed addition of ing them o, p-directing deactivators.
Cl₂ to benzene.

21. Which of the following substituents

18. Which group is an activating sub- is an ortho and para director and ring de-
stituent? activating?

- (a) -NO₂ (a) -NH₂
- (b) -Cl (b) -Cl
- (c) -OH (c) -OCH₃
- (d) -F (d) -OH

Answer: (c)

Answer: (b)

Explanation: -OH donates lone-pair

Explanation: Halogens are inductively electron-withdrawing (deactivating) but resonance-donating, which directs to ortho and para positions.

o,p positions.

19. Which reacts most rapidly with

22. Which of the following will undergo

$\text{HNO}_3 / \text{H}_2\text{SO}_4$?

substitution in the ortho and para positions

(a) toluene rather than in the meta position?

(b) anisole (a) Nitrobenzene

(c) nitrobenzene (b) Benzoic acid

(d) benzonitrile (c) Acetanilide

Answer: (b) (d) Benzaldehyde

Answer: (c)

Explanation: Anisole's $-\text{OCH}_3$ group

is a strong electron-donating resonance ac-

tivator, making it the most reactive. Explanation: Acetanilide's $-\text{NHCOCH}_3$

group donates electrons via res-

20. In electrophilic aromatic substitution, activating the ring and directing to ortho and para positions, a chlorine substituent is a meta-director.

(a) is a deactivator and a m-director 23. Phenol is an ortho-para director be-

(b) is a deactivator and an o, p-director cause the hydroxyl group:

(c) is an activator and a m-director (a) donates electrons that increase electron density at ortho and para posi-

(d) is an activator and an o, p-director tions favoring nucleophilic attack.

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(b) donates electrons that increase electron density at ortho and para positions favoring electrophilic attack. carry strong deactivating groups.

(c) donates electrons to the ortho and para positions and attracts electrons is most readily sulfonated?

away from meta positions favoring nu-

cleophilic attack. (a) Benzene

(d) donates electrons to the ortho and (b) Chlorobenzene

para positions and attracts electrons

away from meta positions favoring (c) Toluene electrophilic attack.

(d) Nitrobenzene

Answer: (b)

Answer: (c)

Explanation: The -OH lone pairs donate electron density via resonance to o,p positions, attracting electrophiles. Explanation: The activating methyl group makes toluene the most electron-rich, hence most reactive toward sulfonation.

24. Compared to benzene, nitration of toluene takes place at 27. p-Nitrotoluene on treatment with chlorine in the presence of FeCl₃ gives

(a) Same rate

(a) m-Chlorotoluene

(b) Faster rate

(c) Slower rate (b) 2-Chloro-4-nitrotoluene

Answer: (b) (c) p-Chlorotoluene

(d) 2-Nitro-4-chlorotoluene

Explanation: The methyl group is electron-donating (hyperconjugation), activating the ring and speeding up electrophilic attack.

Explanation: -CH₃ directs o,p; -NO₂

25. Which of the following compounds directs m. Both point to C-2, giving 2-chloro-4-nitrotoluene.

(a) Benzene 28. How many isomeric dichlorobenzenes are there?

(b) Toluene

(a) 3

(c) Nitrobenzene

(b) 6

(d) Benzoic acid

(c) 9

Answer: (b)

(d) 12

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Answer: (a) (c) Acetone + NH₂ NH₂

(d) Ethyl bromide + aq KOH

Explanation: Three positional iso-

mers: ortho (1,2), meta (1,3), and para (1,4).
 Answer: (a)

Explanation: Benzene undergoes elec-

29. Which of the following compounds
 trophilic aromatic substitution; the others
 has maximum dipole moment?

are addition (b), condensation (c), and nu-

(a) p-Dichlorobenzene (b) m-Dichlorobenzene (c) o-Dichlorobenzene (d) Carbon tetrachloride

32. Aryl halides are less reactive to-
 wards nucleophilic substitution compared

(c) o-Dichlorobenzene to alkyl halides due to

(d) Carbon tetrachloride (a) Inductive effect

Answer: (c) (b) Resonance stabilization

(c) Tautomerism

Explanation: Ortho (60°) has the

(d) Stereoisomerism

smallest bond angle between C-Cl dipoles;

vectors add constructively to give the Answer: (b)

largest resultant. p- (180°) and CCl₄ (tetra-

hedral) cancel to zero.

Explanation: Lone-pair resonance

30. Which of the following reagents is gives the C-X bond partial double-bond
 most suitable for preparing chlorobenzene character, strengthening it and resisting nu-
 from benzene? (a) Aqueous chlorine (b) Chlorine in the presence of ultraviolet light (c) Chlorine in acetic acid (d) Chlorine in the presence of FeCl₃

33. Hydrolysis of benzotrichloride gives

(a) Benzophenone (b) Benzoic acid (c) Benzyl alcohol (d) Phenol

light

(b) Benzoic acid

(c) Chlorine in acetic acid

(c) Benzyl alcohol

(d) Chlorine in the presence of FeCl₃ (d) Phenol

Answer: (d) Answer: (b)

Explanation: FeCl₃ is the Lewis acid Explanation: -CCl₃ hydrolyzes to -

catalyst required to generate the Cl⁺ elec- C(OH)₃, which loses H₂O to form the stable
 trophile for electrophilic aromatic substitu- -COOH group (benzoic acid).
 tion.

34. Which reagent converts

31. Which of the reactions proceeds by 1,4-dichlorobenzene to 1,4-dichloro-2-
 an electrophilic substitution mechanism? (a) Benzene + Br₂/FeBr₃ (a) HNO₃ + H₂SO₄ (b) Propene + Br₂ (b) NaNO₂

(a) Benzene + Br₂/FeBr₃ (a) HNO₃ + H₂SO₄

(b) Propene + Br₂ (b) NaNO₂

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(c) NaNO_2 -HCl at 5°C (c) Zn/HCl

(d) NO_2 (d) NH_2 NH_2

Answer: (a) Answer: (b)

Explanation: Nitration requires the Explanation: Allyl chloride has
nitrating mixture (HNO_3 / H_2SO_4) to gener- an alkene double bond that decolorizes
ate the nitronium ion electrophile. Br_2 / CCl_4 ; chlorobenzene's aromatic ring
does not react without a catalyst.

35. The following compound is called?

[Structure: $\text{Cl}-\text{C}_6\text{H}_4-\text{CH}(\text{CCl}_3)-\text{C}_6\text{H}_4-\text{Cl}$] 38. Which reagent can distinguish be-
tween chlorobenzene and benzyl chloride?

(a) Chloral

(a) alcoholic AgNO_3

(b) DDT

(b) Br_2 in CCl_4

(c) Lindane

(c) KCN

(d) BHC

(d) Br_2 in H_2O

Answer: (b)

Answer: (a)

Explanation: The structure is p, p' -

Explanation: Benzyl chloride gives an
dichlorodiphenyltrichloroethane, univer-
immediate white AgCl precipitate with al-
sally known as DDT.

coholic AgNO_3 ; chlorobenzene's C-Cl bond

36. DDT is prepared by the reaction of is unreactive due to resonance.

chlorobenzene with (in the presence of conc.

39. An organic compound (A) was

H_2SO_4)

heated with NaOH solution. After cooling,

(a) Chloral acidifying with dilute HNO_3 , and adding

AgNO_3 , a white precipitate formed. Com-

(b) Chlorine pound (A) may be

(c) Chloroform (a) Chlorobenzene

(d) Carbon tetrachloride (b) Benzyl chloride

Answer: (a) (c) o-Chlorotoluene

(d) p-Chlorotoluene

Explanation: Two equivalents of chlorobenzene condense with chloral Answer: (b) (CCl₃CHO) in an electrophilic aromatic substitution catalyzed by H₂SO₄. Explanation: Only benzyl chloride (a reactive benzylic halide) readily releases

37. Which reagent can distinguish between chlorobenzene and allyl chloride? halides resist nucleophilic substitution.

(a) H₂/Ni

40. Which statement is False/Incorrect

(b) Br₂ in CCl₄ about nitrobenzene?

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(a) is formed by nitration of benzene at Answer: (d) temperatures below 60°C.

(b) can be further nitrated to give m- Explanation: The nitro group is a dinitrobenzene. meta-director; bromination therefore yields m-bromonitrobenzene.

(c) can be oxidized with KMnO₄ to give

43. Nitration of nitrobenzene with benzoic acid.

HNO₃/H₂SO₄ gives:

(d) can be reduced with Sn/HCl to give (a) m-dinitrobenzene aniline.

(b) o-dinitrobenzene

Answer: (c)

(c) p-dinitrobenzene

(d) m-nitrobenzenesulfonic acid

Explanation: KMnO₄ oxidizes alkyl side chains to -COOH, but nitrobenzene Answer: (a) has no alkyl side chain, so it cannot be oxidized to benzoic acid.

Explanation: The existing -NO₂

41. Which of the following compounds group directs the incoming NO₂⁺ reacts least rapidly in electrophilic substitution reactions? dinitrobenzene.

(a) Nitrobenzene 44. Nitrobenzene undergoes reduction with zinc and KOH to give

(b) Phenol

- (a) Aniline
(c) Bromobenzene (b) Hydrazobenzene
(d) Toluene (c) Azobenzene
(d) None of these

Answer: (a)

Answer: (b)

Explanation: The $-\text{NO}_2$ group is

strongly electron-withdrawing (resonance + inductive), drastically deactivating the ring medium, nitrobenzene undergoes bimolecular reduction to hydrazobenzene. Acidic reduction would give aniline.

42. Nitrobenzene reacts with Br_2 in the presence of FeCl_3 to give 45. Reduction of p-nitrotoluene with $\text{Sn} + \text{HCl}$ gives:

- (a) o-chloronitrobenzene
(a) p-methylaniline
(b) o-bromonitrobenzene
(b) Toluene
(c) m-chloronitrobenzene (c) m-methylaniline
(d) m-bromonitrobenzene (d) N-Methylaniline

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Answer: (a) (a) can be prepared by reduction of benzonitrile with LiAlH_4 .

Explanation: Sn/HCl reduces aromatic $-\text{NO}_2$ to $-\text{NH}_2$ without affecting

benzamide with LiAlH_4 .
the methyl group, yielding p-toluidine (p-methylaniline).

(c) reacts with nitrous acid to give benzyl

46. Which of the following statements is false about aniline?

- (d) reacts with KMnO_4 to give benzoic acid.
(a) It is a primary aromatic amine.

(b) It is a weak base which forms salts with acids. Answer: (b)

(c) It liberates nitrogen on treatment with ice-cold nitrous acid. Explanation: Statement (b) is false:

LiAlH_4 can reduce benzamide ($-\text{CONH}_2$) to

(d) It can be prepared by the reduction of benzylamine ($-\text{CH}_2\text{NH}_2$). nitrobenzene with tin and HCl.

Answer: (c) 49. Rank the following compounds in

order of increasing boiling point: [Struc-

tures: (1) Phenol, (2) Dihydroxybenzene,

Explanation: Primary aromatic (3) Non-polar aromatic]

amines form stable diazonium salts at 0-

5°C; they do not liberate N_2 gas like (a) (1) < (2) < (3)

aliphatic amines do.

47. Which of the following statements (b) (3) < (1) < (2)

is correct about aniline?

(c) (1) < (3) < (2)

(a) is weak acid.

(b) diazotizes to benzenediazonium chlo-

ride when treated with ice-cold ni-

trous acid. Answer: (b)

(c) can be nitrated with concentrated

HNO_3 to form p-nitroaniline. Explanation: Boiling point increases

(d) gives an intense green color with a sus- with hydrogen bonding capacity: non-polar

pension of bleaching powder in water. (3) < one $-\text{OH}$ (1) < two $-\text{OH}$ groups (2).

Answer: (b) 50. Phenol is acidic because of

(a) Resonance

Explanation: Aniline reacts with

NaNO_2/HCl at 0-5°C (diazotization) to

form benzenediazonium chloride, a key syn-

thetic intermediate.

(c) Inductive effect

48. Which of the following statements

is false about benzylamine? (d) Peroxide effect

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Answer: (a) Answer: (c)

Explanation: The phenoxide ion's Explanation: Acetylene ($\text{pK}_a \approx 25$) is

negative charge is delocalized into the aro- a very weak acid compared to HCl (strong

matic ring via resonance, stabilizing the acid), picric acid (strongly activated phe-

conjugate base and enhancing acidity. nol), and phenol itself.

51. Which of the following has the most 54. Which of the following compounds

acidic proton? is least acidic?

- (a) Thiol (a) Formic acid
- (b) Alcohol (b) Ethanol
- (c) Phenol (c) Acetic acid
- (d) Ether (d) Phenol

Answer: (c) Answer: (b)

Explanation: Phenol ($pK_a \approx 10$) is more acidic than alcohols ($pK_a \approx 16$) and lacks the resonance stabilization of phenol's thiols due to resonance stabilization of the conjugate base and the carboxylate stabilization of the phenoxide ion. Ethers lack acidic protons. ity of the acids, making it the weakest acid here.

52. Phenol is

55. Which of the following compounds

- (a) stronger acid than acetic acid is most acidic?
- (b) weaker acid than acetic acid (a) Formic acid
- (c) stronger base than methylamine (b) Ethanol
- (d) weaker base than methylamine (c) Acetic acid

Answer: (b) (d) Phenol

Answer: (a)

Explanation: Phenol ($pK_a \approx 10$) is a weaker acid than acetic acid ($pK_a \approx 4.7$); carboxylate ions are stabilized by two electronegative oxygens. Explanation: Formic acid is stronger than acetic acid because it lacks the electron-donating $-CH_3$ group that would

53. Which of the following compounds destabilize the carboxylate anion. is least acidic?

56. Which of the following compounds

- (a) HCl is most acidic?
- (b) Phenol (a) Water
- (c) Acetylene (b) Cyclohexanol
- (d) Picric acid (c) Ethanol

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(d) Phenol (a) $-NO_2$

Answer: (d) (b) $-SO_3H$

(c) $-OH$

Explanation: Only phenol has a (d) $-COOH$ resonance-stabilized conjugate base (phenoxide ion); water and alcohols lack this stabilization. Answer: (a)

Explanation: A chromophore is a

57. Which one is most acidic?

group that absorbs visible light; $-\text{NO}_2$ (ni-

(a) Hexanol (tro) is a classic chromophore. The oth-

ers are auxochromes or non-color-bearing

(b) Phenol groups.

(c) Water 60. Which of the following is an aux-

ochrome?

(d) Diisopropyl ether

(a) $-\text{N}=\text{O}$

Answer: (b)

(b) $-\text{NO}_2$

(c) $-\text{N}=\text{N}-$

Explanation: Phenol ($\text{pK}_a \approx 10$) is

more acidic than water ($\text{pK}_a \approx 15.7$) and (d) $-\text{OH}$

alcohols ($\text{pK}_a \approx 16$); ethers have no acidic

Answer: (d)

proton.

58. The reason why materials appear Explanation: An auxochrome ($-\text{OH}$,

colored is $-\text{NH}_2$) does not produce color alone but in-

(a) The selective absorption of spectral tensifies and shifts the absorption of a chro-
colors mophore when attached.

(b) The interaction between the light and 61. The water-solubility of dyes can be
the electrons of the dye molecules increased by introducing

(a) SO_3Na groups

(c) The composition of the white light

from various spectral colors (b) COOH groups

(d) All of these (c) OH groups

(d) All of these

Answer: (d)

Answer: (d)

Explanation: Color arises from the

interplay of all three factors: white Explanation: All three groups ($-\text{OH}$,

light composition, electronic interactions in SO_3Na , $-\text{COOH}$, $-\text{OH}$) are highly polar or

molecules, and selective wavelength absorp- ionic, enhancing hydrogen bonding and ion-
tion. dipole interactions with water.

59. Which of the following is a chro- 62. Which dyes become linked to the
mophore? fibre by chemical reaction?

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- (a) Acid dyes (a) triple, double, single
 (b) Direct dyes (b) single, double, triple
 (c) Disperse dyes (c) single, triple, double
 (d) None of these (d) triple, single, double

Answer: (a) Answer: (a)

Explanation: Acid dyes form strong ionic bonds (a chemical reaction) with protonated amino groups on protein fibers like wool and silk under acidic conditions. Explanation: Higher bond order pulls atoms closer: $C\equiv C$ (1.20 Å) < $C=C$ (1.34 Å) < $C-C$ (1.54 Å).

66. Which of the following is the correct

63. All carbon atoms in naphthalene order of bond lengths: are

- (a) $C\equiv C < C=C < C-C$
 (a) sp hybridized
 (b) $C-C > C=C > C\equiv C$
 (b) sp² hybridized
 (c) $C=C > C-C > C\equiv C$
 (c) sp³ hybridized
 (d) $C=C < C-C > C\equiv C$
 (d) None of these

Answer: (b) Answer: (a)

Explanation: Naphthalene is fully conjugated and aromatic (10 π electrons); every carbon must have an unhybridized p-orbital, requiring sp² hybridization. Explanation: Triple bonds are short-est, double bonds intermediate, single bonds longest. Both (a) and (b) state this correctly in different notations.

64. Naphthalene undergoes oxidation with Na₂Cr₂O₇/H₂SO₄ to form phthalic acid. 67. Which of the following hydrocarbons has the shortest C-C bond length?

- (a) Phthalic acid (a) $CH_2=CH_2$
 (b) Phenylacetic acid (b) CH_3CH_3
 (c) Tetralin (c) $HC\equiv CH$
 (d) Benzoic acid (d) (Option missing)

Answer: (a) Answer: (c)

Explanation: Vigorous oxidation cleaves one ring, leaving a benzene ring with two adjacent -COOH groups: phthalic acid character (1,2-benzenedicarboxylic acid). Explanation: Acetylene has a triple bond; sp-hybridized carbons (50% s-character) hold electrons closer, giving the shortest bond length.

65. The carbon-carbon bond lengths in rank of increasing bond length is: maximum in

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(a) $\text{CH}_2=\text{CH}_2$ forming sigma bonds with exactly two atoms.

(b) CH_3-CH_3

71. Which statement is true?

(c) $\text{HC}\equiv\text{CH}$

(a) Resonance hybrids are inherently un-

(d) (Option missing) stable.

Answer: (b) (b) Resonance hybrids are more stable than any individual resonance form.

Explanation: Ethane's sp^3-sp^3 single bond has the lowest bond order (1) and least resonance forms resembling the less s-character (25%), making it the longest C-C bond.

(d) Resonance hybrids are averages of all resonance forms resembling the more

69. What is the hybridization of stable forms.

the carbon atoms numbered 1 and 2 respectively in the following structure? Answer: (d)

[Structure: $-\text{CH}=\text{CH}-\text{CH}$ (likely cumulated/conjugated)]

Explanation: The true structure (hy-

(a) sp , sp

3 brid) is a weighted average that most closely resembles the most stable (lowest-energy)

(b) sp^2 , sp^2 contributing resonance structures.

(c) sp , sp 72. Resonance structures of a molecule have

(d) sp^2 , sp

(a) same arrangement of atoms

Answer: (d)

(b) different arrangement of atoms

(c) same number of paired electrons

Explanation: Carbon in a double bond

is sp^2 ; carbon participating in two double bonds or one triple bond is sp .

Answer: (a) and (c)

70. How many atoms are attached to an atom having sp hybridization?

Explanation: Resonance structures

(a) 0 differ only in electron distribution (π and

- non-bonding electrons); atom positions and
 (b) 1 total electron count remain unchanged.
 (c) 2 73. Which of the following compounds
 have planar molecules?
 (d) 3

(a) ethyl alcohol

Answer: (c) (b) formaldehyde

(c) diethyl ether

Explanation: sp hybridization creates
 two hybrid orbitals oriented 180° apart, (d) 1,3-butadiene

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Answer: (b) and (d) (b) have cis and trans isomers

(c) exist as enantiomers

Explanation: Formaldehyde ($C=O$,
 (d) can be detected with a polarimeter
 sp) is trigonal planar. 1,3-Butadiene is
 2

fully sp² hybridized and conjugated, mak- Answer: (b)
 ing it planar. Alcohols and ethers have sp³
 atoms (non-planar).

Explanation: Cis/trans isomerism is

74. Which of the following compounds geometrical isomerism, unrelated to chiral-
 will not show geometrical isomerism? ity (optical isomerism). Many cis/trans
 compounds are completely achiral.

(a) $BrCH=CHBr$

77. An optically active compound

(b) $BrCH=CHCl$

(a) must contain at least four carbons

(c) $CH_3-C(Cl)=N$ (approx. structure)

(b) when in solution rotate the plane of

(d) $Cl-C(CH_3)=CHCH_3$ (approx. struc- polarized light
 ture)

(c) must always contain an asymmetric

Answer: (d) carbon atom

(d) in solution always give a negative

Explanation: Geometrical isomerism reading in polarimeter
 requires two different groups on each car-

Answer: (b)

bon of the double bond. If one carbon has

two identical groups, cis/trans isomerism is impossible. Explanation: Optical activity is defined by rotation of plane-polarized light.

75. A molecule is said to be chiral Chiral allenes and biphenyls lack asymmetric carbons yet are optically active, disproving (c).

(b) if it contains centre of symmetry

78. Plane-polarized light is affected by

(c) if it cannot be superimposed on its

mirror image (a) Identical molecules

(b) All polymers

(d) if it can be superimposed on its mirror

image (c) Chiral molecules

Answer: (c) (d) All biomolecules

Answer: (c)

Explanation: Chirality (handedness)

means the molecule and its mirror image are

non-superimposable. Symmetry elements Explanation: Only chiral (asymmetric

(plane/center) indicate achirality. (c) molecules interact with and rotate the

plane of plane-polarized light. Achiral

76. Which of the statements is false regarding chiral compounds? molecules have no net effect.

garding chiral compounds?

79. It is possible to distinguish between

(a) rotate the plane of polarized light optical isomers

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(a) by using chemical tests 82. Enantiomers have which of the following characteristics?

(b) by mass spectrometry

(a) rotate ordinary light

(c) by IR spectroscopy

(d) by polarimetry (b) have the same melting point

Answer: (d) (c) are superimposable mirror images

(d) react with optically active molecules

Explanation: Enantiomers have identical physical/chemical properties in achiral

environments. Only polarimetry (direction Answer: (b)

of optical rotation) or chiral reagents can

distinguish them.

Explanation: Enantiomers have identical

physical/chemical properties in achiral

80. Optical isomers that are mirror images are called: density) in achiral environments. They differ only in optical rotation and chiral interactions.

(a) Tautomers

(b) Diastereomers 83. Which of the following statements is false about enantiomers?

(c) Enantiomers

(d) Metamers (a) rotate plane-polarized light

Answer: (c) (b) are superimposable mirror images

(c) are nonsuperimposable mirror images

Explanation: Enantiomers are

stereoisomers that are non-superimposable (d) have the same melting point

mirror images. Diastereomers are stereoisomers that are not mirror images. Answer: (b)

81. Optical isomers that are not mirror

images are called Explanation: Enantiomers are non-

superimposable mirror images, like left and

(a) Diastereomers right hands. Claiming they are superimposable is false.

(b) Enantiomers

84. A meso compound:

(c) Metamers

(d) Meso compounds (a) is an achiral molecule which contains chiral carbons

Answer: (a)

(b) contains a plane of symmetry or a centre of symmetry

Explanation: Diastereomers are

stereoisomers that are not mirror images (c) is optically inactive

of each other. Enantiomers are mirror images. (d) is characterized by all of the above

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Answer: (d) (d) always optically inactive

Answer: (d)

Explanation: All statements are true:

meso compounds have chiral centers but

Explanation: The internal plane of

internal symmetry cancels optical activity,

symmetry in meso-tartaric acid causes the

making them achiral and optically inactive.

rotations of the two halves to cancel; it is

85. What is the possible number of optically inactive.

isomers for a compound containing n

88. Which of the following compounds

dissimilar asymmetric carbon atoms?

will be optically active?

(a) n (a) Propanoic acid

(b) $2n$ (b) 3-Chloropropanoic acid

(c) $n + 1$ (c) 2-Chloropropanoic acid

(d) $n + 2$ (d) 3-Chloropropene

Answer: (b) Answer: (c)

Explanation: Maximum number of Explanation: 2-Chloropropanoic acid

stereoisomers = 2^n , where n is the num-

n has a chiral C-2 bonded to $-H$, $-Cl$, $-CH_3$,

ber of dissimilar chiral centers (no internal and $-COOH$ (four different groups).

symmetry).

89. Which of the following compounds

86. What is the possible number of optically active?

isomers for a compound containing 2 (a) Succinic acid

dissimilar asymmetric carbon atoms?

(b) meso-Tartaric acid

(a) 2

(c) Lactic acid

(b) 4

(d) Chloroacetic acid

(c) 6 Answer: (c)

(d) 8

Explanation: Lactic acid (2-

Answer: (b)

hydroxypropanoic acid) has a chiral carbon

with four different groups ($-H$, $-OH$, $-CH_3$,

Explanation: Using $2^n : 2^n = 4$ pos-

sible optical isomers (two pairs of enan-

tiomers). 90. Which of the following isomeric

compounds show optical isomerism:

87. meso-Tartaric acid is (a) 1-Aminopentane

(a) sometimes optically active (b) 2-Aminopentane

(b) always optically active (c) 3-Aminopentane

(c) sometimes optically inactive (d) 2,2-Dimethylpropylamine

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Answer: (b) (c) $\text{CH}_3\text{CH}_2\text{COOH}$

(d) CH_3COH

Explanation: 2-Aminopentane has

Answer: (a)

a chiral C-2 ($-\text{NH}_2$, $-\text{H}$, $-\text{CH}_3$, $-\text{CH}_2\text{CH}_2\text{CH}_3$).

3-Aminopentane has two

identical ethyl groups (achiral). Explanation: Lactic acid has a stereo-center at the central carbon (four different

91. 2-Butanol is optically active because it contains: five.

(a) an asymmetric carbon 94. Lactic acid, $\text{CH}_3-\text{CH}(\text{OH})-\text{COOH}$,

(b) a plane of symmetry is a molecule which shows

(a) Geometrical isomerism

(c) a centre of symmetry

(b) Tautomerism

(d) a centre of symmetry

(c) Optical isomerism

Answer: (a)

(d) Metamerism

Answer: (c)

Explanation: C-2 in 2-butanol is

bonded to $-\text{H}$, $-\text{OH}$, $-\text{CH}_3$, and $-\text{CH}_2\text{CH}_3$

— four different groups = asymmetric (chiral) carbon. Explanation: The chiral central carbon (with four different substituents) gives rise to optical isomerism (D- and L-

92. Optical isomerism is shown by enantiomers).

(a) n-Butyl chloride 95. How many optical isomers are possible for lactic acid?

(b) sec-Butyl chloride

(a) 2

(c) tert-Butyl chloride

(b) 4

(d) Isobutyl chloride

(c) 6

Answer: (b)

(d) 8

Answer: (a) 2 (Chemical correction)

Explanation: sec-Butyl chloride (2-

chlorobutane) has a chiral C-2 with four dif-

ferent substituents. The others lack a chiral center. Explanation: Lactic acid has only one

center. chiral center; 21 = 2 optical isomers (R and S enantiomers). The source document

93. Which of the following compounds listed (d), but chemically the correct answer is an optically active compound? [Struc- is (a).

tures: (a) Lactic acid, (b) Butane, (c)

Propanoic acid, (d) Acetaldehyde] 96. Which of the following represents a racemic mixture?

(a) $\text{CH}_3\text{-CH(OH)-COOH}$

(a) 75% (R)-2-butanol, 25% (S)-2-

(b) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$ butanol

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(b) 25% (R)-2-butanol, 75% (S)-2- (a) All chiral molecules possess a plane of butanol symmetry.

(c) 50% (R)-2-butanol, 50% (S)-2- (b) All achiral molecules are meso.

butanol

(c) All molecules which possess a single

(d) none of the above asymmetric center of the S configura-

Answer: (c) tion are levorotatory.

(d) A mixture of achiral compounds will

Explanation: A racemic mixture be optically inactive.

(racemate) is an exact 1:1 (50:50) mixture

of enantiomers; net optical rotation = 0. Answer: (d)

97. Consider (R) and (S)-2-butanol.

Which physical property distinguishes the Explanation: Achiral compounds cannot rotate plane-polarized light; any mixture of them remains optically inactive.

(a) melting point tive. R/S configuration does not predict

(b) solubility in common solvents (+)/(-) rotation.

(c) Rotation of plane-polarized light 100. Which of the statements below correctly describes an achiral molecule?

(d) Infrared spectrum

(a) The molecule has a nonsuperimpos-

Answer: (c)

able mirror image.

Explanation: Enantiomers have identical melting points, solubilities, and IR when it interacts with plane-polarized spectra. Only the direction of optical rotation differs.

tation differs.

(c) The molecule has an enantiomer.

98. Which of the following is capable of exhibiting cis-trans isomerism?

(a) 1-butene Answer: (d)

(b) 1-pentene

Explanation: Meso compounds con-

(c) ethene tain chiral centers but are overall achi-

(d) 2-butene ral due to internal symmetry. An achiral molecule could indeed be a meso form.

Answer: (d)

101. How many asymmetric centers are present in a molecule of 2,4,6-

Explanation: 2-Butene has different trimethylheptane?

groups on each double-bond carbon (-H and -CH₃ on both). Terminal alkenes (=CH₂) (a) 0 cannot show cis/trans isomerism.

(b) 1

99. Which of the following is a true statement? (c) 2

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(d) 3 104. Which of the following is not true of enantiomers?

Answer: (a)

(a) They have the same melting point.

Explanation: C-2 and C-6 each bear two identical -CH₃ groups; C-4 bears two

identical branched groups. No carbon has four different attachments, so zero chiral

(d) They have the same specific rotation. centers.

102. Which of the following compounds is never chiral? Answer: (d)

Explanation: Enantiomers share iden-

(a) 2,3-dibromobutane

tical scalar properties (m.p., b.p., density)

(b) 1,3-dibromobutane but have opposite specific rotations, not the same.

(c) 1,2-dichlorobutane

105. A and B are stereoisomers. They

(d) 1,4-dibromobutane are nonsuperimposable and are mirror images of one another. Which of the following

Answer: (d) best describes the relationship between A and B?

Explanation: 1,4-Dibromobutane

(a) structural isomers

(BrCH₂ CH₂ CH₂ CH₂ Br) is symmetrical

with no chiral center; it can never be chiral. (b) enantiomers

103. Which of the following statements (c) cis-trans isomers is correct concerning a pair of enantiomers?

(d) diastereomers

(a) They rotate the plane of polarized light by exactly the same amount and Answer: (b) in opposite directions.

(b) They rotate the plane of polarized light by differing amounts and in opposite directions. Explanation: Non-superimposable mirror-image stereoisomers are, by definition, enantiomers.

(c) They rotate the plane of polarized light by differing amounts and in the same direction. 106. Which of the statements is correct about diastereomers?

(a) They are stereoisomers that are not

(d) They have different melting points. enantiomers.

Answer: (a) (b) They are a pair of identical isomers.

(c) They are a pair of isomers that are

Explanation: Enantiomers have equal mirror images.

magnitude but opposite sign of specific rotation: one (+), the other (–), by exactly

(d) All their asymmetric centers are the same. the same amount. same.

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Answer: (a) (c) both (a) and (b)

(d) None of these

Explanation: Stereoisomers are di-

Answer: (c)

vided into enantiomers (mirror images) and

diastereomers (non-mirror images). Di-

astereomers = stereoisomers that are not enantiomers. Explanation: Hydrocarbons are non-

polar (C and H have similar electronegativities), making them insoluble in polar water,

107. Which of the following is/are optically inactive and contain only C and H.

110. Which of the following statements is

(a) a 50-50 mixture of R and S enantiomers false about propane?

isomers

(a) all bond angles are 109.5°

(b) a meso compound

(b) each carbon is sp^3 hybridized

(c) a racemic mixture (c) the compound is combustible

(d) all the above (d) the compound undergoes polymerization to give polypropylene.

Answer: (d)

Answer: (d)

Explanation: A 50:50 mixture

(racemic mixture) has net rotation = 0. Explanation: Propane is a saturated

Meso compounds have internal symmetry alkane (no π bonds), so it cannot un-

canceling rotation. All are optically inactive. undergo addition polymerization. Polypropylene comes from propene (alkene).

108. Hydrocarbons are 111. Which of the following is a correct name according to the IUPAC rules?

(a) Composed of carbon and hydrogen (a) 2-Methylcyclohexane

(b) Composed of carbon, hydrogen, and oxygen (b) 2-Ethyl-2-methylpentane

(c) 3,4-Dimethylpentane

(c) Composed of carbon and oxygen

(d) 3-Ethyl-2-methylpentane

(d) Composed of carbon and nitrogen

Answer: (d)

Answer: (a)

Explanation: (d) correctly identifies

the longest chain (pentane), gives lowest loc-

Explanation: By definition, hydrocar-

bons contain only carbon and hydrogen

bons contain only carbon and hydrogen

(Ethyl before Methyl).

atoms.

112. A tertiary carbon is bonded di-

109. Hydrocarbons are

rectly to:

(a) Insoluble in water (a) 2 hydrogens

(b) composed of carbon and hydrogen (b) 3 carbons

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(c) 2 carbons (c) C7 H16

(d) 4 carbons (d) C9 H18

Answer: (b) Answer: (a)

Explanation: Carbon classification: 1° Explanation: Hexane (6C) + three (bonded to 1C), 2° (2C), 3° (3C), 4° (4C). methyls (3C) = 9C total. Alkane formula: Tertiary = bonded to three other carbons. C9 H20 .

113. What type of an alkyl group is an 116. How many isomers are possible for isobutyl group? butane?

(a) 2

(a) primary

(b) 3

(b) secondary

(c) 4

(c) tertiary

(d) 5

(d) none of these

Answer: (a)

Answer: (a)

Explanation: C4 H10 has two struc-

Explanation: Isobutyl = -CH2 - tural isomers: n-butane (straight chain)

CH(CH3)2 . The attachment point (-CH2 -) and isobutane (2-methylpropane).

is bonded to only one other carbon in the

group, making it 1° (primary). 117. How many isomers are possible for

2-methylpropane?

114. Which molecular formula indicates (a) 2

2-methylpentane?

(b) 3

(a) C5 H12

(c) 4

(b) C6 H14

(d) 5

(c) C7 H16

Answer: (a)

(d) C6 H12

Answer: (b) Explanation: The molecular formula

C4 H10 (associated with 2-methylpropane)

has exactly 2 structural isomers.

Explanation: Pentane (5C) + methyl

(1C) = 6C total. Saturated alkane: 118. How many isomers are possible for

$C_n H_{2n+2} = C_6 H_{14}$. pentane?

(a) 2

115. Which molecular formula indicates

2,2,4-trimethylhexane? (b) 3

(a) $C_9 H_{20}$ (c) 4

(b) $C_8 H_{18}$ (d) 5

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Answer: (b) Answer: (b)

Explanation: $C_5 H_{12}$ has three isomers: Explanation: Marsh gas is the historical name for methane (CH_4), produced by anaerobic decomposition of organic matter in wetlands.

119. How many isomers are possible for

hexane? 122. Which statement is false?

(a) 4 (a) Many alkanes are soluble in water.

(b) 5 (b) All alkanes have a lower density than water.

(c) 6

(c) At room temperature some alkanes

(d) 7 are liquids, some solids, some gases.

Answer: (b) (d) All alkanes burn.

Answer: (a)

Explanation: $C_6 H_{14}$ has five structural isomers: n-hexane, 2-methylpentane,

Explanation: Alkanes are non-polar

3-methylpentane, 2,2-dimethylbutane, and

(C and H have similar electronegativities)

2,3-dimethylbutane.

and are completely insoluble in polar water.

120. How many isomers are possible for ter-

heptane?

123. As the molecular weight of alka-

(a) 9 nes increases, how do the boiling point and melting point change?

(b) 10

(a) boiling point increases; melting point

(c) 11 increases.

(d) 12 (b) boiling point increases; melting point

increases.

Answer: (a)

(c) boiling point increases; melting point decreases.

Explanation: C₇H₁₆ (heptane) has exactly 9 distinct structural isomers by mathematical enumeration. (d) boiling point increases; melting point increases sequentially for alkanes over four carbons.

121. Marsh gas mainly contains

Answer: (d)

(a) CH₂=CH₂

Explanation: Both b.p. and m.p. generally increase with molecular weight due

(b) CH₄

(c) H₂S to stronger London dispersion forces. M.p. shows slight odd-even alternation based on

(d) CHCl₃ crystal packing efficiency.

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124. The branching of alkanes that produces symmetrical structures: Answer: (d)

(a) raises the boiling point; raises the melting point highly branched, nearly spherical) has minimal surface area → weakest intermolecular

(b) raises the boiling point; lowers the forces → lowest b.p. melting point

127. Methane can be prepared by:

(c) lowers the boiling point; lowers the melting point (a) the reaction of iodomethane with sodium in dry ether

(d) lowers the boiling point; raises the

(b) the reaction of methanol with concentrated H₂SO₄

trated H₂SO₄

Answer: (d) (c) the reaction of sodium methanoate with soda-lime

Explanation: Branching reduces surface area → weaker van der Waals forces soda-lime

→ lower b.p. Symmetrical branching allows

tighter crystal packing → higher m.p. Answer: (d)

125. Which of the following alkanes will have the highest boiling point? Explanation: Sodium ethanoate (CH_3COONa) undergoes decarboxylation

(a) n-Octane with soda-lime (NaOH/CaO) to yield CH_4 and Na_2CO_3 .

(b) Isopentane

128. Which of the following reactions

(c) n-Butane can be used to prepare alkanes?

(d) Neopentane (a) Corey-House synthesis

(b) Williamson synthesis

Answer: (a)

(c) Friedel-Crafts reactions

Explanation: n-Octane (C_8H_{18}) has the highest molecular weight and largest

surface area, giving the strongest London forces and highest b.p. Answer: (a)

126. Which of the following alkanes will have the lowest boiling point? Explanation: Corey-House synthesis couples lithium dialkyl cuprates (Gilman reagents) with alkyl halides to form new

(a) n-Heptane alkanes. Williamson makes ethers; Friedel-Crafts makes alkylbenzenes.

(b) Isopentane

129. Which of the following reactions

(c) n-Hexane can be used to prepare alkanes?

(d) Neopentane (a) Wurtz reaction

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(b) Wolf-Kishner reduction (a) n-Hexane

(c) Kolbe's electrolysis (b) Diethyl ether

(d) All of these (c) 1-Butene

Answer: (d)

(d) Aniline

Explanation: All three are valid alkane Answer: (a)

syntheses: Wurtz ($\text{R-X} + \text{Na}$), Wolff-

Kishner (reduces C=O to CH_2), Kolbe

Explanation: n-Hexane (alkane) can-

(electrolysis of RCOONa).

not be protonated by H_2SO_4 . Ether (O lone

130. Kolbe's electrolysis of sodium butyrate gives bond) all protonate and dissolve.

(a) CH₄ 133. 2,3-Dimethyl-2-butene undergoes catalytic hydrogenation to give

(b) C₆ H₁₄

(c) C₈ H₁₈ (a) 2,3-Dimethylbutane

(d) C₄ H₁₀ (b) 2-Methylpentane

Answer: (b) (c) 2,2-Dimethylbutane

(d) 3-Methylpentane

Explanation: Sodium butyrate (4C

chain) loses CO₂ at the anode, forming Answer: (a)

propyl radicals (3C). Two propyl radicals

couple → hexane (C₆ H₁₄).

Explanation: Catalytic hydrogenation

131. Which of the following classes adds H₂ across the double bond without of compounds is unreactive toward sulfuric skeletal rearrangement; the carbon skeleton acid? remains 2,3-dimethylbutane.

(a) Alkanes

134. Chlorine free radicals react with

(b) Alcohols methane by:

(c) Alkenes (a) donating their free-radical electron to methane to form chloromethane.

(d) Alkynes

Answer: (a) (b) abstracting a hydrogen atom from methane, and producing HCl and a methyl radical.

Explanation: Alkanes have no π bonds

or lone pairs to protonate; they are inert to (c) forming a carbanion intermediate H₂ SO₄. Alcohols, alkenes, and alkynes all that rapidly dissociates to produce react with strong acids. chloromethane.

132. Which of the following compounds (d) forming a carbonium ion intermedi- does not dissolve in concentrated H₂ SO₄ on ate that rapidly dissociates to form warming? chloromethane.

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Answer: (b) (c) with equal rates

(d) with equal or different rate depending

Explanation: Propagation step: Cl· upon the source of alkane abstracts H· from CH₄ → HCl + · CH₃

(methyl radical). This is a radical, not Answer: (b)

ionic, mechanism.

Explanation: Chlorine radicals are

135. Which halogen does not appreciably react with methane in a free-radical substitution reaction?

and faster for chlorination.

(a) chlorine

138. How many monochlorinated iso-

(b) bromine mers would result from the reaction of chlorine with n-butane in the presence of UV

(c) iodine light?

(d) fluorine (a) 2

Answer: (c) (b) 4

(c) 6

Explanation: Iodine radical is too sta-

(d) 8

ble; H-abstraction from methane is highly endothermic, so the reaction does not proceed under standard conditions.

136. What product is formed in the free-radical bromination of methane? unique hydrogen environments: 1° (C-1/C-4) → 1-chlorobutane, and 2° (C-2/C-3) →

(a) bromomethane 2-chlorobutane. Total = 2 isomers.

(b) dibromomethane 139. How many monochlorination products are possible in the reaction of 2,2-

(c) tribromomethane

dimethylbutane with chlorine in the pres-

(d) all of these ence of ultraviolet light?

(a) 2

Answer: (d)

(b) 3

Explanation: Free-radical halogena-

tion is non-selective; once CH₃ Br forms,

it can be further brominated to CH₂ Br₂ , (d) 5

CHBr₃ , and CBr₄ . All are produced.

Answer: (b) 3 (Chemical correction)

137. Chlorination of an alkane as compared to bromination proceeds

Explanation: Three unique H environ-

(a) at a slower rate ments: 9 equiv. 1° H on three -CH₃ at C-2;

2 equiv. 2° H on C-3; 3 equiv. 1° H on C-4.

(b) at a faster rate Gives 3 distinct monochlorinated isomers.

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140. A compound of formula C_4H_8 does Answer: (c)

not react with bromine in CCl_4 in the dark.

The compound could be

Explanation: Cracking (pyrolysis)

(a) Alkane breaks long-chain alkanes into smaller, more useful alkanes and alkenes by heating in the

(b) Cycloalkane absence of oxygen.

143. LPG (Household cooking gas) is

(c) Alkene

mainly a mixture of

(d) Cycloalkene (a) Methane + Ethane

Answer: (b) (b) Acetylene + O_2

(c) Butane + Isobutane

Explanation: $C_4H_8 = C_nH_{2n}$ (alkene

or cycloalkane). No reaction with Br_2/CCl_4 (d) Acetylene + H_2

in the dark = no double bond, so it must Answer: (c)

be a cycloalkane (cyclobutane or methylcyclopropane).

Explanation: Liquefied Petroleum

141. The combustion of pentane pro- Gas (LPG) consists primarily of butane

duces: (n-butane) and isobutane, sometimes with

propane.

(a) Pentene

144. What type of bonding is most im-

(b) $HCl + H_2O$ portant in $CH_3CH_2CH_2CH_2CH_3$?

(c) Pentyne (a) ionic

(b) hydrogen

(d) $CO_2 + H_2O$

(c) covalent

Answer: (d)

(d) polar

Explanation: Complete combustion of Answer: (c)

any hydrocarbon in excess O_2 yields carbon

dioxide and water: $C_5H_{12} + 8O_2 \rightarrow 5CO_2$

Explanation: Hexane is an alkane; all

+ $6H_2O$.

C-C and C-H bonds involve electron shar-

142. The thermal decomposition of alkane (covalent bonding). The bonds are essentially non-polar.

145. What is the name given to a compound

(a) Combustion

containing only carbons and hydrogens and having the maximum number of

(b) Oxidation

hydrogens in the molecule?

(c) Cracking (a) alkene

(d) Hydrogenation (b) unsaturated hydrocarbon

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(c) saturated hydrocarbon Explanation: For isomers: more branching → more spherical → less surface

(d) alkyne area → weaker London forces → lower b.p.

Most branched (2,3-dimethylbutane) boils

Answer: (c) lowest; straight chain (n-hexane) boils highest.

Explanation: A saturated hydrocarbon

148. The eclipsed and staggered forms

of ethane are said to differ in

of ethane are said to differ in

holds the maximum possible number of H

atoms ($C_n H_{2n+2}$). (a) molecular formula

146. Which of the following is the (b) configuration

best description of propane, $CH_3-CH_2-CH_3$,

at room temperature? (c) conformation

(a) liquid, soluble in H_2O (d) constitution

(b) gas, soluble in gasoline Answer: (c)

(c) liquid, soluble in gasoline

Explanation: Different spatial arrangements

(d) gas, soluble in water arrangements arising from rotation about a

C-C single bond are called conformations

Answer: (b) (conformational isomers).

149. Octane number is related to

Explanation: Propane (C_3H_8 , low

MW) is a gas at RT. Non-polar → soluble (a) Gasoline

in non-polar solvents (gasoline), insoluble in

(b) Kerosene

polar water.

(c) Diesel oil

147. Consider the three isomeric alkanes n-hexane, 2,3-dimethylbutane, and 2-methylpentane. Which of the following correctly lists these compounds in order of increasing boiling point?

(a) 2,3-dimethylbutane < 2-methylpentane < n-hexane
Explanation: Octane number (octane rating) measures the antiknock performance of gasoline in internal combustion engines.

(b) 2-methylpentane < n-hexane < 2,3-dimethylbutane

150. A knocking sound is produced in

(c) 2-methylpentane < 2,3-dimethylbutane < n-hexane

(a) Burns fast

(d) n-hexane < 2-methylpentane < 2,3-dimethylbutane (b) contains lubricating oil

(c) Burns slow

Answer: (a)

(d) contains water

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Answer: (a) (d) 2,2,4-Trimethylpentane

Answer: (c)

Explanation: Knocking occurs when the fuel-air mixture detonates prematurely and burns too rapidly, creating a shockwave before the spark plug fires. Explanation: n-Heptane, a straight-chain alkane, causes severe knocking and was arbitrarily assigned an octane rating of 0 as the reference standard.

151. The octane rating of gasoline provides information on

154. Which of the following compounds

(a) its antiknock properties is assigned the Octane Number of 100:

(b) its ignition properties (a) n-Heptane

(c) its percentage proportion of octane (b) 2,3,3-Trimethylpentane

(d) its percentage content of isooctane (c) n-Octane

(d) 2,2,4-Trimethylpentane

Answer: (a)

Answer: (d)

Explanation: Octane rating directly

measures the fuel's ability to resist knocking Explanation: 2,2,4-Trimethylpentane (premature detonation) under compression. ("isooctane"), a highly branched alkane, resists knocking excellently and was assigned

152. In any fuel, the percentage by volume of isooctane in a mixture of isooctane

and n-heptane which will knock under same

155. Gasoline with an octane number

conditions as the fuel being tested, is called: of 90 is equivalent in knocking characteristics to a mixture of heptane and isooctane

(a) Cracking

of the following composition:

(b) Iodine number

(a) 20% heptane + 80% isooctane

(c) Aromatization (b) 90% heptane + 10% isooctane

(d) Octane number (c) 80% heptane + 20% isooctane

Answer: (d) (d) 10% heptane + 90% isooctane

Answer: (d)

Explanation: This is the definition of

octane number: isooctane = 100 (no knock-

ing), n-heptane = 0 (severe knocking). Explanation: Octane number = %

isooctane in reference blend. 90 octane =

153. Which of the following compounds 90% isooctane + 10% n-heptane.

is assigned the Octane Number of zero:

156. A fuel with octane number 90

(a) n-Octane means it is as good as a mixture of:

(b) 2,3,3-Trimethylpentane (a) 90% Isooctane + 10% n-heptane

(c) n-Heptane (b) 90% n-Heptane + 10% isooctane

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(c) 2 Litre of 90% Isooctane + 2 litre of 159. Which of the following methods

benzene can be used to increase the octane rating of gasoline?

(d) 2 Litre of 90% n-heptane + 2 litre of

benzene (a) Adding branched-chain alkanes

Answer: (a) (b) Adding tetraethyllead

(c) Adding aromatic hydrocarbons

Explanation: Same concept: octane

(d) All of these

number 90 = performance equivalent to

90% isooctane + 10% n-heptane reference Answer: (d)

blend.

157. Which hydrocarbon molecule with Explanation: All are valid strategies: branched alkanes, aromatics (benzene, eight carbons would produce the lowest octane number (i.e., the most knocking) when used as a fuel in automobile? crease octane rating.

(a) 2-Methyl-3,3-dimethylpentane 160. The carbon atoms involved in the double bond of an alkene are

(b) n-Octane

(a) sp hybridized

(c) 2-Isopropylpentane

(b) sp² hybridized

(d) 2-Methylheptane

(c) sp³ hybridized

Answer: (b)

(d) None of these

Explanation: Straight-chain alkanes Answer: (b)

have the poorest antiknock properties. n-

Octane (unbranched) would have the lowest

Explanation: Alkene C atoms use

octane rating among these C₈ isomers.

three sp² hybrid orbitals for sigma bonds

158. Which of the following compounds (trigonal planar, 120°) and one unhybridized p orbital for the π bond.

(a) Ethyllithium 161. Which of the following compounds will show geometrical isomerism?

(b) Tetraethyllead

(a) Propene

(c) Ethyl acetate

(b) 2-Butene

(d) Lead acetate

(c) Propyne

Answer: (b) (d) 2-Butyne

Answer: (b)

Explanation: Tetraethyllead (TEL),

Pb(C₂H₅)₄, was historically used in "leaded

gasoline" to boost octane ratings; now Explanation: 2-Butene (CH₃ -

phased out due to lead toxicity. CH=CH-CH₃) has different groups on each

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double-bond carbon (-H and -CH₃), al- (III) Alkenes have higher boiling points

lowing cis/trans isomerism. Propene has than alkanes
 $=CH_2$; alkynes are linear.

(IV) Alkenes have lower boiling points

162. Which of the following compounds than alkanes
 is most stable?

(a) I and III

(a) Ethylene

(b) II and IV

(b) 2,3-Dimethyl-1-butene

(c) I and IV

(c) Propylene

(d) II and III

(d) 2-Butene

Answer: (a)

Answer: (b)

Explanation: Alkenes generally have

Explanation: Alkene stability in-

slightly lower m.p. (rigid double bond dis-
 creases with alkyl substitution (hypercon-
 rupts crystal packing) and slightly higher
 jugation). 2,3-Dimethyl-1-butene is disub-

b.p. (π electrons increase polarizability).
 stituted with additional electron-donating

alkyl groups, making it the most stable 165. How do the melting points of
 among the options. trans isomers compare to the cis isomers for
 alkenes?

163. In which solvent are alkenes most

soluble? (a) cis isomers have higher melting points

(a) water (b) trans isomers have higher melting
 points

(b) ethyl alcohol

(c) both have similar melting points

(c) ammonia

(d) no consistent trend is observed

(d) carbon tetrachloride

Answer: (b)

Answer: (d)

Explanation: T rans isomers are more

Explanation: "Like dissolves like":

symmetrical and linear, packing more effi-
 non-polar alkenes dissolve best in non-polar

ciently in the crystal lattice → stronger intermolecular forces → higher m.p.

ter, alcohol, and ammonia are polar.

166. Why do trans isomers of alkenes

164. Which statements about alkenes

have lower boiling points than cis isomers?

and alkanes of corresponding chain lengths

is true? (a) trans isomers have better symmetry

(I) Alkenes have slightly lower melting points than alkanes (b) cis isomers have better symmetry

(c) trans isomers are less polar

(II) Alkenes have slightly higher melting

points than alkanes (d) cis isomers are less polar

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Answer: (c) (c) 1-Butene

(d) 1-Butyne

Explanation: Cis isomers have a net

dipole moment (polar); trans dipoles cancel (non-polar). Polarity increases dipole-

dipole interactions, raising the b.p. of cis

isomers. highly substituted alkene (2-butene) is the

major product over the less substituted 1-

167. The major product of acid-

butene.

butene.

catalyzed dehydration of 3-pentanol is:

170. In the reaction of propene with

(a) 1-Pentene

HCl, H⁺ ion acts as the:

(b) 2-Methyl-1-butene (a) electrophile

(c) 2-Pentene (b) carbonium ion

(d) 3-Methyl-1-butene (c) nucleophile

Answer: (c) (d) carbanion

Answer: (a)

Explanation: 3-Pentanol eliminates

H₂O to form a double bond between C-

3 and C-2 (or C-4, identical), yielding 2- Explanation: H⁺ is electron-deficient

pentene. (positive charge); it accepts electrons from

the π bond of propene, acting as an elec-

168. Ethylene is obtained from ethyl trophile.

bromide by:

171. When bromine attacks the double

(a) Simple heating bond in propene, which of the following ions is formed in the first stage of the attack?

(b) Hydrolysis

(a) Cyclic bromonium ion

(c) Dehydrohalogenation

(b) Primary carbocation

(d) Nucleophilic substitution

(c) Secondary carbocation

Answer: (c)

(d) Carbanion

Explanation: Alcoholic KOH removes Answer: (a)

H and Br from $\text{CH}_3\text{CH}_2\text{Br}$ (elimination of

HBr), forming $\text{CH}_2=\text{CH}_2$. This is dehydro- Explanation: Br_2 addition proceeds

halogenation. through a three-membered cyclic bromo-

onium ion intermediate, not an open carbo-

169. The dehydrohalogenation of 2-

cation. The Br bridges both carbons of the

bromobutane with alcoholic KOH gives

former double bond.

mainly

(a) 2-Butene 172. In the reaction of ethylene with

H_2O in the presence of sulfuric acid, which

(b) 2-Butyne one adds across the double bond first?

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(a) H^+ Answer: (b)

(b) H^-

Explanation: The π bond (nucle-

(c) HO^- (ophile) attacks Br_2 (electrophile); two Br

atoms add across the double bond. This is

(d) sulfate ion electrophilic addition.

Answer: (a) 175. Baeyer's reagent is

(a) dilute KMnO_4

Explanation: Acid-catalyzed hydra- (b) $\text{HCl} + \text{ZnCl}_2$

tion follows electrophilic addition: the π

bond attacks H^+ first, forming a carboca- (c) Br_2 in CCl_4

tion, which is then attacked by H_2O . (d) NH_2NH_2

173. Why does a H^+ ion attacking a Answer: (a)

carbon-carbon double bond add to the carbon with the least number of substituents?

Explanation: Baeyer's reagent = cold,

(a) the reaction is resonance stabilized. alkaline, dilute KMnO_4 . It tests for unsaturation; purple color disappears and brown

(b) the hybrid geometry favors this process. MnO_2 precipitates with alkenes/alkynes.

176. In the addition of HX to a double

(c) nucleophile tends to attack stable center, the hydrogen goes to the carbon that has more hydrogens is a statement of

(d) a more stable carbonium ion is generated. (a) Hund's rule

(b) Markovnikov's rule

Answer: (d) (c) Hückel rule

(d) Saytzeff rule

Explanation: Markovnikov's rule explained:

adding H^+ to the less substituted carbon places the positive charge on the

more substituted carbon, creating a more stable carbocation ($3^\circ > 2^\circ > 1^\circ$).

Explanation: Markovnikov's rule: in

stable carbocation ($3^\circ > 2^\circ > 1^\circ$).

HX addition to an asymmetric alkene, H

174. Propene reacts with bromine to add to the carbon with more H's (less substituted), X to the more substituted carbon.

Example of 177. Markovnikov's addition of HBr is

not applicable to

(a) Nucleophilic addition

(a) Propene

(b) Electrophilic addition

(b) 1-butene

(c) Nucleophilic substitution (c) 1-pentene

(d) Electrophilic substitution (d) 2-butene

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Answer: (d) (b) Grignard test

(c) Baeyer test

Explanation: 2-Butene is symmetrical; both double-bond carbons have one H

(d) Wurtz test and one CH_3 . HBr addition gives the same

Answer: (c)

product regardless of regiochemistry, so the rule is moot.

Explanation: The Baeyer test uses

178. In the reaction of cold dilute KMnO_4 (purple); alkenes reduce

$\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ with HCl , the H of the MnO_4^- to brown MnO_2 , causing the purple

HCl will become attached to which carbon? color to disappear.

(a) C-1

181. Which of the following will give a

(b) C-2 negative test when treated with bromine in carbon tetrachloride?

(c) C-3

(a) Butane

(d) C-4

(b) 2-Butene

Answer: (a)

(c) 1,3-Butadiene

Explanation: 1-Butene: C-1 ($=\text{CH}_2$) (d) 2-Butyne

has 2 H's, C-2 ($=\text{CH}-$) has 1 H.

Answer: (a)

Markovnikov's rule $\rightarrow$ H adds to C-1 (more H's), Cl to C-2 (more substituted).

Explanation: Butane is a saturated

179. Which of the following compounds alkane (no π bonds); it will not react with or will react most readily with bromine in decolorize Br_2/CCl_4 (negative unsaturation CCl_4 ? test).

(a) $\text{CH}_3\text{CH}_2\text{CH}_3$ 182. Ethylene reacts with HI to give

(b) $(\text{CH}_3)_3\text{CH}$ (a) Iodoethane

(c) $\text{CH}_3\text{CH}=\text{CH}_2$ (b) 2,2-Diiodoethane

(d) $(\text{CH}_3)_4\text{C}$ (c) 1,1-Diiodoethane

Answer: (c) (d) None of these

Answer: (a)

Explanation: Only propene (alkene)

has a π bond that undergoes rapid elec-

trophilic addition with Br_2/CCl_4 in the Explanation: Ethylene ($\text{CH}_2=\text{CH}_2$) is

dark. Alkanes require UV light or heat. symmetrical; HI addition across the double bond yields iodoethane ($\text{CH}_3\text{CH}_2\text{I}$).

180. The disappearance of the purple

color of KMnO_4 in its reaction with alkene 183. Which of the following reagents is known as will react with propene?

(a) Markovnikov test (a) Hot alkaline KMnO_4

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- (b) Sodium metal (a) 1-Bromobutane
(c) Cold dilute HNO_3 (b) 2,3-Dibromobutane
(d) LiAlH_4 (c) 2-Bromobutane
(d) 2,2-Dibromobutane

Answer: (a)

Answer: (c)

Explanation: Hot alkaline KMnO_4 is a strong oxidant that cleaves the $\text{C}=\text{C}$ double bond of propene. The other reagents do not react with alkenes under normal conditions. HBr addition yields only 2-bromobutane regardless of which carbon receives H vs. Br.

184. Which of the following compounds will have zero dipole moment? 187. Which of the following alkenes reacts with HBr in the presence of a peroxide

- (a) cis-1,2-dibromoethylene to give anti-Markovnikov's product?
(b) 1,1-dibromoethylene (a) 1-Butene
(c) trans-1,2-dibromoethylene (b) 2,3-Dimethyl-2-butene
(d) all of these (c) 2-Butene

(d) 3-Hexene

Answer: (c)

Answer: (a)

Explanation: In the trans isomer, the two C-Br bond dipoles are equal, opposite, and cancel completely $\rightarrow$ net dipole = 0. The cis isomer and geminal isomer are polar. The others are symmetrical. Explanation: Anti-Markovnikov (peroxide effect) is only meaningful for asymmetrical alkenes. 1-Butene is asymmetric; the others are symmetrical.

185. 2-Methylpropene reacts with HBr to give tert-Butyl bromide 188. Propene reacts with HBr in the presence of a peroxide to give Allyl bromide

- (a) tert-Butyl bromide (a) n-Propyl bromide
(b) Allyl bromide
(b) Isobutane
(c) Isopropyl bromide
(c) n-Butyl bromide
(d) Vinyl bromide
(d) None of these

Answer: (a)

Answer: (a)

Explanation: Peroxide effect $\rightarrow$ free-

Explanation: Markovnikov addition: radical mechanism. $\text{Br}\cdot$ adds to the less H adds to $=\text{CH}_2$, forming a 3° carbocation; substituted carbon (C-1), giving n-propyl Br^- attacks to give $(\text{CH}_3)_3\text{CBr}$ (tert-butyl bromide (1-bromopropane) bromide).

189. 2-Methylpropene reacts with HBr

186. 2-Butene reacts with HBr to give in the presence of peroxide to give

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(a) A primary alkyl bromide 192. A compound reacts with steam in the presence of concentrated H_2SO_4 to give

(b) A secondary alkyl bromide isopropyl alcohol. The compound could be

(c) A tertiary alkyl bromide (a) Alkane

(d) A vicinal dibromide (b) Alkyne

Answer: (a) (c) Alkene

(d) Allene

Explanation: Anti-Markovnikov ad-

Answer: (c)

dition places Br on the terminal $=\text{CH}_2$

(less substituted), producing 1-bromo-2-

methylpropane — a 1° alkyl bromide. Explanation: Acid-catalyzed hydration of propene ($\text{CH}_3\text{CH}=\text{CH}_2$) yields iso-

190. 2-Methyl-2-butene reacts with propyl alcohol, $(\text{CH}_3)_2\text{CHOH}$. This is a

3° HBr in the presence of peroxide to give characteristic alkene reaction.

(a) A primary alkyl bromide 193. Propene reacts with Cl_2 in H_2O to

(b) A secondary alkyl bromide give

(a) 1-Chloro-2-propanol

(c) A tertiary alkyl bromide

(b) 2-Chloro-2-propanol

(d) A vicinal dibromide

(c) 1-Chloro-1-propanol

Answer: (b)

(d) 2-Chloro-1-propanol

Explanation: Anti-Markovnikov: Br Answer: (a)

adds to the less substituted $=\text{CH}-$ carbon

(secondary), producing a secondary alkyl

bromide (2-bromo-3-methylbutane). Explanation: Halohydrin formation:

Cl adds to less substituted C (C-1), OH

191. Hydration of 2-methyl-1-propene adds to more substituted C (C-2) $\rightarrow$ 1- (with $\text{H}_2\text{O}/\text{H}_2\text{SO}_4$) gives: chloro-2-propanol.

(a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ 194. 1-Butene reacts with Cl_2 in H_2O to give

(b) $(\text{CH}_3)_3\text{COH}$

(a) 1-Chloro-2-butanol

(c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

(b) 2-Chloro-2-butanol

(d) $(\text{CH}_3)_2\text{CHOH}$

(c) 1-Chloro-1-butanol

Answer: (b) (d) 2-Chloro-1-butanol

Answer: (a)

Explanation: Markovnikov hydration:

H^+ adds to $=\text{CH}_2$, forming a 3° carbocation.

H_2O attacks to give tert-butyl alcohol, Explanation: Halohydrin regiochemistry: Cl on C-1 (less substituted), OH

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on C-2 (more substituted) $\rightarrow$ 1-chloro-2- Answer: (c) butanol.

195. Catalytic hydrogenation of 3- Explanation: Ozonolysis of symmetrical 2-butene cleaves it into two identical molecules of acetaldehyde (CH_3CHO).

(a) Isobutane

198. Which of the following alkenes will

(b) 2,2-Dimethylbutane give a mixture of acetone and formaldehyde on ozonolysis?

(c) 2-Methylbutane

(a) 2-butene

(d) 2,3-Dimethylbutane (b) 2-methyl-2-butene

Answer: (c) (c) 1-butene

(d) 2-methylpropene

Explanation: H_2 /catalyst saturates Answer: (d)

the double bond without skeletal re-

arrangement. 3-Methyl-1-butene $\rightarrow$ 2-

methylbutane (isopentane). Explanation: Retrosynthetic analysis: acetone = $(\text{CH}_3)_2\text{C}=\text{O}$; formalde-

196. Which of the following alkenes hydro = $\text{H}_2\text{C}=\text{O}$. Joining fragments gives gives only acetic acid on oxidation with hot $(\text{CH}_3)_2\text{C}=\text{CH}_2$ (2-methylpropene). concentrated KMnO_4 ?

199. Which of the following alkenes will

(a) Ethylene give a mixture of acetone and acetaldehyde

on ozonolysis?

(b) 1-Butene (a) 1-butene

(c) Propene (b) 2-methyl-2-butene

(d) 2-Butene (c) 2-butene

(d) 2-methylpropene

Answer: (d)

Answer: (b)

Explanation: Hot KMnO_4 cleaves

$\text{C}=\text{C}$. Symmetrical 2-butene ($\text{CH}_3\text{CH}=\text{CHCH}_3$) Explanation: Acetone fragment = splits into two identical acetic acid (CH_3COOH) 2 $\text{C}=\text{}$; acetaldehyde fragment = molecules (CH_3CHO). $\text{CH}_3\text{CH}=\text{}$. Joined: (CH_3COOH) 2 $\text{C}=\text{CHCH}_3$ (2-methyl-2-butene).

197. Which of the following compounds

will give only acetaldehyde on ozonolysis? 200. A hydrocarbon, C_6H_{12} , on ozonolysis gives only one product which does not

(a) 1-Butene give silver mirror with Tollens' reagent. The hydrocarbon is

(b) Acetylene

(a) 2,3-Dimethyl-2-butene

(c) 2-Butene (b) 2-Hexene

(d) Ethylene (c) 2-Methyl-2-pentene

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(d) 3-Hexene (a) 1

Answer: (a)

(b) 2

Explanation: Only one product = (c) 3

symmetrical alkene. Negative Tollens'

test = product is a ketone, not aldehyde. (d) 4

hyde. 2,3-Dimethyl-2-butene $\rightarrow$ two ace-

tone molecules (ketone, no silver mirror). Answer: (d)

201. Combustion of an alkene with suf-

ficient oxygen will produce Explanation: A double bond = one σ

(a) carbon dioxide and water bond (2 electrons) + one π bond (2 electrons) = 4 electrons total.

(b) carbon monoxide and water

204. Which of the following statements

(c) only carbon dioxide about ethene, C_2H_4 , is incorrect?

(d) only carbon monoxide

(a) The H-C-H bond angles are approxi-

Answer: (a) mately 109.5° .

Explanation: Complete combustion of (b) There is a total of five sigma bonds.

any hydrocarbon in excess O_2 yields CO_2

and H_2O . Incomplete combustion (limited (c) The carbon atoms are sp^2 hybridized.

O_2) would produce CO and/or C .

(d) The $H-C-H$ bond angles are approxi-

120° . Which of the following are correct mately 120° .

statements concerning unsaturated hydro-

carbons? Answer: (a)

(a) All unsaturated hydrocarbons are in-

soluble in water.

Explanation: Ethene carbons are sp^2

(b) All unsaturated hydrocarbons are sol- hybridized (trigonal planar), giving $H-C-H$

soluble in nonpolar solvents. bond angles $\approx 120^\circ$, not 109.5° (sp^3). State-

ment (a) is false.

(c) All unsaturated hydrocarbons are

more dense than water. 205. Which of the following statements

(d) (a) & (c) about propene, $CH_3CH=CH_2$, is correct?

(a)(b)

Answer: (d) per source; chemically (a) & All are

incorrect

atoms lie in the same plane.

Explanation: Hydrocarbons are non- (b) The compound has a cis and trans

polar $\rightarrow$ insoluble in water (a) and solu- isomer.

soluble in nonpolar solvents (b). They are less

dense than water, so (c) is false. (c) There is a total of eight sigma bonds.

203. How many electrons are involved (d) All the carbon atoms are sp^2 hy-

bridized in a carbon-carbon double bond?

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Answer: (c) (a) 1-chloropentane

(b) 2-chloropentane

Explanation: Propene: 3 $C-H$

(c) 3-chloropentane

(methyl) + 1 $C-C$ + 1 $\sigma(C=C)$ + 3 $C-$

H (sp^2 carbons) = 8 σ bonds. The methyl (d) 1-chloro-2-methylbutane

carbon is sp^3 , not sp^2 , and the methyl hy-

Answer: (c)

hydrogens are out of plane.

206. Which of the following is capable Explanation: 3-Chloropentane is sym-

of exhibiting cis-trans isomerism? metrical; elimination of HCl from either ad-

(a) 1-butene jacent carbon yields identical 2-pentene. 2-

Chloropentane would give a mixture of 1-

(b) 1-pentene pentene and 2-pentene.

(c) ethene 209. What is the major product from

the acid-catalyzed hydration of 2-methyl-2-

(d) 2-butene pentene?

Answer: (d) (a) 2-methylpentane

(b) 2-methyl-1-pentanol

Explanation: 2-Butene has -H and -

(c) 2-methyl-2-pentanol

CH₃ on each double-bond carbon (different

groups). 1-Butene, 1-pentene, and ethene (d) 2-methyl-3-pentanol

all have =CH₂ (two identical H's), prevent-

Answer: (c)

ing cis/trans isomerism.

207. Which of the following is the most Explanation: Markovnikov hydration:

stable alkene? H adds to less substituted C (C-3), form-

(a) trans-3-hexene ing 3° carbocation at C-2. H₂ O attacks C-2

→ 2-methyl-2-pentanol (tertiary alcohol).

(b) cis-3-hexene

210. 1,2-Butadiene has

(c) 1-hexene (a) only sp hybridized carbon atoms

(d) (Z)-3-hexene (b) only sp² hybridized carbon atoms

Answer: (a) (c) only sp³ hybridized carbon atoms

(d) sp, sp², sp³ hybridized carbon atoms

Explanation: Internal alkenes > ter- Answer: (d)

minal alkenes (hyperconjugation). T rans

> cis (less steric hindrance between bulky

alkyl groups). trans-3-hexene is most sta- Explanation: CH₂=C=CH-CH₃ : C1

ble. = sp², C₂ = sp (central allene carbon), C₃

= sp², C₄ (methyl) = sp³. All three hy-

208. Which of the alkyl chlorides listed bridizations present.

below undergoes dehydrohalogenation in

the presence of a strong base to give 2- 211. How many sigma bonds are there

pentene as the only alkene product? in CH₂=CH-CH=CH₂ ?

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(a) 3 (a) C₄ H₁₀

(b) 6 (b) C₄ H₈

(c) 9 (c) C₄ H₆

(d) 12 (d) C₄ H₄

Answer: (c)

Answer: (c)

Explanation: 1,3-Butadiene: 6 C-H σ

Explanation: Alkane: C_n H_{2n+2}. Each

bonds + 3 C-C σ bonds = 9 sigma bonds.

double bond (or ring) removes 2 H's. Two

Every bond (single or multiple) contains ex-

double bonds remove 4 H's: C₄ H₁₀ - 4H =

actly one σ bond.

C₄ H₆ (diene or alkyne).

212. Propadiene, CH₂=C=CH₂, is

215. 1,3-Butadiene reacts with bromine

(a) a planar compound to mainly give

(b) a cumulated diene (a) 3,4-Dibromo-1-butene

(c) an isolated diene (b) 4-Bromo-1-butene

(d) a conjugated diene (c) 1,4-Dibromo-2-butene

Answer: (b) (d) 1-Bromo-2-butene

Answer: (c)

Explanation: Cumulated dienes (al-

lenes) have double bonds sharing a single

carbon: C=C=C. They are not planar; ter- Explanation: Conjugated dienes un-

minal CH₂ groups are perpendicular. dergo 1,4-addition (thermodynamic con-

trol). Br adds to C-1 and C-4; double bond

213. Which of the following compounds shifts to C-2/C-3 $\rightarrow$ 1,4-dibromo-2-butene.

have planar molecules?

(a) 1,3-Butadiene 216. What descriptive term is applied

to the type of diene represented by 1,5-

(b) Dimethyl ether octadiene?

(c) 1-Butene (a) conjugated diene

(d) Allene (b) cumulated diene

Answer: (a) (c) isolated diene

(d) alkynyl diene

Explanation: 1,3-Butadiene is fully

sp hybridized; all atoms lie in one plane

2

Answer: (c)

for maximum π conjugation. Allene is non-

planar (perpendicular π systems).

Explanation: Double bonds separated

214. Which of the following molecular by two or more single bonds (sp^3 carbons) formulas will correspond to an alkene with cannot interact via resonance; they are iso- two double bonds? lated dienes.

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217. What descriptive term is applied Answer: (a)
to the type of diene represented by 2,4-
hexadiene?

Explanation: Central carbon in

(a) conjugated diene $CH_2 = C = CH_2$: two σ bonds, no lone pairs

→ steric number = 2 → sp hybridized. Uses

(b) cumulated diene two p orbitals for the two π bonds.

(c) isolated diene

220. What compound results from the

(d) alkynyl diene 1,4-addition of one equivalent of HBr to 1,3-
butadiene?

Answer: (a)

(a) 1-bromo-1-butene

Explanation: Alternating double-

single-double bonds allow continuous p - (b) 2-bromo-2-butene
orbital overlap and electron delocalization

— this is a conjugated diene. (c) 4-bromo-1-butene

218. Which of the following state- (d) 1-bromo-2-butene
ments are correct concerning delocalized
electrons?

Answer: (d)

(a) Electrons do not belong to a single
atom

Explanation: 1,4-Addition: H^+ adds

(b) Electrons are not confined to a bond to C-1, Br attacks at C-4 (allylic carbo-
between two atoms cation resonance). Double bond shifts to
C-2/C-3 → 1-bromo-2-butene.

(c) Electrons are shared by three or more
atoms 221. Rank the following dienes

in order of increasing stability: trans-

(d) (a), (b) and (c) 1,3-pentadiene, cis-1,3-pentadiene, 1,4-
pentadiene, and 1,2-pentadiene.

Answer: (d)

(a) 1,2-pentadiene < 1,4-pentadiene

Explanation: All statements correctly < cis-1,3-pentadiene < trans-1,3-

describe delocalized electrons: they are pentadiene spread over multiple atoms in extended π systems (e.g., conjugated dienes, aromatic (b) 1,4-pentadiene < 1,2-pentadiene rings). < trans-1,3-pentadiene < cis-1,3-pentadiene

219. What is the hybridization of the central carbon of allene (1,2-propadiene)?

- (c) trans-1,3-pentadiene < cis-1,3-pentadiene
 (a) sp pentadiene < 1,4-pentadiene < 1,2-pentadiene
 (b) sp²
 (c) sp³ (d) 1,2-pentadiene < cis-1,3-pentadiene < 1,4-pentadiene < trans-1,3-pentadiene
 (d) p pentadiene

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Answer: (a) Answer: (d)

Explanation: Stability order: cumu- Explanation: Triple bond = 3 shared lated (least stable) < isolated < conjugated pairs of electrons. 3×2 electrons/pair = 6 cis < conjugated trans (most stable due to electrons total. resonance + less steric hindrance).

225. Which of the following impro-

222. A triple bond consists of erly describes the physical properties of an alkyne?

- (a) 2 sigma bonds and 1 pi bond
 (a) relatively nonpolar
 (b) 3 sigma bonds
 (b) nearly insoluble in water
 (c) 1 sigma bond and 2 pi bonds
 (c) less dense than water
 (d) 3 pi bonds
 (d) insoluble in most organic solvents

Answer: (c)

Answer: (d)

Explanation: First bond = σ (head-on overlap). Second and third bonds = π (side- Explanation: Alkynes are non-polar by-side p-orbital overlap). Triple bond = 1σ hydrocarbons; "like dissolves like" → they + 2π . are soluble in non-polar organic solvents.

Statement (d) is false.

223. The bond angles associated with

the hybrid orbitals of a carbon involved in 226. Which of the following statements a triple bond is correctly describes the general reactivity of alkynes?

(a) 180°

(a) An alkyne is an electron-rich molecule

(b) 120° and therefore reacts as a nucleophile.

(c) 109° (b) The pi bonds of alkynes are higher in energy than the sigma bonds and are

(d) 45°

thus more reactive.

Answer: (a) (c) Unlike alkenes, alkynes fail to undergo electrophilic addition reactions.

Explanation: sp-Hybridized carbon:

(d) Alkynes are generally more reactive

two sp orbitals oriented 180° apart $\rightarrow$ linear

than alkenes.

geometry.

Answer: (a)

224. How many electrons are involved in a carbon-carbon triple bond?

Explanation: The exposed π electrons

(a) 1 of the triple bond create a region of high electron density; alkynes act as nucleophiles

(b) 2

toward electrophiles.

(c) 3

227. Which of the following describes a

(d) 6 triple bond?

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(a) two sigma bonds and two pi bonds 230. Which is the correct order of decreasing acidity in the following com-

(b) one sigma bond and one pi bond pounds?

A. H_2O B. CH_3CH_3 C. NH_3 D.

(c) two sigma bonds and one pi bond

$\text{CH}_2=\text{CH}_2$ E. $\text{HC}\equiv\text{CH}$

(d) one sigma bond and two pi bonds

(a) $\text{A} > \text{E} > \text{C} > \text{D} > \text{B}$

Answer: (d)

(b) $\text{A} > \text{E} > \text{D} > \text{B} > \text{C}$

Explanation: Same as Q222: triple (c) $\text{E} > \text{A} > \text{C} > \text{B} > \text{D}$

bond = 1 σ bond (head-on) + 2 π bonds

(d) $A > C > E > D > B$

(side-by-side p-orbital overlap).

228. What two atomic orbitals overlap to form the

carbon-carbon sigma bond in ethyne? Explanation: pK_a values: $H_2O \approx 15.7$

$> HC \equiv CH \approx 25 > NH_3 \approx 38 > CH_2 = CH_2$

$\approx 44 > CH_3CH_3 \approx 50$. Acidity decreases as

(a) sp^2-sp^2

(b) sp^3-sp^3 conjugate base stability decreases.

(c) s-s 231. Which of the following compounds

is the major product when 1-hexyne is

(d) $sp-sp$ treated with excess HBr?

Answer: (d) (a) 1,1-dibromohexane

(b) 1,1-dibromohexene

Explanation: Ethyne ($HC \equiv CH$): both

carbons are sp hybridized. The C-C σ bond (c) 1,2-dibromohexane

= $sp-sp$ head-on overlap.

(d) 2,2-dibromohexane

229. How many distinct internal

alkynes exist with a molecular formula of Answer: (d)

C_6H_{10} ?

(a) 1 Explanation: Markovnikov addition

(twice): 1st HBr $\rightarrow$ 2-bromo-1-hexene; 2nd

(b) 2 HBr adds H to C-1, Br to C-2 $\rightarrow$ 2,2-

dibromohexane (geminal dihalide).

(c) 3

232. The reagent needed to convert 2-

(d) 4-butyne to cis-2-butene is:

Answer: (c) (a) H_2/Pt

(b) H_2 /Lindlar's catalyst

Explanation: Internal C_6H_{10} alkynes:

2-hexyne, 3-hexyne, and 4-methyl-2- (c) Li/ NH_3

pentyne. Total = 3 distinct internal

alkynes. (d) Na/ NH_3

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Answer: (b) (c) vinyl radical

(d) carbene

Explanation: Lindlar's catalyst (poi-

soned Pd) gives syn addition of $H_2 \rightarrow$ cis- Answer: (b)

alkene. Li or Na in NH_3 gives trans-alkene.

H_2 /Pt fully reduces to alkane. Explanation: Electrophilic addition:

π electrons attack H^+ , leaving a positive

233. Among the compounds water, 1-

charge on the adjacent sp-carbon = vinyl

butyne, 2-butyne, and ethane, which are

cation intermediate.

stronger acids than ammonia?

(a) 1-butyne and ethane 236. Starting with 2-butene, which of

the following is the best method for prepar-

(b) water and 1-butyne ing 2-butyne?

(c) water and ethane (a) HBr ; H_2 /Ni; Zn/H^+

(d) 1-butyne and 2-butyne (b) HBr ; Zn/H^+ ; H_2 /Ni

Answer: (b) (c) Br_2 ; H_2 /Ni

(d) Br_2 ; 2 $NaNH_2$

Explanation: NH_3 $pK_a \approx 38$. H_2O

Answer: (d)

($pK_a \approx 15.7$) and terminal alkyne 1-butyne

($pK_a \approx 25$) are stronger acids. 2-Butyne

(internal alkyne) and ethane lack acidic pro- Explanation: Alkene $\rightarrow$ alkyne: (1)

tons. Br_2 addition $\rightarrow$ 2,3-dibromobutane; (2)

double dehydrohalogenation with strong

234. What is the product when 3- base ($NaNH_2$) $\rightarrow$ 2-butyne.

heptyne reacts with the Lindlar catalyst?

237. In the hydrogenation reaction, the

(a) 2-methyl-2-heptene hybrid state of the carbon atom changes

(b) trans-3-heptene from

(c) cis-2-methyl-3-hexene (a) sp^3 - sp^2 - sp

(b) sp - sp^2 - sp^3

(d) cis-3-heptene

(c) sp^2 - sp^3 - sp

Answer: (d)

(d) sp - sp

Explanation: Lindlar's catalyst $\rightarrow$ syn Answer: (b)

addition of H_2 to internal alkyne $\rightarrow$ cis-3-

heptene specifically.

Explanation: Complete hydrogenation

235. In the addition of hydrogen chloride: alkyne (sp) $\rightarrow$ alkene intermediate

ride to alkynes, which of the following (sp^2) $\rightarrow$ alkane (sp^3).

species is believed to be an intermediate?

238. Which of the following hydrocar-

(a) vinyl anion has acidic hydrogens?

(b) vinyl cation (a) 1-Butene

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(b) 1-Butyne (a) Alkynes are reduced more readily

than alkenes.

(c) 2-Butene

(b) Alkynes are reduced less readily than

(d) 2-Pentyne

alkenes.

Answer: (b)

(c) Alkynes and alkenes are reduced with

equal speed.

Explanation: Only terminal alkynes

have acidic H ($pK_a \approx 25$). 1-Butyne has (d) Alkynes and alkenes cannot be re-

a terminal $\equiv C-H$; 2-pentyne is an internal alkyne (no H on sp carbon).

alkyne (no H on sp carbon).

Answer: (a)

239. Lindlar's catalyst is

(a) $LiAlH_4$ Explanation: The linear geometry of

alkynes allows stronger adsorption onto the

(b) $Pd/BaSO_4$ in Quinoline metal catalyst surface (Pt, Pd, Ni), making

them reduce faster than alkenes.

(c) $NH_2 NH_2$

(d) $HCl/ZnCl_2$ 242. Which of the following sequences

regarding acid-strength is correct?

Answer: (b)

(a) $HC\equiv CH > CH_3 CH_2 OH >$

$CH_3 COOH$

Explanation: Lindlar's catalyst = Pd

deposited on $CaCO_3$ or $BaSO_4$, poisoned (b) $H_2 O > HC\equiv CH > CH_3 COOH$

with lead/sulfur and treated with quinoline.

Partially reduces alkynes to cis-alkenes. (c) $HC\equiv CH > CH_3 CH_2 OH$

240. The higher reactivity of an alkene (d) $CH_3 COOH > H_2 O > CH_3 CH_2 OH >$

or alkyne, as compared to an alkane, is due $HC\equiv CH$

to

Answer: (d)

(a) sigma bonds

(b) pi bonds Explanation: pKa values: CH_3COOH

(4.8) > H_2O (15.7) > $\text{CH}_3\text{CH}_2\text{OH}$ (16) >

(c) hydrogen bonds

$\text{HC}\equiv\text{CH}$ (25). Carboxylic acid > water >

(d) None of these alcohol > terminal alkyne.

Answer: (b) 243. 1,2-Dichloroethane reacts with excess of NaNH_2 to form

Explanation: π bonds have electron (a) Vinyl chloride density above/below the internuclear axis

— exposed, higher energy, and more accessible to electrophiles than buried σ bonds.

(c) Ethyl chloride

241. Which of the following statements about alkenes and alkynes is correct? (d) Acetylene

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Answer: (d) (d) CuCl_2

Answer: (b)

Explanation: Excess NaNH_2 (strong base) causes double dehydrohalogenation:

Explanation: Mg_2C_3 contains the C_4^-

$\text{ClCH}_2\text{CH}_2\text{Cl} \rightarrow \text{CH}_2=\text{CHCl} \rightarrow \text{HC}\equiv\text{CH}$ 3

(propargylide) ion. Hydrolysis: $\text{Mg}_2\text{C}_3 +$ (acetylene).

$4\text{H}_2\text{O} \rightarrow \text{CH}_3\text{C}\equiv\text{CH} + 2\text{Mg(OH)}_2$. Al_4C_3

244. Propyne is formed by gives methane.

(a) Polymerization of acetylene 247. Which one of the following compounds will (a) decolorize dilute cold

(b) Reaction of acetylene with methane KMnO_4 (b) decolorize bromine water; and

(c) give a white precipitate with ammonia-

(c) Reaction of acetylene with methyl cal AgNO_3 solution?

chloride

(a) 1-Hexene

(d) Reaction of sodium acetylide with methyl chloride (b) 1-Hexyne

Answer: (d) (c) 2-Hexene

(d) 2-Hexyne

Explanation: $\text{HC}\equiv\text{C Na}$ (sodium — +

Answer: (b)

SN 2

acetylide) + $\text{CH}_3\text{Cl} \longrightarrow \text{HC}\equiv\text{C}-\text{CH}_3$

(propyne) + NaCl .

Explanation: (a) and (b) = unsatura-

245. Which of the following compounds tion tests (all pass). (c) = terminal alkyne on hydrolysis gives acetylene? test (Tollens' reagent). Only 1-hexyne has a terminal $\equiv\text{C}-\text{H}$ to form the silver acetylide

(a) CaC_2 precipitate.

(b) Mg_2C_3 248. Ethylene and acetylene can be dis-

(c) Al_4C_3 tinguished by using

(a) Bromine in CCl_4

(d) CuCl_2

(b) Tollens' reagent

Answer: (a)

(c) Baeyer's reagent

Explanation: Calcium carbide: CaC_2 (d) Phenylhydrazine

+ $2\text{H}_2\text{O} \rightarrow \text{HC}\equiv\text{CH} \uparrow + \text{Ca}(\text{OH})_2$. This is

Answer: (b)

the classic industrial method for acetylene production.

Explanation: Both decolorize

246. Which of the following compounds Br_2/CCl_4 and KMnO_4 (unsaturation).

on hydrolysis gives propyne? Only acetylene (terminal alkyne) reacts with Tollens' reagent to form a silver

(a) CaC_2

acetylide precipitate.

(b) Mg_2C_3

249. 1-Butyne can be distinguished

(c) Al_4C_3 from 2-butyne by using

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(a) potassium permanganate 252. 2-Butyne undergoes catalytic hydrogenation in the presence of Lindlar's cat-

(b) bromine in CCl_4 alyst to give

(c) Tollens reagent (a) 2-Butene

(d) chlorine in CCl_4 (b) Butane

Answer: (c) (c) 1-Butene

(d) 2-Methylpropene

Explanation: Both are alkynes and

Answer: (a)

react with halogens and oxidizers. Only

1-butyne (terminal alkyne) has an acidic

$\equiv\text{C-H}$ and gives a precipitate with Tollens' Explanation: Lindlar's catalyst $\rightarrow$ syn

reagent. addition of H_2 to internal alkyne $\rightarrow$ cis-2-

butene specifically.

250. 1-Butyne reacts with

253. Addition of two moles of HCl to

(a) NaNH_2

propyne gives:

(b) $\text{Dil H}_2\text{SO}_4$ and HgSO_4 (a) 2,2-Dichloropropane

(c) HBr (b) 1,3-Dichloropropane

(d) All of these (c) 1,2-Dichloropropane

Answer: (d) (d) None of these

Answer: (a)

Explanation: 1-Butyne: terminal

alkyne reacts with NaNH_2 (acid-base); un-

Explanation: Markovnikov addition

dergoes HgSO_4 -catalyzed hydration $\rightarrow$ 2-

(twice): both H 's add to C-1, both

butanone; adds HBr (electrophilic addi-

Cl 's to C-2 $\rightarrow$ geminal dihalide 2,2-

tion). Reacts with all.

dichloropropane, $(\text{CH}_3)_2\text{CCl}_2$.

251. Propyne can be converted to

254. Propyne reacts with aqueous

propene by using

H_2SO_4 in the presence of HgSO_4 to form

(a) H_2 + Lindlar's catalyst (a) acetone

(b) NH_2NH_2 (b) 1-propanol

(c) H_2 + Pt catalyst (c) acetaldehyde

(d) NH_2NH_2 + KOH (d) 2-propanol

Answer: (a) Answer: (a)

Explanation: Lindlar's catalyst (poi- Explanation: Markovnikov hydra-

soned Pd) partially hydrogenates alkynes to tion of alkyne: enol intermediate $(\text{CH}_3 -$

cis-alkenes without over-reduction to alka- $\text{C}(\text{OH})=\text{CH}_2$) tautomerizes to the more sta-

nes. H_2/Pt would fully reduce to propane. ble ketone, acetone $(\text{CH}_3\text{COCH}_3)$.

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255. Acetylene reacts with water in the Answer: (d)

presence of sulfuric acid and mercuric sulfate to give

Explanation: 2-Butyne ($\text{CH}_3\text{C}\equiv\text{CCH}_3$)

(a) Acetone is an internal alkene — no terminal $\equiv\text{C}-\text{H}$, therefore no acidic proton to react with

(b) Acetic acid Tollens' reagent.

258. Which of the following compounds

(c) Formaldehyde

reacts with ammoniacal cuprous chloride to

(d) Acetaldehyde give a precipitate?

(a) 1-Butene

Answer: (d)

(b) 1-Butyne

(c) 2-Butene

Explanation: Hydration of acetylene:

vinyl alcohol ($\text{CH}_2=\text{CHOH}$) enol tautomer- (d) 2-Butyne
izes to acetaldehyde (CH_3CHO). Only two

Answer: (b)

carbons, so product is an aldehyde.

256. Which of the following compounds Explanation: Ammoniacal Cu_2Cl_2
will react with ammoniacal silver nitrate? is another terminal alkyne test. 1-
Butyne forms a red precipitate of copper(I)

(a) 1-Butene acetylide.

(b) 1-Butyne 259. Ozonolysis of 2-butyne gives

(a) Formic acid

(c) 2-Butene

(b) Propanoic acid

(d) 2-Butyne

(c) Acetic acid

Answer: (b) (d) Butanoic acid

Answer: (c)

Explanation: Ammoniacal AgNO_3

(Tollens' reagent) specifically tests for ter-

minal alkynes ($\equiv\text{C}-\text{H}$). Only 1-butyne has Explanation: Ozonolysis cleaves
this acidic proton. the triple bond. Symmetrical 2-butyne

($\text{CH}_3\text{C}\equiv\text{CCH}_3$) splits into two identical

257. Which of the following compounds acetic acid molecules (CH_3COOH).

does not react with ammoniacal AgNO_3 so-

lution? 260. Which alkyne yields propanoic

acid as the only product upon treatment

(a) Acetylene with ozone followed by hydrolysis?

- (a) 1-Butyne
- (b) 1-Butyne
- (b) 2-Hexyne
- (c) Propyne (c) 1-Pentyne
- (d) 2-Butyne (d) 3-Hexyne

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Answer: (d) (c) Chloroprene

(d) 1,3-butadiene

Explanation: Only one product

= symmetrical alkyne. Propanoic Answer: (c)

acid = 3C fragment. Two 3C frag-

ments joined by triple bond = 3-hexyne

(CH₃ CH₂ C≡CCH₂ CH₃). Explanation: Neoprene (polychloroprene) is a synthetic rubber made by free-

261. When acetylene is passed through radical polymerization of chloroprene (2-hot iron tube at 400°C it gives chloro-1,3-butadiene).

(a) Benzene 264. A compound, C₄ H₆ , reacts with bromine and forms a white precipitate with

(b) Toluene

ammoniacal silver nitrate solution. It re-

(c) o-Xylene acts with dilute H₂ SO₄ in the presence of mercuric sulfate to form 2-butanone. The

(d) Mesitylene compound could be

Answer: (a) (a) 1-Butyne

(b) 1-Butene

Explanation: Cyclic polymerization:

hot Fe tube, 400°C

3 HC≡CH -----→ C₆ H₆ (benzene). (c) 2-Butyne

Three acetylene molecules trimerize to form

one benzene ring. (d) 2-Butene

262. When propyne is passed through

Answer: (a)

hot iron tube at 400°C it gives

(a) Benzene Explanation: C₄ H₆ = alkyne/diene.

(b) Toluene White ppt with AgNO₃ = terminal alkyne (≡C-H). Hydration to 2-butanone confirms

(c) m-Xylene Markovnikov addition of water to 1-butyne.

(d) Mesitylene 265. Which of the following correctly ranks the cycloalkanes in order of increas-

Answer: (d) ring strain per methylene?

(a) cyclopropane < cyclobutane < cyclo-

Explanation: Trimerization of

hexane < cycloheptane

propyne ($\text{CH}_3\text{C}\equiv\text{CH}$) yields 1,3,5-

trimethylbenzene, commonly called mesityl-

lene. Each propyne contributes one $-\text{CH}_3$ group.

group.

(c) cyclopentane < cyclobutane < cy-

263. The monomer for Neoprene is

cyclopentane < cyclopropane

(a) Isoprene

(d) cyclopentane < cyclopropane < cy-

(b) acrylonitrile cyclobutane < cyclohexane

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Answer: (b) (a) trans-1,2-dimethylcyclopentane

(b) cis-1,2-dimethylcyclopentane

Explanation: Ring strain increases as

bond angles deviate from ideal 109.5° : cy-

(c) a mixture of trans and cis-1,2-

cyclohexane (strain-free chair) < cyclopentane

(slight) < cyclobutane (90°) < cyclo-

propane (60° , highest strain). (d) 1,1-dimethylcyclopentane

266. Which of the following has two

Answer: (b)

equatorial alkyl substituents in its most sta-

ble conformation?

Explanation: Catalytic hydrogenation

(a) 1,1-dimethylcyclohexane

= syn addition; both H atoms add to the

(b) cis-1,2-dimethylcyclohexane same face of the double bond, forcing the

two $-\text{CH}_3$ groups to the same side $\rightarrow$ cis

(c) cis-1,3-diethylcyclohexane isomer exclusively.

(d) cis-1,4-diethylcyclohexane 269. Which of the following is not a

possible reaction of a carbocation?

Answer: (c)

(a) addition of a nucleophile

Explanation: cis-1,3-disubstituted cy-

cyclohexane can place both substituents in (b) rearrangement to a more stable car-

equatorial positions (both up or both down)

down), avoiding 1,3-diaxial interactions.

(c) addition of a proton to form an alkane

267. Which one of the following is not a

metal catalyst for the hydrogenation of an (d) loss of a β -hydrogen to form an alkene alkene?

Answer: (c)

(a) Pd

(b) Pt Explanation: A carbocation (R_3C^+)

is electron-deficient and positively charged.

(c) Na It cannot bond with another positively

(d) Ni charged H^+ due to electrostatic repulsion.

Answer: (c) 270. Addition of HCl to 3-methyl-1-

pentene gives two products. One of these

is 2-chloro-3-methylpentane. What is the

Explanation: Pd, Pt, and Ni are stan- other product?

dard transition metal hydrogenation cata-

lysts. Na is an alkali metal used in dis- (a) 1-Chloro-3-methylpentane

solving metal reductions (e.g., Na/NH₃ for

trans-alkenes), not catalytic hydrogenation. (b) 3-Chloro-3-methylpentane

268. What is(are) the product(s) (c) 3-Chloro-2-methylpentane

in the Pd-catalyzed hydrogenation of 1,2-

dimethylcyclopentene? (d) 2-Chloro-2-methylpentane

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Answer: (b) 273. A compound, C₁₅H₂₄, is reacted

with excess hydrogen using a metal cat-

alyst. One equivalent of the compound

Explanation: Initial 2° carbocation at consumed three equivalents of hydrogen.

C-2 gives 2-chloro-3-methylpentane. 1,2- How many rings did the original compound

Hydride shift (H from C-3 to C-2) forms have?

more stable 3° carbocation at C-3 $\rightarrow$ 3-

chloro-3-methylpentane (major, rearranged (a) 1 only

product).

(b) 2 only

271. Predict which of the following (c) 3 only

alkenes reacts the fastest with HCl?

(d) None of these

(a) CH₃CH₂CH₂CH₂CH=CH₂

Answer: (a)

(b) cis-CH₃CH₂CH=CHCH₂CH₃

(c) trans-CH₃CH₂CH=CHCH₂CH₃ Explanation: Saturated acyclic:

C₁₅H₃₂. Missing 8 H's = 4 degrees of un-

(d) (CH₃)₂C=CHCH₂CH₃ saturation. 3 equivalents H₂ consumed = 3

π bonds. Remaining 1 degree = 1 ring.

Answer: (d)

274. The reaction of 1-butene with

bromine, Br₂, in aqueous solution gives pri-

Explanation: Rate depends on car- marily 1-bromo-2-butanol. Identify the nu-
bocation stability in the rate-determining cleophilic species in the reaction.

step. (d) forms a 3° carbocation (most sta-

ble) → fastest reaction. Others form 2° or (a) Br₂

1° carbocations.

(b) Br-

272. The hydroboration-oxidation re- (c) H₂O

action can be characterized as the _____

to an alkene. (d) HOBr

(a) anti-Markovnikov syn addition of wa- Answer: (c)

ter

Explanation: Bromonium ion forms

(b) anti-Markovnikov anti addition of wa-

first. In aqueous solution, H₂O (vast ex-
ter

cess) acts as the nucleophile, attacking the

(c) Markovnikov syn addition of water more substituted carbon → bromohydrin
(1-bromo-2-butanol).

(d) Markovnikov anti addition of water

275. Which brief statement most accu-

Answer: (a) rately describes why alkenes react the way
they do?

Explanation: Hydroboration: BH₃ (a) C=C double bonds are weak.

adds syn (same face). B attaches to less (b) A π bond is lost but a stronger σ bond
hindered C (anti-Markovnikov). Oxidation is gained.

replaces B with OH, retaining stereochem-

istry → syn, anti-Markovnikov hydration. (c) C=C double bonds are unstable.

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(d) C-C σ bonds are attacked by nucle- (b) CCl₂F₂

ophiles.

(c) CF₄

Answer: (b)

(d) CH_2Cl_2

Answer: (b)

Explanation: Thermodynamic driving

force: weak π bond ($\approx 264 \text{ kJ/mol}$) breaks;

two new stronger σ bonds ($\approx 350 \text{ kJ/mol}$ each) form. Net stabilization. Explanation: CFCs like CCl_2F_2 release $\text{Cl}\cdot$ radicals upon UV photolysis in the stratosphere. These radicals catalyze ozone

276. Which of the following is responsible for the destruction of ozone? $\text{Cl}\cdot + \text{O}_3 \rightarrow \text{ClO}\cdot + \text{O}_2$

gas is formed when chloroform is exposed to

light and air? 279. Which of the following reagents

cannot be used to prepare an alkyl chloride

(a) Mustard gas from an alcohol?

(b) Carbon monoxide (a) $\text{HCl} + \text{ZnCl}_2$

(c) Phosgene (b) SOCl_2

(d) Chlorine (c) NaCl

Answer: (c) (d) PCl_5

Answer: (c)

$h\nu$

Explanation: $\text{CHCl}_3 + \text{O}_2 \rightarrow \text{COCl}_2$

(phosgene, carbonyl chloride) + HCl . Phos-

gene is highly toxic; chloroform is stored in dark bottles to prevent this. Explanation: $-\text{OH}$ is a poor leaving

group. HCl/ZnCl_2 , SOCl_2 , and PCl_5 activate

the $-\text{OH}$ to make it a good

leaving group. NaCl alone cannot do this;

277. Freon-12, CCl_2F_2 , is used as a

refrigerant. no reaction occurs.

(a) Local anesthetic

280. 2-Propanol reacts with KBr and

(b) Dry-cleaning agent concentrated H_2SO_4 to give

(c) Refrigerant (a) 1-Bromopropane

(d) Disinfectant (b) 1,2-Dibromopropane

(c) 2-Bromopropane

Answer: (c)

(d) 2,2-Dibromopropane

Explanation: Freon-12 (dichlorodifluoromethane) is a CFC historically used as

a refrigerant and aerosol propellant. Now

banned due to ozone depletion. Explanation: H_2SO_4 protonates $-\text{OH}$

$\rightarrow \text{H}_2\text{O}$ (good leaving group). Br^- (from

278. Which of the following compounds is attacked via $\text{S}_\text{N}2$ or $\text{S}_\text{N}1$ to give 2-

bromopropane (isopropyl bromide). has been suggested as causing depletion of bromopropane (isopropyl bromide).

the ozone layer in the upper stratosphere?

281. The best way to prepare 1,2-

(a) CH₄ dibromoethane is to treat:

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(a) Ethylene with Br₂ CH₃ CH₂ CH₂ I → CH₃ CH₂ OCH₂ CH₂ CH₃

(ethyl propyl ether) + NaI.

(b) Acetylene with excess of HBr

284. 1-Bromobutane reacts with alco-

(c) Ethylene with excess of HBr holic KOH to mainly give

(d) 1,2-Dichloroethane with Br₂ (a) 1-Butene

Answer: (a) (b) 2-Butene

(c) 1-Butanol

Explanation: Direct electrophilic

addition of Br₂ across the C=C of (d) 2-Butanol

ethylene gives clean, quantitative 1,2-

dibromoethane. Acetylene + HBr gives Answer: (a) 1-Butene (Chemical correction)

geminal dihalide; ethylene + HBr gives

ethyl bromide.

Explanation: 1-Bromobutane is a 1°

282. Alkyl halides undergo alkyl halide; β-hydrogens only on C-2.

Elimination yields 1-butene, not 2-butene.

(a) Electrophilic substitution reactions The source document's answer (b) contains a chemical error.

(b) Electrophilic addition reactions

(c) Nucleophilic substitution reactions 285. 2-Bromobutane reacts with alcoholic KOH to mainly give

(d) Nucleophilic addition reactions

(a) 1-Butene

Answer: (c)

(b) 2-Butene

Explanation: The polar C-X bond (c) 1-Butanol

makes the carbon electron-deficient (δ⁺),

attracting nucleophiles. Halide is a good (d) 2-Butanol

leaving group → nucleophilic substitution

(S_N 1 or S_N 2). Answer: (b)

283. n-Propyl iodide reacts with

Explanation: Zaitsev's rule: elimina-

sodium ethoxide to give:

tion from 2-bromobutane gives the more

(a) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ substituted, more stable 2-butene as the major product over 1-butene.

(b) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_2\text{CH}_3$

286. Isopropyl bromide reacts with al-

(c) $\text{CH}_3\text{CH}_2\text{OCH}_3$ alcoholic KOH to give

(d) $\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$ (a) Propene

Answer: (b) (b) Isopropyl alcohol

(c) Propane

Explanation: Williamson ether

synthesis ($\text{S}_\text{N}2$): $\text{CH}_3\text{CH}_2\text{O}^- \text{Na}^+ +$ (d) n-Propyl alcohol

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Answer: (a) (c) Ethyl bromide

(d) 2-Bromopropane

Explanation: Alcoholic KOH =

strong base $\rightarrow$ E_2 elimination dominates.

$(\text{CH}_3)_2\text{CHBr} \rightarrow \text{CH}_3\text{CH}=\text{CH}_2$ (propene) + Answer: (a)

HBr.

287. 2,2-Dichloropropane reacts with Explanation: Aryl halides (bromoben- aqueous KOH to give zene) have C-Br bonds strengthened by resonance; Br^- is not released. Alkyl halides

(a) 2,2-Propanediol (b-d) readily release halide ions $\rightarrow$ AgBr precipitate.

(b) Propanal

290. Which of the following compounds

(c) Acetone

will give a white precipitate with alcoholic

(d) Propene AgNO_3 ?

Answer: (c) (a) Vinylbenzene

Explanation: Aqueous KOH $\rightarrow$ sub- (b) Chlorobenzene

stitution. Geminal diol $(\text{CH}_3)_2\text{C}(\text{OH})_2$

is unstable; loses H_2O to form acetone (c) Vinyl chloride $(\text{CH}_3\text{COCH}_3)$.

(d) Allyl chloride

288. 1,1-Dichloropropane reacts with

aqueous KOH to give Answer: (d)

(a) 1,1-Propanediol

(b) Acetaldehyde Explanation: Allyl chloride

$(\text{CH}_2=\text{CHCH}_2\text{Cl})$ forms a resonance-

(c) Propanone stabilized allylic carbocation, enabling rapid SN 1 and AgCl precipitation. Oth-

(d) Propyne ers have Cl on sp² carbon (unreactive).

Answer: Propanal (CH₃ CH₂ CHO); source291.

listsCarbylamine

(b) but chemically should

test involves be propanal

heating

a mixture of

Explanation: 1,1-Dichloropropane

(3C) → gem-diol → propanal (3C alde- (a) alcoholic KOH, methyl iodide, and hyde), not acetaldehyde (2C). The source sodium metal

document's answer contains a structural

error. (b) alcoholic KOH, methyl iodide, and

primary amine

289. Which of the following compounds

will not give a white precipitate with alco- (c) alcoholic KOH, chloroform, and pri- holic AgNO₃ ? mary amine

(a) Bromobenzene

(d) alcoholic KOH, methyl alcohol, and

(b) 1-Bromopropane primary amine

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Answer: (c) (c) Concentration of reagents

(d) Nature of the nucleophile

Explanation: Carbylamine reaction:

Δ (e) All of these

RNH₂ + CHCl₃ + 3KOH (alc.) → RNC

(isocyanide, foul odor) + 3KCl + 3H₂O. Answer: (e)

Test for 1° amines.

292. When chloroform is heated with Explanation: All factors matter: sub- aqueous NaOH, it gives strate structure (1°/2°/3°), solvent polarity

(protic vs. aprotic), reagent concentration,

(a) Formic acid

and nucleophile/base strength collectively

(b) Sodium formate determine the mechanism.

(c) Acetic acid 295. Which compound reacts most rapidly by an SN 1 mechanism?

(d) Sodium acetate

(a) Methyl chloride

Answer: (b)

(b) Isopropyl chloride

Explanation: $\text{CHCl}_3 + 4\text{NaOH} \rightarrow$ (c) Ethyl chloride

HCOONa (sodium formate) + 3NaCl +

$2\text{H}_2\text{O}$. Formic acid is initially formed but (d) tert-Butyl chloride

immediately neutralized by NaOH to the

Answer: (d)

formate salt.

293. Which alkyl halides react most Explanation: SN_1 rate depends on

readily by nucleophilic substitution? carbocation stability: $3^\circ > 2^\circ > 1^\circ > \text{CH}_3$.

(a) $\text{CH}_3\text{CH}_2\text{Cl}$ tert-Butyl chloride forms a stable 3° carbocation and reacts fastest via SN_1 .

(b) $\text{CH}_3\text{CH}_2\text{I}$

296. Which of the following compounds

(c) $\text{CH}_3\text{CH}_2\text{Br}$ would react most rapidly in an SN_2 reaction?

(d) $\text{CH}_3\text{CH}_2\text{F}$

(a) $\text{CH}_3\text{CH}_2\text{I}$

Answer: (b)

(b) $\text{CH}_2=\text{CH-I}$

Explanation: Reactivity order: $\text{RI} >$ (c) $(\text{CH}_3)_2\text{CHI}$

$\text{RBr} > \text{RCI} > \text{RF}$. I^- is the best leaving

group (weakest base, most stable anion, (d) $(\text{CH}_3)_3\text{CI}$ weakest C-X bond).

Answer: (a)

294. Which of the following factors influence whether a reaction will proceed by

Explanation: SN_2 requires backside attack; steric hindrance is the key factor. 1°

(a) Structure of the alkyl halide alkyl halides $(\text{CH}_3\text{CH}_2\text{I})$ are least hindered and react fastest. 3° (d) is too hindered for

(b) Solvent SN_2 .

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297. There are 8 isomers that have the (b) polar; δ^+ at carbon and δ^- at chlorine molecular formula $\text{C}_5\text{H}_{11}\text{Br}$. How many of

these are tertiary alkyl bromides? (c) polar; δ^- at carbon and δ^+ at chlorine

(a) 1 (d) ionic

(b) 2 Answer: (b)

(c) 3

Explanation: Cl is more electronegative

(d) 8 times than C (3.16 vs. 2.55). Cl pulls electron density toward itself $\rightarrow$ $\text{C}\delta^+ - \text{Cl}\delta^-$ polar

Answer: (a) covalent bond.

Explanation: Only one 3° 300. Which of the following will have bromide: 2-bromo-2-methylbutane, the lowest boiling point?

$(\text{CH}_3)_2\text{CBrCH}_2\text{CH}_3$. The Br is on a car-

(a) CH_3Cl

bon attached to three other carbons.

(b) CH_4

298. Which of the following statements

is correct for a saturated alkyl halide? (c) CH_2Cl_2

(a) the C-X bond results from overlap of

(d) CHCl_3

the s orbital of carbon and the p or-

bital of the halogen (X)

Answer: (b)

(b) the C-X bond results from overlap of

the p orbital of carbon and the p or- Explanation: CH_4 is non-polar with

bital of the halogen (X) only weak London dispersion forces (lowest

MW = 16). Chlorinated methanes are po-

(c) the C-X bond results from overlap of

lar (dipole-dipole) and have higher MW $\rightarrow$

the sp^3 orbital of carbon and the p

higher b.p.

orbital of the halogen (X)

(d) the C-X bond results from overlap of 301. Which of the following is not cor-

the sp^2 orbital of carbon and the p rect concerning substitution and elimina-

orbital of the halogen (X) tion reactions of alkyl halides?

Answer: (c) (a) The electrophile replaces the leaving group.

Explanation: Saturated alkyl halide (b) Compounds containing electron-carbon is sp^3 hybridized; the σ bond forms donating groups bonded to an sp^3 by overlap of $\text{C}(\text{sp}^3)$ with a halogen p or- hybridized carbon undergo substitution and elimination reactions.

299. Which of the following best de- (c) The electronegative atom is replaced scribes the carbon-chlorine bond of an alkyl by another atom or group in substitution reactions.

(a) nonpolar; no dipole (d) (a) and (b)

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Answer: (d) than ordinary alkyl ones. Vinyl carbocations ($R_2C=CR^+$) are exceptionally unstable. Order: 2° allylic $>$ 2° $>$ vinyl.

Explanation: (a) is false: a nucleophile (not electrophile) replaces the leaving group. 304. Which of the following species is most reactive in an S_N2 reaction? (b) CH_3CH_2Br due to electron-withdrawing halogens, not electron-donating groups. Both incorrect. (a) CH_3CH_2Cl

302. Which of the following statements concerning S_N2 reactions of alkyl halides is not correct? (c) CH_3CH_2I

(a) The rate of reaction depends on the concentration of the nucleophile. (d) CH_3CH_2F

Answer: (c)

(b) The rate of reaction depends on the concentration of the alkyl halide. Explanation: S_N2 rate depends on leaving group ability: $I^- > Br^- > Cl^- >$

(c) The rate of reaction of a particular alkyl bromide depends on the steric hindrance of the carbon of the $C-Br$ bond. F^- is the weakest base, most stable anion, best leaving group $\rightarrow$ fastest S_N2 .

accessibility of the carbon of the $C-Br$ bond. 305. Which of the following is the best leaving group? (d) All alkyl iodides react more rapidly than all alkyl chlorides. (a) F^-

Answer: (d) (b) Cl^-

(c) Br^-

Explanation: Not universally true: a bulky 3° alkyl iodide is severely hindered (d) I^- and reacts slower via S_N2 than an unhindered 1° alkyl chloride. Steric effects can override leaving group ability.

303. Which of the following correctly reflects relative stabilities of carbocations? weak bases (stable anions). I is the largest, most polarizable, and most stable halide ion (a) 3° allylic $>$ 2° $>$ 1° benzylic — the best leaving group among halogens. (b) methyl $>$ 2° benzylic $>$ 3° 306. Which of the following is the strongest nucleophile in an aqueous solution? (c) 3° benzylic $>$ vinyl $>$ 1°

Explanation: Best leaving groups are weak bases (stable anions). I is the largest, most polarizable, and most stable halide ion

(a) 3° allylic $>$ 2° $>$ 1° benzylic — the best leaving group among halogens.

(b) methyl $>$ 2° benzylic $>$ 3° 306. Which of the following is the strongest nucleophile in an aqueous solution? (c) 3° benzylic $>$ vinyl $>$ 1°

(a) 3° allylic $>$ 2° $>$ 1° benzylic — the best leaving group among halogens.

(d) 2° allylic $> 2^\circ >$ vinyl (a) H_2O

Answer: (d) (b) Cl^-

(c) Br^-

Explanation: Resonance-stabilized allylic/benzylic carbocations are more stable (d) I^-

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Answer: (d) (c) $\text{Rate} = k[\text{Alkyl Halide}]$

(d) $\text{Rate} = k_1 [\text{Alkyl Halide}] +$

Explanation: In polar protic solvents $k_2 [\text{Nucleophile}]$

(water), larger ions are less solvated (less

“caged” by H-bonds). I^- , being large and Answer: (c)

highly polarizable, is the strongest nucle-

ophile in water. Explanation: $\text{SN} 1 = \text{unimolecu-}$

307. Which of the following is the best 1 $^\circ$ nucleophilic substitution. The rate-determining step (carbocation formation)

involves only the alkyl halide. Rate is first-

(a) I^- order, independent of nucleophile concentration.

(b) CH_3SCH_3

310. Which of the following factors has

(c) CH_3OCH_3 no effect on the rate of $\text{SN} 1$ reactions?

(d) Cl^- (a) the nature of the alkyl halide

Answer: (a) (b) the nature of the leaving group

(c) the concentration of the alkyl halide

Explanation: I^- is a strong nucle-

ophile in water (large, polarizable, weakly (d) the concentration of the nucleophile

solvated). Neutral ethers (b, c) are weak

nucleophiles; Cl^- is more tightly solvated Answer: (d)

than I^- .

Explanation: The nucleophile does

308. Which of the following compounds

not participate in the rate-determining step

will undergo an $\text{SN} 2$ reaction most readily?

(RDS) of $\text{SN} 1$. The RDS is solely the ion-

(a) $(\text{CH}_3)_3\text{CCH}_2\text{I}$ ionization of the alkyl halide to form a carbocation.

(b) $(\text{CH}_3)_3\text{CI}$

311. Which is the most reactive alkyl

(c) $(\text{CH}_3)_2\text{CHI}$ halide in an $\text{SN} 1$ reaction?

(d) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{I}$ (a) CH_3F

Answer: (d) (b) CH_3Br

(c) CH_3I

Explanation: $\text{S}_\text{N}2$ requires unhindered backside attack. (d) is a 1° alkyl iodide with the least steric hindrance. (a) is neopentyl

Answer: (c)

(extremely hindered); (b) is 3° ; (c) is 2° .

309. Which of the following is the rate law for $\text{S}_\text{N}1$ mechanisms? group, $\text{S}_\text{N}1$ rate depends on leaving group ability. C-I bond is weakest; I is the best leaving group $\rightarrow \text{CH}_3\text{I}$ reacts fastest via

(a) $\text{Rate} = k[\text{Alkyl Halide}][\text{Nucleophile}]$

(b) $\text{Rate} = k[\text{Nucleophile}]$

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312. Which of the following halides is most reactive in an $\text{E}2$ reaction with sodium methoxide? Answer: (d)

Explanation: Strong base (e.g., alkoxide) + 2° alkyl halide $\rightarrow$ bimolecular elimination ($\text{E}2$) predominates. Concerted removal of $\beta\text{-H}$ and departure of Br^- .

(a) $(\text{CH}_3)_3\text{CCH}_2\text{I}$

(b) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{I}$

(c) $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$

(d) $(\text{CH}_3)_2\text{CHCH}_2\text{Cl}$

315. Which of the alkyl chlorides listed below undergoes dehydrohalogenation in the presence of a strong base most rapidly?

Answer: (b) (a) 1-chlorobutane
(b) 2-chlorobutane

Explanation: (b) is a 1° alkyl iodide — unhindered and with the best leaving group (I $^-$) among the options. (a) is neopentyl (no $\beta\text{-H}$ accessible for $\text{E}2$). (c, d) have poorer leaving groups. Answer: (c)

313. Which of the following halides is least reactive in an $\text{E}2$ reaction with sodium methoxide? $\text{E}2$ elimination most rapidly with strong bases; the product alkene is highly substituted and thermodynamically stable, lowering the activation energy.

(a) $(\text{CH}_3)_3\text{CCH}_2\text{I}$

(b) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{I}$

3 2 2 2

(c) $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$

316. Predict the two most likely mech-

anisms which occur when 2-bromobutane is

(d) $(\text{CH}_3)_2\text{CHCH}_2\text{Cl}$ treated with sodium ethoxide in ethanol.

Answer: (a) (a) $\text{S}_\text{N}1$ and $\text{E}1$

(b) $\text{S}_\text{N}2$ and $\text{E}2$

Explanation: $(\text{CH}_3)_3\text{CCH}_2\text{I}$ is a

(c) $\text{S}_\text{N}2$ and $\text{E}1$

neopentyl halide. The β -carbon is quater-

nary (no β -hydrogens), and the backside is (d) $\text{E}1$ and $\text{S}_\text{N}1$

extremely sterically hindered — virtually

unreactive toward $\text{E}2$. Answer: (b)

314. Dehydrohalogenation of 2-

Explanation: NaOEt is a strong base

bromobutane in the presence of a strong

and good nucleophile reacting with a 2°

base proceeds via which of the following

alkyl halide. These conditions favor bi-

mechanistic pathways?

molecular pathways: $\text{S}_\text{N}2$ (substitution)

(a) $\text{S}_\text{N}1$ and $\text{E}2$ (elimination) compete.

(b) $\text{S}_\text{N}2$ 317. Predict the two most likely mech-

anisms which occur when 2-iodohexane is

(c) $\text{E}1$ heated in ethanol.

(d) $\text{E}2$ (a) $\text{S}_\text{N}2$ and $\text{S}_\text{N}1$

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(b) $\text{E}1$ and $\text{E}2$ (a) Ethane

(c) $\text{S}_\text{N}2$ and $\text{E}2$ (b) 3-Ethyl-3-pentanol

(d) $\text{E}1$ and $\text{S}_\text{N}1$ (c) Propanoic acid

Answer: (d) (d) All of these

Answer: (d)

Explanation: Heating a 2° alkyl halide

in a weak, neutral, protic solvent (ethanol)

without strong base favors unimolecular Explanation: Grignard versatility: +

pathways: slow carbocation formation $\rightarrow \text{H}_2\text{O} \rightarrow$ ethane (a); + diethyl ketone $\rightarrow 3$ -

$\text{S}_\text{N}1$ and $\text{E}1$ mixture. ethyl-3-pentanol (b); + CO_2 then $\text{H}_3\text{O}^+ \rightarrow$

propanoic acid (c). All are possible.

318. Grignard reagents do not show any

reaction with 320. Which is the best reagent to ac-

complish the following conversion?

(a) Alkoxyalkanes

$(\text{CH}_3)_2\text{CHBr} \rightarrow (\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$

(b) Alkanones

(a) Conc. H_2SO_4

(c) Alkyl alkanoates

(b) Na

(d) Acyl halides

(c) Conc. HCl

Answer: (a)

(d) Mg, then appropriate electrophile

Explanation: Grignard reagents Answer: (d)

(RMgX) are strong nucleophiles/bases and

react vigorously with carbonyl compounds

(b, c, d). Ethers (alkoxyalkanes) are inert Explanation: Carbon-carbon bond

to Grignard reagents — this is why ethers formation requires organometallic chem-

are used as solvents for their preparation. istry. Form the Grignard reagent

$(\text{CH}_3)_2\text{CHMgBr}$ using Mg metal, then react

319. The Grignard reagent, with an appropriate electrophile to extend

$\text{CH}_3\text{CH}_2\text{MgBr}$, can be used to prepare the carbon chain.

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CH102 Practice Exam #1 WAY longer than actual exam, answers are at the end

True/False

Indicate whether the sentence or statement is true or false.

- ____ 1. Organic chemistry can be defined as the chemistry of carbon-containing compounds.
- ____ 2. Organic compounds can contain other elements such as oxygen or nitrogen.
- ____ 3. Organic compounds are always derived from a living source.
- ____ 4. An unlimited number of carbon atoms can bond together to form chains.
- ____ 5. Most organic compounds contain ionic bonds.
- ____ 6. There are more inorganic compounds than organic compounds.
- ____ 7. Most inorganic compounds have a higher melting point than organic compounds.
- ____ 8. Inorganic compounds are never found in living systems.
- ____ 9. Isomers differ from each other in the number of neutrons present in each.
- ____ 10. Isomers always have the same empirical (molecular) formula.
- ____ 11. Isomers have identical physical and chemical properties.
- ____ 12. Carbon-carbon bonds are usually strong and covalent.
- ____ 13. A characteristic of the alkanes is that they burn in air.
- ____ 14. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2$ is the formula for a saturated hydrocarbon.
- ____ 15. Alkenes must have at least two carbon atoms.
- ____ 16. Alkenes only have one double bond.
- ____ 17. The addition of bromine (Br_2) to an alkene results in an alkane because one bond of the multiple bond is broken.
- ____ 18. One of the halogenation reactions occurs when a halogen, a member of group VIIA, reacts with alkene.
- ____ 19. Cyclic compounds do not undergo halogenation reactions.
- ____ 20. Markovnikov's rule states that when a hydrogen and a different molecule (OH^+ , Br^+) add to a double bond the hydrogen goes to the end of the double bond that already had more hydrogens.
- ____ 21. An alkene with one multiple bond can be converted to an alkane by hydration.
- ____ 22. Polymers are compounds that are composed of repeating units chemically bound to each other.
- ____ 23. The chemical properties of alkynes are very different from those of alkenes.
- ____ 24. 2-butyne can exist as *cis*- and *trans*- isomers.
- ____ 25. Two moles of hydrogen gas would be required to convert one mole of 2-butyne into butane.
- ____ 26. Benzene is an alkene with more than one multiple bond.
- ____ 27. Phenyl is the name given to benzene when it is a substituent of a larger molecule.
- ____ 28. Aromatic compounds dissolve well in a nonpolar solvent.

Multiple Choice

Identify the letter of the choice that best completes the statement or answers the question.

- ____ 29. What is the reason why there are so many organic compounds?
 - a. Organic compounds are produced by living things.
 - b. There are cyclic organic compounds, and none of the inorganic compounds are cyclic.

- c. Carbon atoms can form stable covalent bonds with other carbon atoms.
d. Organic compounds can be larger than inorganic compounds.

30. Which applies to the manner in which carbon bonds in alkanes?

- a. The bonds are formed with carbon using an sp^3 configuration.
b. The bonds between carbon and other atoms are ionic.
c. carbon to carbon bonds are expected to be double bonds.
d. The bonds formed by carbon all use p orbitals.

31. Which statement referring to functional groups is correct? Functional groups are

- a. where nothing happens in a chemical reaction
b. the parts of molecules used to classify organic molecules
c. where most chemical reaction occur
d. both b and c

32. Identify the correct statement: The alkanes represent one family of organic compounds composed of carbon compounds

- a. in which the halogens are a major component
b. in which there is at least one double bond between carbons
c. that are composed only of carbon and hydrogen
d. in which hydrogens can be bonded to other hydrogens

33. Which of the following formulas represents an organic compound?

- a. CH_4
b. $CH_3CH_2CH_3$
c. $CH_3CH_3CH_3$
d. they all do

34. Which of the following is an alcohol?

- a. CH_3CH_2COOH
b. CH_3CH_2OH
c. CH_3COCH_3
d. CH_3CH_2Br

35. What is meant by the term "saturated" when describing hydrocarbons?

- a. The hydrocarbons are saturated with water and are wet.
b. The hydrocarbons are saturated with carbons in its parent chain.
c. The hydrocarbons are saturated with hydrogens and have only single bonds between carbons.
d. The compound is pure

36. The naming of organic compounds is specific. Which is an incorrect statement?

- a. The smaller of the compounds have been known long enough to have common names.
b. The naming system of organic compounds is set out by IUPAC rules.
c. The naming system for hydrocarbons requires the identification of the longest carbon chain.
d. The naming system for organic compounds requires and IUPAC name and a common name.

37. Identify the correct statement. Cycloalkanes a. can have as few as two carbons.

- b. are unsaturated hydrocarbons.
c. have more hydrogens than straight chain alkanes with the same number of carbons
d. these are all wrong

38. Which of the following terms applies to the alkanes?

- a. hydrophobic
b. hydrophilic
c. soluble
d. bases

39. What is a characteristic shared by the alkanes? a. solids

- b. polar
c. flammable
d. colored

40. Which of the following is a used for alkanes? a. fuel

- b. lubricant
c. solvent
d. all of the above

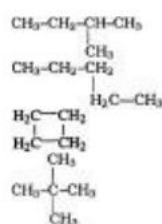
____ 41. Which of the following formulas represents a saturated hydrocarbon?

- a. C_2H_6
- b. C_2H_4
- c. C_3H_6
- d. C_3H_{10}

____ 42. All of the members of the alkane series of hydrocarbons have the general formula

- a. C_nH_{2n}
- b. C_nH_n
- c. C_nH_{2n+2}
- d. C_nH_{2n-2}

43. Which of the following structural formulas represents an unbranched alkane? a.



- b.
- c.
- d.

____ 44. Which of the following compounds exhibits structural isomerism?

- a. propane
- b. methane
- c. butane
- d. ethane

____ 45. How many structural isomers are possible for butane (molecular formula C_4H_{10})?

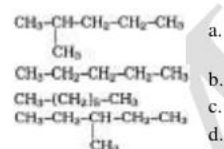
- a. 2
- b. 3
- c. 4
- d. 5

____ 46. Which of the following compounds is a

structural isomer of 2-methylbutane?

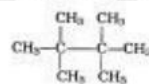
- a. propane
- b. w-methylpropene
- c. butane
- d. pentane

____ 47. A structural isomer of $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}-\text{CH}_3$ is



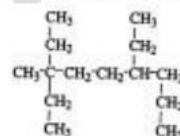
- a.
- b.
- c.
- d.

____ 48. The name for the following structure will end with



- a. ethane
- b. butane
- c. hexane
- d. octane

____ 49. Which is the longest carbon chain?



- a. 9
- b. 6
- c. 7
- d. 10

____ 50. The IUPAC name for $\text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}_3$ is

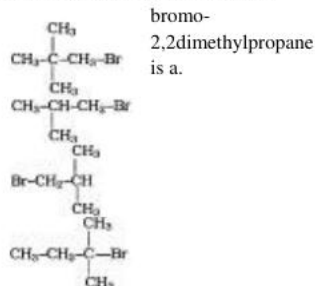
- a. 2-ethylbutane
- b. 2-methylbutane
- c. 3-methylpentane
- d. 3-ethylbutane

____ 51. The compound $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$ has

the IUPAC name

- 4-isopropyloctane
- 5-isopropyloctane
- 5-sec-butyloctane
- 4-propyloctane

52. The structural formula of 1-



b.

c.

d.

53. A general formula for cycloalkanes is

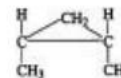
- C_nH_{2n+2}
- C_nH_{2n}
- C_nH_{2n-2}
- C_nH_{2n-4}

54. The IUPAC name of $CH_3CH_2CH_2CH(CH_2CH_3)CH_2CH_3$ contains an ethyl group located at position

- 1
- 2
- 3
- 4

55. A correct IUPAC name for $H_2C(Br)-CH(CH_3)-CH_2CH_3$ is

- 1-bromo-3-ethyl-1-methylcyclopropane
- 1-bromo-2-ethyl-1-methylcyclopropane
- 1-bromo-1-ethyl-2-methylcyclopropane
- 2-bromo-1-ethyl-2-methylcyclopropane



56. The component 1,2-dimethylcyclopropane. a. cis

- trans
- tis
- crans

57. An alkane with 10 carbon atoms is a _____ at room temperature. a. solid

- liquid
- gas
- not possible to tell

58. Alkanes are _____ in water and _____ than water.

- insoluble, less dense
- soluble, less dense
- insoluble, more dense
- soluble, more dense

59. Complete combustion of pentane produces a. carbon monoxide and water

- carbon dioxide and water
- carbon dioxide and carbon monoxide
- carbon monoxide and carbon

60. Alkanes react readily with a. sodium hydroxide

- hydrochloric acid
- sulfuric acid
- none of the above

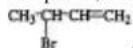
61. Name a difference between a saturated and an unsaturated hydrocarbon.

- Saturated hydrocarbons are composed of only carbon and hydrogen, and unsaturated hydrocarbons include other atoms than just carbon and hydrogen.
- Saturated hydrocarbons do not contain multiple bonds between carbons, but unsaturated hydrocarbons do contain multiple bonds.
- Unsaturated hydrocarbons are flammable, but saturated hydrocarbons aren't flammable.
- Saturated hydrocarbons are essentially insoluble. Unsaturated hydrocarbons are soluble.

____ 62. What number would be used to indicate the double bond position in the IUPAC name for $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$?

- a. 1
- b. 2
- c. 3
- d. 4

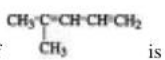
____ 63. In the IUPAC name for the following compound, the -Br group is located at position



- a. 1
- b. 2
- c. 3
- d. 4

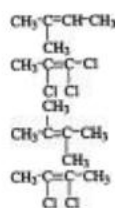
____ 64. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$ is

- a. 3-ethyl-1-butene
- b. 2-ethyl-2-butene
- c. 3-ethyl-3-butene
- d. 2-ethyl-1-butene

- ____ 65. The IUPAC name of  is
- 2-methyl-1,4-pentadiene
 - 2-methyl-2,4-dipentene
 - 4-methyl-1,3-pentadiene
 - 4-methyl-2,4-pentadiene

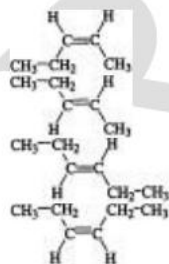
- ____ 66. A correct IUPAC name for  is
- 5-bromo-1,3-cyclohexadiene
 - 2-bromo-1,3-cyclohexadiene
 - 2-bromo-1,4-cyclohexadiene
 - 3-bromo-1,5-cyclohexadiene

- ____ 67. Which of the following compounds could exist in the form of two geometric isomers? a.



-
-
-
-

- ____ 68. Which of the following compounds is *trans*-3hexene?



-
-

-
-

- ____ 69. Which of the following represents an addition reaction?

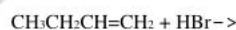
- $HX + C_4H_8 \rightarrow C_4H_9X$
- $X_2 + C_3H_6 \rightarrow C_3H_5X + HX$
- $C_4H_8 \rightarrow C_4H_6 + H_2$
- more than one response is correct

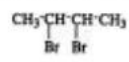
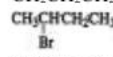
- ____ 70. Select the major product that would result from the reaction:



- 
- $CH_3CH_2CH_2-OH$
- $CH_3CH_2CH_3$
- $CH_3CH_2CH_2-SO_4$

- ____ 71. Select the major product that would result from the reaction:

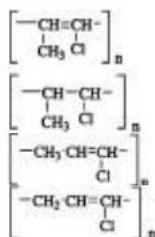


- 
- $CH_3CH_2CH_2CH_3$
- 
- $CH_3CH_2CH_2CH_2Br$

- ____ 72. The conversion of cyclohexene to cyclohexane requires the reagent(s)

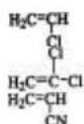
- HCl
- H_2O and H_2SO_4
- H_2 and H_2SO_4
- H_2 and Pt

- ____ 73. The addition polymer produced from $CH_3-CH=CH-Cl$ is



- a.
b.
c.
d.

74. The monomer of teflon (polytetrafluoroethene) is a.



- b.
c.
d. $\text{CF}_2=\text{CF}_2$

75. Which is the formula for an alkene?

- a. CH_3CHCH_2
b. $\text{CH}_3\text{CH}_2\text{CH}_2$
c. $\text{CH}_3\text{CH}_2\text{CH}_2$
d. all of these.

76. Which is the formula for an alkyne?

- a. $\text{CH}_3\text{CH}_2\text{CCH}_2$
b. $\text{CH}_3\text{CH}_2\text{C}_2\text{CH}_3$
c. $\text{CH}_3\text{CH}_2\text{CHCH}$
d. $\text{CH}_3\text{CH}_2\text{CCH}_2$

77. Which is a difference between butane and butene?

- a. butane burns and butene does not
b. the presence of a double bond

- c. they are isomers
d. the presence of a triple bond

78. Which is a difference between butene and cyclobutene?

- a. They are isomers.
b. Cyclobutene has 2 double bonds, butene does not.
c. The phase variance of the infrared transconductance is asymmetric.
d. Cyclobutene has fewer hydrogens than butene.

79. Which is a difference between butyne and cyclobutyne?

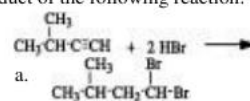
- a. Cyclobutyne can't exist!
b. Butyne's multiple bond is not as strong as that of cyclobutyne.
c. Cyclobutyne burns much hotter than butyne because of the greater unsaturation.
d. Both b and c are differences between the molecules.

80. The correct IUPAC name for the compound

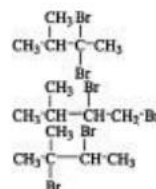
$\text{CH}_2=\text{C}=\text{C}-\text{CH}_2-\text{CH}_2-\text{Br}$ is a.

- 4-bromopentyne
b. 1-bromo-2-pentyne
c. 1-bromo-3-pentyne
d. 5-bromo-2-pentyne

81. Select the product of the following reaction:



- b.
c.
d.



___ 82. Acetylene (ethyne gas) is commercially useful

as:

- a starting material for plastics
- a fuel for torches
- an ingredient in pesticides
- a component in paint formulations

b.

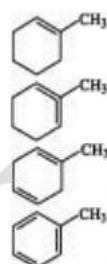
c.

d.

___ 83. What is the characteristic of aromatic compounds that is responsible for them being named aromatic compounds?

- The compounds have a pleasant smell.
- These compounds contain a benzene ring.
- It must contain a hydrocarbon chain that is either saturated or unsaturated and at least 3 carbons long.
- There is more than one correct response.

___ 84. Which of the following compounds is not considered to be aromatic? a.



___ 85. Which of the following structures is wrong?

a.

b.

c.

d.

42. C
43. B

64. D
65. C

TRUE/FALSE

Answer Section

21. T

22. T

1. T

2. T

3. F

4. T

5. F

6. F

7. T

8. F

9. F

10. T

11. F

12. T

13. T

14. F

15. T

16. F

17. T

19. F

23. T

24. F

25. T

26. T

27. T

28. T

MULTIPLE CHOICE

29. C

30. A

31. D

32. C

33. D

34. B

35. C

36. D

37. D

38. A 60. D 82. B 18. T 39. C 61. B 83. B

40. D

44. C

45. A

46. D

47. D

48. B

49. A

50. C

51. A

52. A

53. B

54. C

55. B

56. A

57. B

58. A

59. B

62. B

66. A

67. D

68. C

69. A

70. A

71. C

72. D

73. B

74. D

75. A

76. C

77. B

78. D

79. A

80. D

81. B

84. C

20. T 41. A 63. C 85. B

ROUTING

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THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE *Department of Chemistry*

CHE 515–Wood Pulp and Paper Chemistry

COURSE PARTICULARS

Course Code: CHE515

Course Title: Wood Pulp and Paper Chemistry

No. of Units: 2

Course Duration: Two hours of theory per week for 15 weeks. **Status:** Compulsory

Course Email Address: Course

Webpage:

Prerequisite: NIL

COURSE INSTRUCTORS

Professor L. Lajide

Room 3, 3rd Floor,

Academic Building, Obanla

Federal University of Technology, Akure, Nigeria.

Phone: +2348034721396

Email: llajide@futa.edu.ng

and

O. O.Oluwasina (PhD)

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Dept. of Chemistry, Oba Kekere,

Federal University of Technology, Akure, Nigeria.

Phone: +2348035462618

Email: oooluwasina@futa.edu.ng

COURSE DESCRIPTION

This course seeks to introduce the undergraduate students in the department of chemistry and other related fields to the science and technology involved in the utilization of wood as chief raw materials for the production of wood pulp, paper, wood- and pulp- based value-added products. Basic topics to be covered in this course include: an overview of wood production from different types of forests; wood as renewable natural resource; wood chemistry; wood pulping processes; pulp bleaching; paper production and recycling.

COURSE OBJECTIVES

The objectives of this course are to:

- introduce students to the use of wood, non-woody and agricultural waste as viable industrial raw material;
- introduce students to the basic chemistry of wood and separation of basic wood components, through various chemical, mechanical and chemo mechanical methods; and
- take the students through various chemical reaction methods for wood components modification and their uses.

COURSE LEARNING OUTCOMES / COMPETENCIES

Upon successful completion of this course, the student will be able to:

(Knowledge based)

- develop know-how to provide technology for the adoption of alternative raw materials for industrial purposes as they face future industrial challenges;
- know the basic chemical structures of wood and being able to separated them;
- learn the details of the chemistry of pulping and bleaching; and
- understand the purpose and procedures for chemical modification of wood based raw materials for industrial uses

GRADING SYSTEM FOR THE COURSE

This course will be graded as follows:

Class Attendance	05%
Assignments	15%
Test(s)	20%
<u>Final Examination</u>	<u>60%</u>
<u>TOTAL</u>	<u>100%</u>

GENERAL INSTRUCTIONS

Attendance: It is expected that every student will be in class for lectures. Attendance records will be kept and used to determine each person's qualification to sit for the final examination. In case of illness or other unavoidable cause of absence, the student must communicate as soon as possible with any of the instructors, indicating the reason for the absence.

Academic Integrity: Violations of academic integrity, including dishonesty in assignments, examinations, or other academic performances are prohibited. You are not allowed to make copies of another person's work and submit it as your own; that is plagiarism. All cases of academic dishonesty will be reported to the University Management for appropriate sanctions in accordance with the guidelines for handling students' misconduct as spelt out in the Students' Handbook.

Assignments and Group Work: Students are expected to submit assignments as scheduled. Failure to submit an assignment as at when due will earn you zero for that assignment. Only under

extenuating circumstances, for which a student has notified any of the instructors in advance, will late submission of assignments be permitted.

Code of Conduct in Lecture Rooms and Laboratories: Students should turn off their cell phones during lectures. Students are prohibited from engaging in other activities (such as texting, watching videos, etc.) during lectures. Food and drinks are not permitted in the laboratories.

READING LIST

- ¹Robert, J. C., (1991). Paper Chemistry. Published by Blackie Academic and Professional, Chapman and Hall, WesterCleddens Road, Bishopbriggs, Glasgow.
- ¹Shukla, S.D. and Pandey, G.N., (1979): Published by Vikas Publishing House PVT LTD. New Delhi. 268p.
- ³Sharma, B.K. (2006): Industrial Chemistry. KRISHA Prakashan Media (P) Ltd., Darya Ganj, Delhi. 1191p
- ⁴Philip, J. C., (2002): Survey of Industrial. Kluwe Academic / Plenum Publishers, New York, Boston, Dordrecht, London, Moscow.
- ⁴Martin, B. H., (2006): Handbook of Chemical Technology and Pollution Control. 3rd Edition. Elsevier Science and Technology Book

Legend

- 1- Available in the University Library 2- Available in Departmental/School Libraries 3- Available on the Internet.
4- Available as Personal Collection
5- Available in local bookshops.

COURSE OUTLINE

Week	Topic	Remarks
1	Introduction and Course Overview Forest Conservation, Exploitation and Afforestation	During this first class, the students are introduced to course and were taught on the importance and adverse effects of conservation of forest, exploitation and afforestation.
2	Wood : A renewable resource - Different types of wood - Paper making characteristics of soft- and hard-wood - Sources of promising non-woody species as alternative fiber source	Students will know types of wood that exist and what makes a wood to be good raw material for paper making. Students should be able to provide alternative raw materials for existing

		ones in their future industrial challenges
4	Wood Chemistry • Holocellulose(cellulose and hemicellulose) • Lignin • Extractives • Inorganic matters	Students would be taught the different chemical compositions of wood, their sub-monomer and their functions as related to wood.
5	Wood Preparation for Pulping • Wood Preparation • debarking • Chipping • Sieving	Wood harvesting , transportation, drying, debarking, grinding, sieving and storage as a method for preparing wood for pulping would be introduced to students
6&7	Wood pulping processes :Part A Mechanical pulping • Stone ground-wood • Chip refiner ground-wood • Thermo-mechanical pulping • Chemo-mechanical pulping	Students will be expected to understand the concept of mechanical pulping, differences in the various methods and the end use of the obtained products.
		MID-SEMESTER TEST
8& 9	Wood pulping processes :Part B Chemical pulping • Sulphite pulping • Kraft pulping • Alkaline pulping • Organosolv pulping	Students will be expected to understand the concept of chemical pulping, differences in the various methods and the end use of the obtained products.
10 &11	Pulping Screening and Cleaning Chemical recovery in Kraft pulping	To teach students the method of isolating and purifying pulp fiber after pulping processes and the environmental methods of spent liquor treatment for recovery of
		useful chemical and reuseable of the recovered chemicals

12 & 13	Pulp Bleaching • Mechanical pulp bleaching • Chemical pulp bleaching	Students are expected to know the: the aim of bleaching; chemistry of chemical reaction of bleaching technology; and the end use of product for which a particular bleaching method must be used
14	Revision and Test	Students are expected to ask questions on aspect of the course that is not well understood and test would be given to evaluate the student
15	Solving the test questions	Test question would be treated in order to clear any grey area and to prepare the students for the coming Examination.

Paper 7 of 7 — BAMBOY 100L 2nd semCHE 102 (BAMBOY)Che 102 BAMBOY .pdf

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THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
DEPARTMENT OF CHEMISTRY

SECOND SEMESTER CBE

2018/19 SESSION

COURSE CODE: CHE 102

COURSE TITLE: GENERAL CHEMISTRY II

INSTRUCTIONS:

A. ANSWER SIXTY QUESTIONS (60) ONLY

B. THERE ARE FIVE (5) SECTIONS, ANSWER AT LEAST TWELVE (12) QUESTIONS FROM EACH SECTION

TIME ALLOWED: 30 MINS

SECTION 1

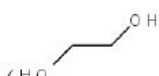
1. Which of the following is otherwise known as grain alcohol?
 - a) Propanol
 - b) Butanol
 - c) Ethanol+++
 - d) Methanol
2. Alcohols have higher boiling points than do alkanes of similar molecular weight because
 - a) Oxygen is present in alcohols which is deficient in alkanes
 - b) Strong ionic bonding is present in alcohols while covalent bonding is present in alkanes
 - c) Interaction between oxygen of the hydroxyl group with other molecules of alcohols which is absent in alkanes+++
 - d) Alkanes are largely hydrocarbons which mostly are in gaseous states
3. The carbon atom bonded to a hydroxyl group is called

- a) *carbol*
 - b) *carbocation*
 - c) *carbinol+++*
 - d) *carbene*
4. Glycols have——— hydroxyl group(s).
- a) *one*
 - b) *two+++*
 - c) *three*
 - d) *four*
5. Which of the alcohols is otherwise known as wood spirit?
- a) *Ethanol*
 - b) *Propanol*
 - c) *Methanol+++*
 - d) *Butanol*
6. Which of these is not another name for 2-propanol?
- a) *Isopropanol*
 - b) *Isopropyl alcohol*
 - c) *Rubbing alcohol*
 - d) *Vinyl propanol+++*
7. Alcohols react with active metals (Na, K, etc.) to produce ——gas.
- a) *oxygen*
 - b) *hydrogen+++*
 - c) *nitrogen*
 - d) *helium*
8. Which of these chemicals would you recommend for a person who has a dry skin?
- a) *Ethane 1, 2-diol*
 - b) *Propane 1, 2, 3-triol+++*
 - c) *Glycol*
 - d) *Glyoxal*
9. The green engine coolant in an automobile radiator will likely contain one of these

- a) Glycerol
- b) Ethane 1, 2-diol+++
- c) Ethylene
- d) 2-propanol

10. An example of a polyhydric alcohol is

- a) Glycerol
- b) Glycol
- c) Cellulose+++
- d) Polymer

11. The formation of ethylene glycol () from ethylene (= $\text{C}=\text{C}$) is an example of

- a) Oxidation+++
- b) reduction
- c) dehydration
- d) hydrogenation

12. Which of these is a secondary alcohol?

- a) $\text{CH}_3\text{CH}_2\text{OH}$
- b) $\text{CH}_3(\text{CH}_3)\text{CHOH}+++$
- c) $\text{CH}_3(\text{CH}_3)_2\text{COH}$
- d) $\text{CH}_3\text{CH}_2\text{COH}$

13. An example of a compound that contains a carbonyl group is

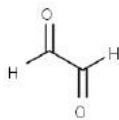
- a) $\text{H}_2\text{CO}+++$
- b) CH_3OH
- c) CH_2CH_2
- d) CH_3OCH_3

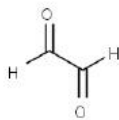
14. One mole of NaBH_4 reacts with 1 mole of acetone, $\text{CH}_3\text{C}(\text{O})\text{CH}_3$, to give, after treatment with aqueous acid,–

- a) Propanoic acid
- b) 2-propanone
- c) 2-propanol+++
- d) 2-propanal

15. Which of these is not true about alcohols?

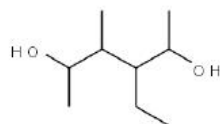
- a) The melting and boiling points of butanol is higher than ethanol
- b) Ethanol is more soluble in water than octanol
- c) Hexanol is more miscible with water than Methanol+++
- d) The boiling point of methanol is higher than that of ethane



16. LAH reacts with glyoxal, , to give

- a) $\text{CH}_3\text{CO}_2\text{H}$
- b) $\text{HOCH}_2\text{CH}_2\text{OH}+++$
- c) H_2CCH_2
- d) $\text{HOCH}_2\text{CO}_2\text{H}$

17. The correct name for the structure drawn is



- a) 4-isopropoxyl-3-methyl hexan-2-ol
- b) 4-ethyl-3-methyl-hexan-2, 5-diol+++
- c) Nonan-2, 5- diol
- d) 4-ethyl-2, 3, 5-trimethyl hexan-1, 6-diol

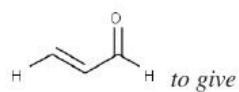
18. Which of these will not mix with water in all proportions?

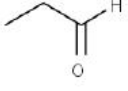
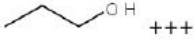
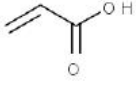
- a) Ethanol
- b) Methanol
- c) *t*-butanol
- d) Nonanol+++

19. LAH reacts with $\text{H}_2\text{CCHC(O)H}$ to give

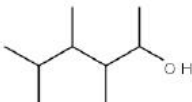
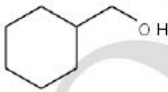
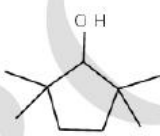
- a) $\text{CH}_3\text{CH}_2\text{C(O)H}$
- b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- c) $\text{H}_2\text{CCHCO}_2\text{H}$
- d) $\text{H}_2\text{CCHCH}_2\text{OH}+++$

20. Hydrogen (H_2), in the presence of Raney Ni catalyst, reacts with



- a) 
- b)  +++
- c) 
- d) 

21. Which of these is a tertiary alcohol?

- a) 
- b)  +++
- c) 
- d) 

22. Alcohol can be prepared by all the following ways except

- a) Addition of acid to alkene
- b) Addition of bromine water to alkene
- c) Addition of borane tetrahydrofuran to alkene followed by alkaline hydrogen peroxide
- d) Addition of Grignard reagent to alkene+++

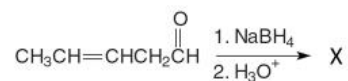
23. Which of these compounds has no carbonyl group present in it?

- a) Alkanals
- b) Amides

- c) Acid chlorides
d) Nitriles+++
24. In a carbonyl group, the oxygen atom is said to be
a) Electrophilic and reacts with nucleophiles
b) Nucleophilic and reacts with electrophiles+++
c) Electronegative and reacts with bases
d) Nucleophilic and reacts with bases
25. In the reduction of ketone to a secondary alcohol by either NaBH_4 or LAH, why will you prefer not to add aqueous acid in the first step?
a) It will reduce a ketone to a primary alcohol instead
b) There will be no reaction
c) Acid will have reacted with the reagent before the carbonyl has any chance to undergo reduction+++
d) It causes a serious explosion that can set the whole laboratory on fire
26. Reaction involving the use of lithium aluminium hydride (LAH) must always be in inert solvents because
a) LAH is inactive in any other solvents
b) The presence of reactive solvents will produce large amount of side reaction products
c) LAH will react with solvents having acidic hydrogen atoms+++
d) LAH is very reactive and catches fire easily
27. Why will you prefer LAH to NaBH_4 as reducing agent for carbonyl group?
a) LAH is cheaper than NaBH_4
b) NaBH_4 is too slow to reduce carbonyl group+++
c) NaBH_4 is not as selective as LAH
d) LAH is more environment-friendly than LAH
28. What are the products of the reaction of $\text{CH}_3\text{CH}_2\text{C}(\text{O})\text{OCH}_3$ with LAH followed by treatment with acid?
a) 1-propanol and methanol+++
b) Propanal and methanol
c) Ethanol and ethanoic acid
d) 2-propanol and methanol
29. Which of these reagents will give a saturated alcohol with an unsaturated aldehyde?

- a) $\text{NaBH}_4/\text{H}_3\text{O}^+$
- b) $\text{LAH}/\text{H}_3\text{O}^+$
- c) $\text{H}_2/\text{Raney Nickel}+++$
- d) CH_3MgBr

30. What is X in this reaction?



- a) $\text{CH}_3\text{CHCHCH}_2\text{COH}+++$
- b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COH}$
- c) $\text{CH}_3\text{CHCH}_2\text{CH}_2\text{COH}$
- d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHCOH}$

31. What is the function of H_3O^+ in the reaction of RMgX with carbonyl compounds?

- a) To produce magnesium salt of an alkoxide ion
- b) To convert the alkoxide ion to alcohol+++
- c) To neutralize the alkaline reaction medium
- d) To reduce the carbonyl group to alcohol

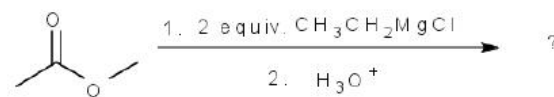
32. Reactions involving RMgX must be carried out in inert solvents like ether because

- a) Solvents having acidic protons will react with and destroy the reagent before it could react with the carbonyl compound+++
- b) Reactive solvents will interact with the reagent to form undesired products with the carbonyl compounds
- c) RMgX , being a form of salt, is very reactive toward solvents having acidic protons
- d) RMgX is a very sensitive reagent and becomes deactivated upon contact with reactive solvents.

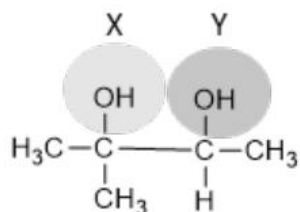
33. Which of these groups does not possess an acidic proton?

- a) $-\text{C}\equiv\text{CH}$
- b) $-\text{OH}$
- c) $-\text{O}-\text{CH}_3+++$
- d) $-\text{NH}_2$

34. What is(are) the product(s) of this reaction?



- a) 3-methyl-3-pentanone
 - b) 3-methyl-3-pentanone and methanol
 - c) 3-methylpentan-3-ol and methanol+++
 - d) 3-methylpentan-3-ol
35. What is the product of the reaction of 1 equivalent of a Grignard reagent with an ester, acyl chloride or carboxylate salt?
- a) Aldehyde
 - b) Alkanone+++
 - c) Alkanoic acid
 - d) Alcohol
36. Which of these reagents will oxidize primary alcohols directly to aldehydes?
- a) $\text{Cr}_2\text{O}_7^{2-}$
 - b) MnO_4^-
 - c) PCC+++
 - d) HIO_4
37. Oxidation of secondary alcohols in either acidic or basic conditions will give
- a) Aldehydes
 - b) Ketones+++
 - c) Ozonides
 - d) Carboxylic acids
38. What is the product you would obtain if you oxidize tertiary alcohol in either dichromate or permanganate?
- a) Alkanone only
 - b) No product+++
 - c) Alkanone and Alkanal
 - d) Alkanal only
39. Identify X and Y as different classes of alcohols present in the structure drawn



- a) X is 2° alcohol, Y is 3° alcohol
 - b) X is 2° alcohol, Y is 1° alcohol
 - c) X is 3° alcohol, Y is 2° alcohol+++
 - d) X is 3° alcohol, Y is 1° alcohol
40. A terminal ketone is otherwise known as
- a) Aldehyde+++
 - b) Alkanone
 - c) Alkanoic acid
 - d) Freon
41. The carbonyl group is capable of undergoing the following reactions except
- a) Oxidation
 - b) Reduction
 - c) Nucleophilic addition
 - d) Elimination+++
42. The preservative used for biological specimens contains
- a) Ethanol
 - b) Methanal+++
 - c) Formic acid
 - d) Trioxane
43. To prepare formalin I would dissolve
- a) 73 g formaldehyde in 17 g water
 - b) 37 g formaldehyde in 63 g water+++
 - c) 37 g formaldehyde in 100 g water
 - d) 63 g formaldehyde in 37 g water
44. A pure source of methanal is
- a) Dry formaldehyde
 - b) Trioxane+++

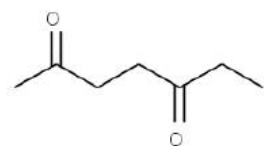
- c) *Dioxane*
d) *Acetone*
45. *Acetone is used in organic laboratories for cleaning glassware because it is*
a) *Very cheap*
b) *Highly volatile and polar+++*
c) *Colourless and non-viscous*
d) *Colourless and soapy*
46. *Which of these could wood varnishes contain?*
a) *Propanol*
b) *2-butanone+++*
c) *Benzene*
d) *Ethanol*
47. *CH₃CHO reacts first with ethyl lithium (CH₃CH₂Li) and then with dilute acid to give*
a) *ethane*
b) *butan-2-ol+++*
c) *butane*
d) *no reaction*
48. *Lithium reacts with n-butyl chloride to give*
a) *n-butane*
b) *n-butyllithium+++*
c) *chlorine*
d) *hydrogen*
49. *Formaldehyde (HCHO) reacts first with CH₃MgCl and then with dilute acid to give a*
a) *primary alcohol+++*
b) *secondary alcohol*
c) *tertiary alcohol*
d) *quaternary alcohol*
50. *The oxidation of CH₃CH₂OH with acidic dichromate (Cr₂O₇²⁻) gives*
a) *a ketone*
b) *ethyl alcohol*
c) *an aldehyde*

- d) an acid+++
51. $\text{CH}_3\text{CH}_2\text{OH}$ reacts with pyridinium chlorochromate (PCC) to give
- a) a ketone
 - b) an alcohol
 - c) an aldehyde+++
 - d) an acid
52. PBr_3 reacts with ethanol to give
- a) tribromoethane
 - b) bromoethanol
 - c) bromoethane+++
 - d) no reaction
53. Thionyl chloride, ClS(O)Cl , reacts with isopropyl alcohol, $(\text{CH}_3)_2\text{CHOH}$, to give
- a) 2-chloropropane+++
 - b) 1-chloropropane
 - c) 2-propanol
 - d) acetone (a ketone)
54. Isopropyl alcohol, $(\text{CH}_3)_2\text{CHOH}$, reacts with hot concentrated H_2SO_4 to give
- a) Propane
 - b) 2-propanol
 - c) Propene+++
 - d) Propyne
55. Aldehydes have ——— boiling points than/as alcohols of comparable molecular weight.
- a) higher
 - b) lower+++
 - c) about the same
 - d) exactly the same
56. Acetone is the common name for
- a) 1-butanone
 - b) 1-propanone
 - c) 2-propanone+++
 - d) 2-butanone

57. Which material is most reactive toward nucleophilic addition?
- a) Acetaldehyde+++
 - b) dimethyl ketone
 - c) diisopropyl ketone
 - d) di-tert-butyl ketone
58. The oxidation of 1-propanol with chromic acid yields
- a) an alcohol
 - b) an aldehyde
 - c) a ketone
 - d) an acid+++
59. The oxidation of 2-propanol with chromic acid yields
- a) an alcohol
 - b) an aldehyde
 - c) a ketone+++
 - d) an acid
60. The oxidation of 1-propanol with PCC gives
- a) an alcohol
 - b) an aldehyde+++
 - c) a ketone
 - d) an acid
61. The oxidation of ——— with ozone followed by reduction with $(\text{CH}_3)_2\text{S}$ yields one product, 1,5-pentanedial.
- a) 1,5-pentanediol
 - b) 2,4-pentanedione
 - c) Cyclopentene+++
 - d) cyclopentane
62. The acid-catalyzed hydration of cyclodecyne yields
- a) Cyclodecanone+++
 - b) cyclodecadiene
 - c) cyclodecanediol
 - d) cyclodecanedial
63. Reduction of acetic acid with $\text{Li}(\text{tri-tert-butoxy})\text{AlH}$ yields
- a) an acid
 - b) an aldehyde+++

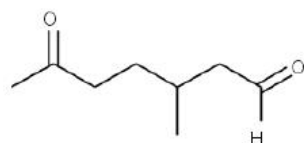
- c) an alcohol
d) a ketone
64. The reaction of acetyl chloride with excess Grignard reagent followed by acidification yields
- a) a ketone
b) an ester
c) a secondary alcohol
d) a tertiary alcohol+++
65. Lithium dimethylcuprate (Gilman reagent) reacts with acetyl chloride to give
- a) a ketone+++
b) an ester
c) a secondary alcohol
d) a tertiary alcohol
66. The cyanide ion $\text{C}\equiv\text{N}^-$ reacts with a ketone to give
- a) an acid
b) a cyanohydrin+++
c) an aldehyde
d) an amine
67. The acid-catalyzed addition of water to formaldehyde gives
- a) a ketal
b) a hemiacetal
c) an acetal
d) a hydrate+++
68. The acid-catalyzed addition of 2 mol of alcohol to 1 mol of aldehyde yields
- a) a ketal
b) a hemiacetal
c) an acetal+++
d) a hydrate
69. Ammonia reacts with an aldehyde to give an
- a) alcohol
b) amide
c) imine+++
d) amine
70. The reaction of a Wittig reagent with ketones give

- a) Alkenes+++
 - b) ketones
 - c) aldehydes
 - d) alkanes
71. Silver ion, Ag^+ , reacts with aldehydes to form
- a) a red precipitate
 - b) a silver "mirror"+++
 - c) a blue solution
 - d) there is no reaction
72. Dimethylamine reacts with a ketone to give an
- a) amide
 - b) imide
 - c) amine
 - d) enamine+++
73. Carbocations are
- a) carbon atoms with a negative charge
 - b) carbon atoms with a positive charge+++
 - c) carbon atoms with no charge
 - d) any atom with a positive charge
74. A—— carbocation has the greatest stability.
- a) methyl
 - b) primary
 - c) secondary
 - d) tertiary+++
75. Carbocations are stabilized by adjacent groups that—— the carbocation.
- a) donate electrons to+++
 - b) withdraw electrons from
 - c) donate protons to
 - d) withdraw protons from
76. What is the name of this compound?



- a) 2,5-heptanedione+++
- b) 2,5-heptadione
- c) 3,6-heptadione
- d) 3,6-heptanedione

77. The correct name for this compound is



- a) 3-methyl-6-oxo heptanal+++
- b) 3-methyl-1,6-heptanedione
- c) 2-keto-5-methyl-heptanal
- d) 5-methyl-2,7-heptanedione

78. What is responsible for the high boiling points of aldehydes and ketones?

- a) Hydrogen bonding existing between the molecules
- b) Intermolecular forces of attraction between molecules
- c) Polar interaction between molecules+++
- d) Dispersion forces between molecules

79. The Markovnikov rule states that a hydrogen atom bonds to the carbon atom that already is bonded to—— hydrogen atoms.

- a) an equal number of
- b) the greatest number of+++
- c) the least number of
- d) no

80. Carbocations undergo rearrangement reactions

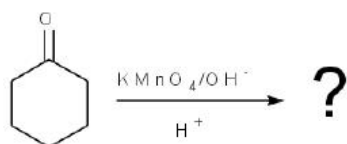
- a) to give more stable carbocations
- b) by hydride shifts
- c) by methyl shifts
- d) for all of the above reasons+++

81. Alkenes undergo an acid-catalyzed addition of water to give
- alkynes
 - alkanes
 - ethers
 - alcohols+++
82. Borane reacts with alkenes
- to give the anti-Markovnikov addition product
 - by a concerted mechanism
 - to give syn addition
 - all of the above+++
83. Diols are synthesized by reacting an alkene with
- water
 - borane
 - ozone
 - osmium tetroxide+++
84. Ozone reacts with 2,3-dimethyl-2-butene to give
- one ketone+++
 - two different ketones
 - one aldehyde
 - one ketone and one aldehyde
85. The rapid conversion between an enol and its keto form is called
- resonance
 - cationic rearrangement
 - tautomerization+++
 - hydrogenation
86. The solubility of some aldehydes and ketones in water is due to
- Hydrogen bonding between the molecules of water and those of aldehydes or ketones
 - Polar interaction and hydrogen bonding between the molecules of water and those of aldehydes or ketones+++
 - The intermolecular forces between the molecules of water and those of aldehydes or ketones
 - Dispersion and Van der Waal forces between the molecules of water and those of aldehydes or ketones

87. Aldehydes are said to be more reactive than ketones because
- a) The carbonyl carbon in the former is more polar than that of the latter
 - b) The carbonyl carbon in the latter is more sterically hindered from nucleophilic attack than in the former+++
 - c) The carbonyl carbon in the former is a terminal one hence better contact with nucleophiles unlike in the latter where the carbonyl carbon is not easily available for attack
 - d) The tetrahedral structure of the former possesses higher potential energy than the latter
88. Which of these has the most stable central carbon?
- a) $RCOR'$ +++
 - b) $RCOH$
 - c) RCH_3
 - d) CH_4
89. The intermediate formed during the reaction of ozone with unsaturated compounds is called
- a) Oxidizing agent
 - b) Ozonide+++
 - c) Trioxane
 - d) Peroxide
90. Reaction of alkene with ozone in the presence of hydrogen peroxide will produce
- a) Two acids
 - b) Ketone and acid
 - c) Two ketones
 - d) All of the above+++
91. Ozonolysis is a very important reaction in chemistry because
- a) It is not toxic
 - b) It can be used to characterize complex molecules+++
 - c) It takes place at room temperature
 - d) It is a means of reducing complex molecules
92. Which of these is not a way of preparing aldehydes and ketones?
- a) Hydration of alkynes
 - b) Reduction of acid chlorides

- c) Acylation reaction
- d) Amination reaction+++

93. What is the product of this reaction?



- a) Heptan-1,6-dioic acid+++
- b) Cycloheptan-1,6-dioic acid
- c) Ethanoic acid and pentane
- d) Cycloheptanoic acid

94. How many isomers will this compound have?



- a) 1
- b) 2
- c) 3+++
- d) 4

95. Which of these has the highest boiling point?

- a) Butane
- b) Butanal
- c) Butanol+++
- d) Butanone

96. What is the product you would get from the hydration of $\text{CH}_3\text{C}\equiv\text{CCH}_2\text{CH}_3$ using mercury salt?

- a) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$
- b) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$
- c) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ +++
- d) $\text{CH}_3\text{CH}_2\text{C}(\text{OH})\text{CH}_2\text{CH}_3$

97. Aromatic ketones are prepared by

- a) Wittig reactions
- b) Friedel-Craft acylation+++

- c) Clemmensen reaction
 - d) Mercuration reaction
98. Cyanohydrins can be hydrolysed in the presence of acid to produce
- a) Compounds with both hydroxyl and carboxylic groups+++
 - b) Beta-Amino alcohols
 - c) Aldehydes
 - d) Ketones
99. What is the product of the reaction of water with ketones?
- a) Acetal
 - b) Hydrate+++
 - c) Hemiacetal
 - d) Geraniol
100. What is the final product of the reaction of alcohol with aldehyde called?
- a) Hemiacetal
 - b) Acetal+++
 - c) Hydrate
 - d) Ketal

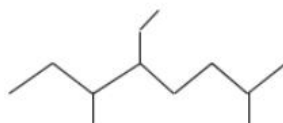
SECTION 2

1.

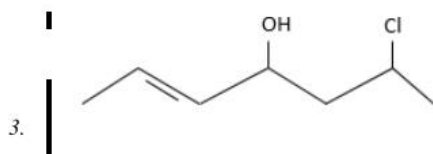


- A. Hex-1-en-6-yne
- B. Hex-1-en-1-yne
- C. Hept-1-en-6-yne+++
- D. Hept-6-en-1-yne

2.



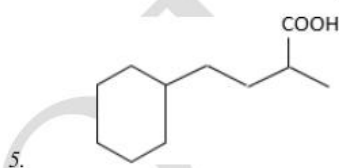
- A. 4-Ethyl-3, 7- dimethyl octane
- B. 5-Ethyl-2, 6-dimethyl octane+++
- C. 4-Ethyl-3, 6-dimethyl heptane
- D. 5-Ethyl-3, 7-dimethyl heptane



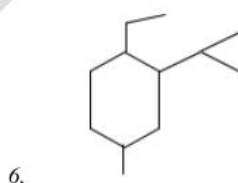
- A. 6-chloro hept-2-en-4-ol
- B. 2-chloro hept-5-en-4-ol+++
- C. 2-chloro hept-6-en-4-ol
- D. 2-chloro-4-hydroxy hept-5-ene



- A. 4-tertButyl hexanal+++
- B. 3-tertbutyl hexanal
- C. 4-Ethyl-5, 5-dimethyl hexanal
- D. 3-tertButyl hexanal

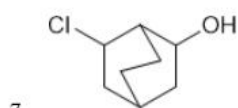


- A. 5-cyclotexyl carboxylic acid
- B. 3-cyclohexyl carboxylic acid
- C. 2-Ethyl-3-cyclohexyl butanoic acid
- D. 2-Methyl-4-cyclohexyl butanoic acid+++

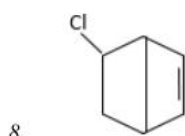


- A. 1-Ethyl-5- methyl-2-isopropyl cyclohexane
- B. 4- Ethyl-3-isopropyl-1-methyl cyclohexane
- C. 2- Ethyl-1-isopropyl-5-methyl cyclohexane

D. 1-Ethyl-2-isopropyl-4-methyl cyclohexane+++



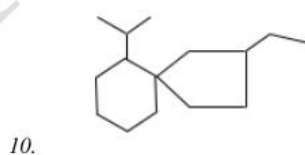
- A. 3-chloro -2-hydroxy bicyclo [2,2,2]-octane
- B. 6-chloro bicyclo [2,2,2] octan-2-ol+++
- C. 2-chloro bicyclo [2,2,2] octan-2-ol
- D. 5-chloro bicyclo [2,2,2] octan-3-ol



- A. 5-chloro bicyclo [2,2,0] hex-2-ene+++
- B. 2-chloro bicyclo [2,2,0] hex-5-ene
- C. 2-chloro bicyclo [2,2,0] hex-2-ene
- D. 5-chloro bicyclo [2,2,0] hex-1-ene



- A. 4-Ethyl spiro[4,5] nonane
- B. 1-Ethyl spiro[4,4]nonane+++
- C. 4-Ethyl spiro[4,4]octane
- D. 1-Ethyl spiro[5,5]nonane



- A. 2-isopropyl-8- Ethyl spiro [5,4] decane
- B. 2-Ethyl-6-isopropyl spiro [4,5] decane+++
- C. 3-Ethyl-7-isopropyl spiro [4,5] decane
- D. 3-Ethyl-10-isopropyl spiro [4,5] decane

11. hexene and 2-hexene are:

- A. Functional group isomers
- B. Chain isomers
- C. Alkene isomers
- D. Position isomers+++

12. These two compounds are:



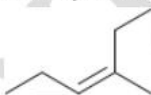
- A. Position isomers
- B. Functional group isomers+++
- C. Ether-alcohol isomers
- D. Chain isomers

13. These are:



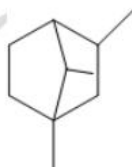
- A. Cycle-alkene isomers
- B. Functional group isomers
- C. Chain-cycle isomers+++
- D. Position isomers

14. The name of this compound is



- A. trans-2- methyl hex-3-ene
- B. Trans-3- methyl hex-3-ene
- C. Cis-3- methyl hex-3-ene+++
- D. Trans -2- methyl hex-2-ene

15. How many chiral centres are there in the structure?



- A. 1
- B. 2
- C. 3
- D. 4+++

16. An optical isomer has all except

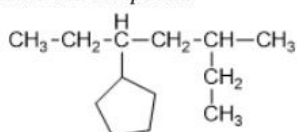
- A. has a chiral centre
- B. rotates plane polarised light

- C. has a mirror image
 D. is super imposable on its mirror image+++

17. A compound has 8 carbon atoms only six of the carbons are methyl radicals. The name of the compound is:

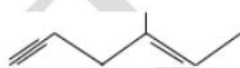
- A. 1,2-hexamethyl ethane
 B. 2,4-tetramethyl butane
 C. 2,2,3,3-tetramethyl butane+++
 D. 1,2,3,4-tetramethyl butane

18. Name the compound



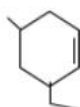
- A. 2-Ethyl-4- cyclopentyl hexane
 B. 3-Methyl-4- cyclopentyl heptane
 C. 3-Cyclopentyl-5-methylheptane+++
 D. 3- Methyl-6- cyclopentyl heptanes

19. Name the compound



- A. 4-Methyl hex-2-en-5-yne
 B. 4-Methylhex -1-yne-4-ene
 C. 4-methyl hex-4-en-1-yne+++
 D. 4-methyl hex-2-en-1-yne

20. This compound is



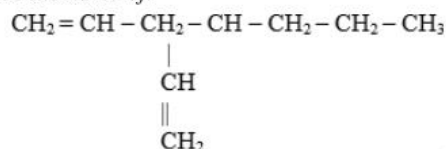
- A. 2-Ethyl-4-methyl cyclohexene
 B. 3-Ethyl-5-methyl cyclohexene+++
 C. 5-Ethyl-3-methyl cylohexene
 D. 1-Ethyl-5-methyl cyclohex-2-ene

21. Chain isomers are: compounds that have the same molecular

formula but differ in ---

- A. Optical arrangement
- B. Functional groups
- C. Spatial arrangement of atoms
- D. Bonding arrangement of radicals+++

22. Give the name of:



- 4-vinyl heptene
- 4-allylhexene
- 3-propylhexa-1, 5-diene+++
- 4-propyl hexa-1, 5-diene

23. How many structural isomers can you write for C_5H_{12}

- A. 4
- B. 3+++
- A. 2
- B. 5

24. How many saturated cycles can you write from C_5H_{10}

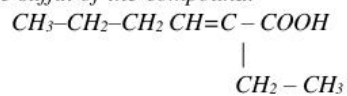
- A. 5
- B. 4
- C. 2
- D. 3+++

25. Which experiment will distinguish between



- A. Ignition
- B. Sodium
- C. Bromination+++
- D. Solubility test

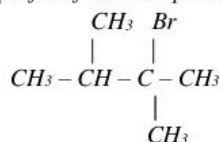
26. The suffix of the compound:



- A. Hexenoic acid
- B. Hex-2-enoic acid+++
- C. Hex-4-enoic acid

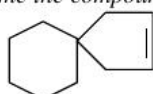
D. Hex-3-enoic acid

27. The prefix of this compound:



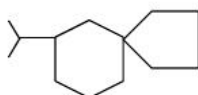
- A. 3-Bromo-1, 3-dimethyl
- B. 2,3-dimethyl-3-bromo
- C. 2-Bromo-2, 3-dimethyl+++
- D. 2-Bromo-2-Isoprophyl

28. Name the compound



- A. Spiro {4,5} dec-2-ene+++
- B. Spiro {4,6} dec -2- ene
- C. Spiro {5,6} dec-2-ene
- D. Spiro {4,5} dec -1- ene

29. Pick the correct name for:



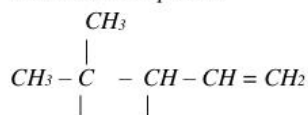
- A. Spiro {4,5} noriane
- B. 2-isopropyl spiro {4,5} nonane
- C. 3-isopropyl spiro {4,5} nonane
- D. 7-isopropyl spiro {4,5} nonane+++

30. This compound is:



- A. 1-ethyl bicyclo [2,1,1]- hexane
- B. 2-Ethyl bicyclo [211]- hexane+++
- C. 5-Ethyl bicyclo [2,1,1]- hexane
- D. 3-Ethyl bicycle [2,1,1] – hexane

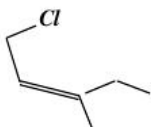
31. Name the compound





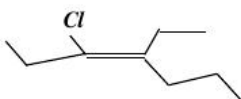
- A. 2,2 – Dimethyl-3-Vinyl pentene
- B. 2,2-Dimethyl-3-Vinyl pent-5-ene
- C. 3-tertbutylpenta-1, 4-diene+++
- D. 4,4-Dimethyl-3-Vinylpentene

32. The name of this compound is:



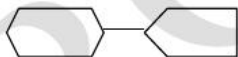
- A. 5-chloro-3-methyl pentene
- B. 5-chloro-3-methyl pent-3-ene
- C. 1-chloro-2-methyl pent-2-ene
- D. 1-chloro-3-methyl pent-2-ene+++

33. This compound is



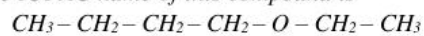
- A. 3-chloro-4-ethyl hept-3-ene+++
- B. 3-chloro-3-ethyl hept-3-ene
- C. 4-chloro-3-propyl hex-3-ene
- D. 5-chloro-4-ethylhept-4-ene

34. What is the IUPAC name of the compound?



- A. Cyclohexane cyclopentane
- B. Cyclohexylcy clopentane
- C. Cyclopentyl cyclohexane+++
- D. Cyclopentane cyclohexane

35. The IUPAC name of this compound is



- A. Ethyl butyl ether
- B. Butyl ethyl ether
- C. Butoxy ethane
- D. Ethoxy butane+++

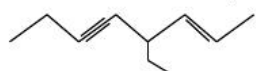
36. Which of these compounds has a chiral carbon?

- A. Chleropentane
- B. 3-chloropentane
- C. 2-chloropentane+++
- D. Chloro-2, 2-dimethylpropane

37. One of these compounds will not be optically active

- A. Chlorobromol odometane
- B. Chloro-2-methyl propane
- C. 2-chloro-2-methyl propane
- D. 2-chloropropane+++

38. What is the IUPAC name of this compound?

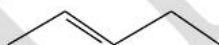


- A. 4-Ethyl oct-6-en-3-yne
- B. 4-Ethyl oct-2-en-5-yne+++
- C. 5-Ethyl oct-2-en-5-yne
- D. 4-Ethyl oct-2-en-3-yne

39.



a.

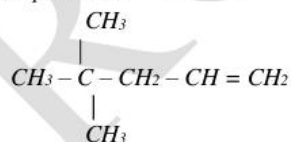


b.

a and b are:

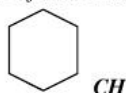
- A. Functional group isomers
- B. Tautomers
- C. Alkene isomers
- D. Position isomers+++

40. This compound is:

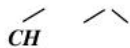


- A. 2,2-dimethyl pentene
- B. 4,4-dimethyl pentene+++
- C. 2,2-dimethyl pent-4-ene
- D. 4,4-dimethyl pent-4ene

41. The name of this compound is:

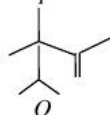


CH



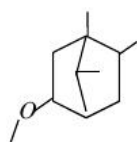
- A. 1,3-dimethyl hexane
- B. 1,3-dimethyl cyclohex-2-ene
- C. 1,3-dimethyl cyclohexene+++
- D. 1,3-dimethyl hex-1-ene

42. Name the compound below:



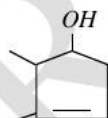
- A. 2-Methyl-2-isopropyl 1 butanone
- B. 2-Methyl-2-isopropyl butan-2-one
- C. 3,3,4-trimethyl pentan-3-one
- D. 3,3,4-trimethyl pentan-2-one+++

43. Name the Compound:



- A. 1,2,7-trimethyl-3-methoxy bicycle [2,2,10] heptane
- B. 1,2,7-trimethyl bicycle [2,2,1] heptanes
- C. 1,2,7-trimethyl-5-methoxy bicyclo [2,2,1] heptanes+++
- D. 1,2,7-trimethyl-4-methoxy bicycle [2,2,1] heptanes

44. What is the name of the compound?



- A. 1,3-dimethyl pent-3-enol
- B. 1,3-dimethylpent-2-enol
- C. 2,3-dimethyl pent-3-enol+++
- D. 1,3-dimethyl penten-3-ol

45. Give the name of the compound.

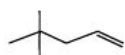


- A. 3 – secbutyl hept -1- en-4-yne+++
- B. 5-isopropyl hept-1-en-3-yne
- C. 3- isopropyl hept -6-en-3-yne
- D. 4-vingl-3-methyloet -3-yne

46. $\text{CH}_2=\text{CHCl}$ is

- A. Allychloride
- B. Ethyl chloride
- C. Vinyl chloride+++
- D. Chloro ethylidene

47. Name the compound

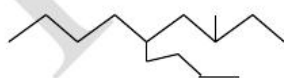


- A. 2,2-dimethyl pentene
- B. 4,4-dimethyl pentene+++
- C. 2,2-dimethyl pent-4-ene
- D. 4,4-dimethyl pent-2-ene

48. The different parts of a systematic name include all of these except

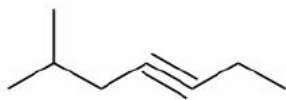
- A. suffix
- B. Principal chain
- C. Straight chain+++
- D. prefix

49. The prefix for this compound



- A. 3-Butyl-5-methyl.
- B. 5-butyl-7-methyl
- C. 3-methyl-4-butyl
- D. 5-butyl-3-methyl+++

50. The name of this compound is:



- A. 2-methylheptyne
- B. 2-methylhept-3-yne
- C. 2-methylhept-4-yne
- D. 6-methylhept-3-yne+++

51. The melting point of members of carboxylic acid homologous series
- A increases up the services
 - B decreases up the series
 - C the even carbon acids have higher melting points than their immediate higher and lower members+++
 - D show no definite trend in the series
52. The melting points of structural isomers of carboxylic acid series aretheir normal chain counterpart
- (A) Lower than+++
 - (B) higher than
 - (C) the same as
 - (D) not defined
53. Carboxylic acids have higher boiling points due to
- (A) the alkyl chain
 - (B) hydrogen bonding+++
 - (C) carboxyl group
 - (D) carboxylic acid bonding
54. Hell Volard Zelinsky reaction is
- (A) regioselective+++
 - (B) regiospecific
 - (C) stereoselective
 - (D) stereospecific
55. Hell-Volard Zelinsky reaction substitutes only
- (A) α -carbon atom
 - (B) β -carbon atom
 - (C) α -hydrogen+++

- (D) β -hydrogen
56. In Hell-Volard-Zelirsky a catalytic amount of.....is needed
- (A) sodium
- (B) potassium
- (C) red phosphorus+++
- (D) bromonium ion
57. Which of these will reduce carboxylic acid to alcohol?
- (A) LiAlH_4 +++
- (B) NaBH_4
- (C) H_2/Pd
- (D) H_2/Ni
58. ClCH_2COOH is more acidic than CH_3COOH by reason of
- (A) resonance effect
- (B) conjugative effect
- (C) cummulative effect
- (D) inductive effect+++
59. When the hydrogen of the $-\text{COOH}$ is replaced by an alkyl group (R) it becomes (A) an amide
- (B) an ester+++
- (C) a ketone
- (D) an alkane
60. The end result of the reduction of carboxylic acid by borane (B_2H_4) in
- (A) an aldehyde
- (B) a ketone
- (C) an alcohol+++
- (D) an alkane
61. Benzene can be differentiated from an alkene by the reaction
- (A) it burns with sooty lumminious flame
- (B) it boils at 78°C
- (C) it does not decolourise bromine water+++
- (D) it melts at 5.5°C
62. Which of these groups will be good meta directing group
- (A) $-\text{NH}_2$

- (B) $-CH_3$
(C) $-NO_2$ +++
(D) $-Cl$
63. Benzene is insoluble in
(A) chloroform
(B) ethanol
(C) water +++
(D) acetone
64. Benzene can be produced from
(A) ethylene
(B) acetylene +++
(C) methylene
(D) propylene
65. Benzene is resistant to oxidation and reduction because double bonds are:
(A) three
(B) cumulated
(C) additive
(D) conjugated +++
66. The name of this compound
- CC(=O)Oc1ccccc1
- (A) acetic benzene acid
(B) benzyl acid
(C) benzoic acid
(D) 2-phenyl acetic acid +++
67. Benzene ring undergoes
(A) substitution reactions +++
(B) oxidation reaction
(C) addition reaction
(D) reduction reaction
68. Benzene is most available from this
(A) petroleum +++

- (B) natural products
(C) wood
(D) saw dust
69. Benzene is oxidized readily with
(A) ozone+++
(B) oxygen
(C) chlorine
(D) potassium permanganate
70. Acetophenone is easily prepared by
(A) Friedel-Crafts benzoylation
(B) Friedel-Cryts alkylation
(C) Friedel-Crafts acylation+++
(D) phenylation of acetone

SECTION 3

1. The periodic table is a chart that arranges element according to
A. Atomic mass m
B. Atomic number Z +++
C. Molecular mass
D. None of the above
2. The principal use of diamond is in
A. Lighting
B. Jewelry +++
C. Lead pencil
D. Saw
3. The ability of metals to conduct electricity is due to
A. Surface tension
B. Smooth surface
C. Presence of free mobile electrons+++

- D. High tension*
4. The reason why graphite is used as electrodes is due to;
- A. Absence of free mobile electrons*
 - B. Presence of free mobile π electrons+++*
 - C. Rigid structure*
 - D. Flexible structure*
5. Which of these elements is not alkaline - earth metal?
- A. Ba*
 - B. Sr*
 - C. Mg*
 - D. Pd+++*
6. Which of these elements is not an alkali metal?
- A. Li*
 - B. Na*
 - C. Ba+++*
 - D. Cs*
7. What is an alloy?
- A. A miscible mixture of two or three metals at their molten states+++*
 - B. Two nonmetals*
 - C. Nonmetals*
 - D. Gases.*
8. The Morseley's equation is;
- A. $E = mc^2$*
 - B. $\sqrt{v} = a(Z - b) +++$*
 - C. $PV = mc^2$*
 - D. $PV/T = \text{constant}$*

9. Metals on ionization become charged with smaller
- A. Sizes +++
 - B. Sites
 - C. Sources
 - D. Momentum
10. Diamond has which hybridization?
- A. Sp
 - B. Sp^2
 - C. Sp^3 +++
 - D. Spd
11. Is graphite convertible to diamond?
- A. Yes +++
 - B. No
 - C. Uncertain
 - D. None of the above
12. Electrolysis of water produces;
- A. Hydrogen and oxygen +++
 - B. Nitrogen and coke
 - C. Colanut
 - D. Bitter cola
13. What is a crystal?
- A. A solid with lattice points occupied by ions, atoms or molecule +++
 - B. Glass
 - C. Cold room
 - D. Liquid
14. Example of amorphous solid is;

- A. Gold
- B. Glass +++
- C. Silver
- D. Mites

15. What is a metal?

- A. A malleable, ductile, lustrous solid that conducts electricity and heat. +++
- B. Insulator
- C. Perforator
- D. Plate

16. The characteristics of alkali metal compounds is that they are;

- A. White and form colourless solution in water +++
- B. Insoluble in water
- C. Black in colour
- D. None of the above

17. Which of these materials is not a covalent crystal?

- A. Copper+++
- B. Graphite
- C. Diamond
- D. Quartz

18. Period 1 contains;

- A. H and He+++
- B. Uranium and other actinides
- C. Lanthanides
- D. Halogens

19. Group 1 elements are;

- A. Copper, silver and gold

- B. Iron, manganese and cobalt*
- C. Lithium, sodium, potassium, rubidium, caesium and francium*
+++
- D. Hydrogen and helium*
20. An isotope has;
- A. Element with two or more different atomic masses* +++
- B. Element with different atomic number*
- C. Element without mass*
- D. Element with high electron affinity*
21. Group 18 elements like Ar, He, Ne and Kr are known as;
- A. Noble gases*+++
- B. Noble metals*
- C. Precious stones*
- D. Alkali*
22. Sucrose, ice cube, sulphur, phosphorus and iodine are known as
- A. Metallic crystals*
- B. Molecular crystal*+++
- C. Jasmine*
- D. Ionic crystals*
23. Group 17 elements like I, Br, F, Cl are known as
- A. Hydrogen*
- B. Halogens*+++
- C. Petrogens*
- D. Androgens*
24. The lattice points of ionic crystals are occupied by;
- A. Molecules*
- B. Tubers*

- C. Charged ions+++*
- D. Chronicles*
25. Which of this is the property of diamond?
- A. Hardness+++*
- B. Softness*
- C. Brittleness*
- D. Ductility*
26. Metallic crystals contain;
- A. Molecules*
- B. Charged ions of the metal in a sea or cloud of electrons +++*
- C. Butter*
- D. Yam*
27. Electrovalent compounds are held together by;
- A. Molecular forces*
- B. Electrovalent forces+++*
- C. Covalent forces*
- D. Military forces*
28. The major use of graphite is;
- A. Electrodes+++*
- B. Portal*
- C. Primer*
- D. Emerald*
29. Diamond is used primarily as;
- A. Jewelry+++*
- B. Pavement*
- C. Food*
- D. Farm*

30. Which of these elements will liberate hydrogen from cold water?

- A. Phosphorus
- B. Sulphur
- C. Potassium+++
- D. Purine

31. Electrolysis is

- A. Use of electrical current to break electrolytic compounds in molten or aqueous states to their constituents elements+++
- B. Use of hammer to break substances
- C. Use of adhesive to bind surfaces
- D. Use of water to spray.

32. Which of these is not an alkali element?

- A. lithium
- B. Gold+++
- C. Sodium
- D. Potassium

33. Which of these elements will displace hydrogen from cold water?

- A. Rubidium+++
- B. Copper
- C. Gold
- D. Tungsten

34. The group in a periodic table shares;

- A. Similar electronic configuration physical and chemical properties+++
- B. Different properties.
- C. Different mobility
- D. Different tools.

35. What force holds covalent compounds together?

- A. Ionic
- B. Van der Waal+++
- C. Cubic
- D. Rectangular
36. Which of these are allotropes of carbon?
- A. Diamond and graphite+++
- B. Procter and Gamble
- C. Silver
- D. Solar
37. Gold is a
- A. Bullet
- B. Malt
- C. Metal+++
- D. Glass
38. Which of these is the characteristic of Transition elements?
- A. Partly filled d-orbitals+++
- B. Fully filled life cycle
- C. Non metals
- D. Polarizable electrodes
39. Which of these is not the characteristic of lanthanides and actinide?
- A. Non metals+++
- B. Radioactive
- C. Partially filled f-orbitals
- D. Synthetic elements
40. Which of these elements is not a transition metal?
- A. Iron
- B. Manganese

C. Zinc

D. Fluorine+++

41. Which of these elements is not a halogen?

A. Fluorine

B. Chlorine

C. Iodine

D. Boron+++

42. Which of these substances is not a good conductor of electricity?

A. Copper

B. Silver

C. Iron

D. Diamond+++

43. Why is graphite a good conductor of electricity?

A. It contains mobile π electrons+++

B. It is rigid

C. It flows

D. It is a gas.

44. When an element with high electron affinity gains an electron to become an anion; its size;

A. Decreases+++

B. Does not change

C. Increases

D. Not certain.

45. The ions in ionic crystal lattice have;

A. Same size

B. Different sizes+++

C. Unknown sizes

- D. Large sizes.*
46. Which of these substances is amorphous?
- A. Glass+++*
 - B. Sodium chloride*
 - C. Calcium nitrate*
 - D. Diamond*
47. The molecular crystals have this property;
- A. Melt at below 100 °C+++*
 - B. Very hard solid*
 - C. Flows*
 - D. Fluid*
48. Can diamond be converted to graphite?
- A. Yes*
 - B. No+++*
 - C. Not certain*
 - D. Possible*
49. Halogens are highly reactive because they;
- A. Have high electro affinity+++*
 - B. Low ionization energy*
 - C. Low boiling point*
 - D. Inert*
50. The reactivity of alkali elements is due to;
- A. Low ionization energy+++*
 - B. High electron affinity*
 - C. Inertness*
 - D. Not sure*
51. Mendeleev's Periodic Table was based on:

- A. Atomic mass+++
B. Ionization energy.
C. Atomic radius
D. Ionic radius.
52. NaCl is an ionic compound because
- A. Ions are held in lattice+++
B. It is covalent molecule
C. The melting point is below freezing point.
D. It is formed by sharing of electrons
53. Morseeley's Periodic Table was rather based on:
- A. Covalent radius
B. Atomic number+++
C. Electron sharing
D. Electrode potentials
54. Metal structure has a sea of delocalized electrons functioning for:
- A. Conduction of electricity+++
B. Covalent bonding
C. Molecule
D. Electron sharing.
55. Why are members of the same group having similar physical and chemical properties?
- A. They have similar electronic configuration+++
B. They conduct heat
C. They are able to conduct electricity
D. None at all
56. Alkali metals are highly electropositive because they
- A. Hold tightly their outermost electrons.

- B. Easily release their outermost electrons for ionization+++*
- C. Have moisture.*
- D. They are covalent.*
57. Ice, Sulphur and iodine are examples of:
- A. Molecular crystals+++*
- B. Solid with high melting points.*
- C. No lattice*
- D. None of the above.*
58. Graphite, diamond and quartz are examples of:
- A. Brittle materials*
- B. Strongly covalent crystals+++*
- C. No rigid structures*
- D. Crude solids*
59. Diamond has:
- A. sp^3 hybridisation+++*
- B. Sp hybridization*
- C. Spd hybridization*
- D. $Spdf$ hybridization.*
60. Graphite is mostly used as:
- A. Glass cutter*
- B. Electrode+++*
- C. Jewelry*
- D. Brittle material.*
61. Diamond is an important:
- A. Jewelry+++*
- B. Lead pencil*
- C. Lubricant*

D. Braille.

62. Which of these elements is not an alkali metal?

A. Gold+++

B. Sodium

C. Caesium

D. Rubidium

63. Which of these elements is not an alkaline earth metals?

A. Sulphur+++

B. Calcium

C. Barium

D. Magnesium.

64. Covalent compounds are held together by:

A. van der waal forces+++

B. Collagen

C. Gum

D. Ionic forces.

65. Chlorine, fluorine, bromine, iodine and astatine belong to the group of:

A. Chalcogen

B. Halogen+++

C. Transition metals

D. Inner transition metals.

66. Lanthanides and Actinides have:

A. Partially filled outermost f-orbitals+++

B. No orbital.

C. S-orbital only

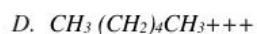
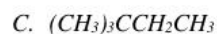
D. None of the above.

67. Noble gases have the following properties:

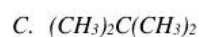
- A. Vacancies in their in their s-orbitals
- B. No vacancies in their outermost s- or p-orbitals+++
- C. Vacancies in p-orbitals
- D. Vacancies in d-orbitals.
68. Elements in Period 7 in the Periodic table are unstable with short half-lives because they are:
- A. Synthetic and elements+++
- B. Naturally occurring
- C. Stable
- D. None of the above
69. Helium, xenon, neon, krypton and argon in Group 18 are called:
- A. Noble gases+++
- B. Chalcogens
- C. Halogens
- D. Actinides
70. Is graphite convertible to diamond?
- A. Not sure
- B. No
- C. Yes+++
- D. Uncertain

SECTION 4

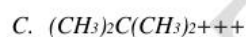
24. Which of the following alkanes has the lowest boiling point?
- A. $(CH_3)_2CHCH(CH_3)_2$
- B. $(CH_3CH_2)_2CHCH_3$
- C. $(CH_3)_3CCH_2CH_3$ +++
- D. $CH_3(CH_2)_4CH_3$
25. Which of the following alkanes has the highest boiling point?



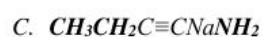
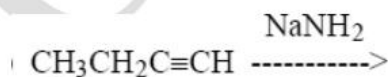
26. Which of the following compounds has the highest boiling point?



27. Which of the following compounds has the lowest boiling point?



28. Predict the product or products of this reaction.

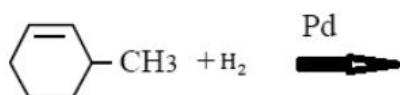


29. Predict the product or products of this reaction.



- A. $(\text{CH}_3\text{CH}_2)_3 \text{Br} + \text{HBr}$
- B. $(\text{CH}_3\text{CH}_2)_3 \text{CBr} + \text{HBr} \text{ +++}$
- C. $(\text{CH}_3\text{CH}_2)_3 \text{HBr} + \text{HBr}$
- D. $(\text{CH}_3\text{CH}_2)_2 \text{Br} + \text{HBr}$

30. Predict the product or products of this reaction



- A. +++
- B.
- C.
- D.

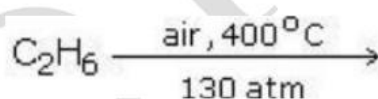
31. What are hydrocarbons?

- A. Compounds containing only the elements carbon called hydrocarbons
- B. Compounds containing only the elements carbon and oxygen are called hydrocarbons
- C. Compounds containing the elements carbon, hydrogen and oxygen are called hydrocarbons
- D. Compounds containing only the elements carbon and hydrogen are called hydrocarbons +++

32. Choose one which does not belong to this set: C_2H_2 , C_3H_4 , C_3H_6

- A. C_3H_6 +++
- B. C_2H_2

- C. C_3H_6
D. C_3H_4
33. An organic compound A with the formula C_2H_6O gives the compound B with the same number of carbon atoms when oxidized, which further undergoes oxidation to acetic acid. Identify the compound A.
- A. Acid
B. Ethanol+++
C. Aldehyde
D. Ketone
34. What are the products of reaction between ethanol and concentrated sulphuric acid at $170^\circ C$?
- A. Ethene
B. Propene and water
C. Ethene and water+++
D. Butene
35. In a molecule of a hydrocarbon, the number of 'C' atoms is '5'. What will be its formula if it is an alkyne?
- A. C_5H_{10}
B. C_5H_{12}
C. C_5H_8 +++
D. C_5H_{14}
36. What would be the product of reaction this reaction below



- A. Ethane
B. C. Benzene
C. C. Acetic acid+++
D. D. Ethanol
37. Which of the following will not give addition reactions; Propane, propyne, propene, benzene.
- A. Propane+++
B. Propyne
C. Propene
D. Benzene
38. Which one of the following compounds is not expected to be completely soluble in water at room temperature.
- A. $CH_3CH_2CH_2OH$

- B. CH_3COOH
- C. $\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$
- D. $\text{CH}_3\text{+++}$

39. The functional group of a molecule does not affect a compound's

- A. solubility
- B. melting point
- C. reaction with other molecules
- D. carbon chain length+++

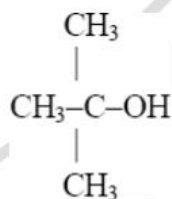
40. The combustion of a hydrocarbon will produce mandatory:

- A. SO_2
- B. $\text{CO}_2\text{+++}$
- C. CO
- D. SO_3

41. Which of the following substances presents only primary carbon atoms in its structure?

- A. methane
- B. acetylene
- C. ethane+++
- D. propane

42. The formula of the compound below represents



- A. primary alcohol
- B. secondary alcohol
- C. tertiary alcohol+++
- D. quaternary alcohol

43. What can result by the reduction of an aldehyde?

- A. a primary alcohol+++
- B. a secondary alcohol
- C. a carboxylic acid
- D. a compound with a mixed function alcohol – aldehyde

44. What can result by the reduction of a ketone?

- A. a primary alcohol
- B. a secondary alcohol+++
- C. a carboxylic acid
- D. a compound with a mixed function alcohol – ketone

45. The alcoholic fermentation is a reaction of:

- A. dehydration
- B. hydrolysis
- C. esterification